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PATTERN AND SCALING OF REGIONAL CEREBRAL GLUCOSE METABOLISM IN MAMMALS

by

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Abstract

The human brain constitutes approximately 2% of total body mass, but it consumes 20%-25% of the body's total energy production. Neuroimaging studies have revealed that the brain is metabolically heterogeneous – different regions of the brain use different amounts of energy. However, little is known about the scaling of metabolism among brain components in different species, and how these components sum to create an overall pattern of energy use in the mammalian brain. For this dissertation, I use experimental measurements of cerebral glucose utilization in humans and eight other mammalian species to analyze the allometric properties of six major subdivisions in the brain: the cortex, thalamus, basal ganglia, hippocampus, cerebellum, and white matter. I also use stereotaxic brain atlases to estimate the volumes of these subdivisions among species. The combination of these data provides a picture of the distribution of energy consumption in the brains of a variety of mammals.

My analysis emphasizes two important methodological issues in comparative neuroimaging: the effects of anesthesia and partial-volume averaging. Anesthesia is a confounding factor in metabolic studies that produces complex changes in cerebral glucose uptake. Partial-volume averaging biases the results of measurements obtained by positron emission tomography (PET).

The results of my analysis indicate a consistent pattern of energy consumption in the mammalian brain. Log-log regressions of regional metabolic rate on brain size yield allometric coefficients of –0.12 (range: –0.08, -0.14). These slopes do not appear to differ between brain regions. The cortex and thalamus metabolize at a high

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rate while white matter exhibits the lowest metabolic rate. Metabolic rates in the basal ganglia, hippocampus, and cerebellum are intermediate. When metabolic data are combined with volumetric data, results indicate that the large volume of white matter in the human brain constitutes a higher metabolic load in humans than in other animals. Humans spend approximately 20% of total brain metabolism on white matter – more than twice the amount of other species measured here.

These results have important implications for the study of human brain evolution. The increasing energy demands of white matter metabolism highlight the problems of maintaining sufficient neuronal interconnectivity as the cortex evolves to a larger size. Additionally, the high cost of neuronal spiking means that only a small fraction of human cortical neurons can be active at any one time. This may tend to favor processing strategies that can code information with a minimal amount of neuronal activity.

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1. Introduction

The human brain is a metabolically "expensive" organ. Though it constitutes approximately 2% of body mass, it consumes 20% - 25% of the body's energy budget (Grande, 1980). But the brain is also a highly differentiated organ. Neurological research during the past three decades has revealed that the brain is metabolically heterogeneous – different regions of the brain use different amounts of energy to perform their work. Despite a rapid accumulation of data on cerebral metabolic rate, little is known about the metabolic scaling of individual brain components and how these components sum to characterize brain metabolism as a whole.

Glucose is the preferred and (nearly) exclusive substrate for brain metabolism in the non-pathological state (Clarke and Sokoloff, 1994). The energy stored in the chemical bonds of glucose is used to synthesize adenosine tri-phosphate, ATP. ATP is the energetic "currency" of the cell, since it can be de-phosphorylated at a later time to fuel a host of reactions throughout the cell. In neural tissue, most of this energy is required to maintain chemo-electric membrane potentials in neurons (Jueptner and Weiller, 1995). The rate of local glucose uptake is proportional to local neuronal activity, which is in turn correlated with the regional cerebral metabolic rate (rCMR) of the tissue (Raichle, 1987). Measurement of rCMR has become an important clinical tool for understanding a variety of neurological states, both normal and pathological. However, little is known about patterns of normal, baseline regional metabolism across different species of mammals.

For anthropologists interested in human brain evolution, the fact that baseline metabolic rate in the mammalian brain is non-uniform generates a host of questions

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concerning the metabolic condition and position of the human brain in relation to those of other mammals. Are regions of the human brain that metabolize at a high rate the same regions that metabolize at a high rate in the brains of other species? How do the metabolic patterns change as brains change in size? Does mosaic brain evolution (the differential expansion of brain regions) distinguish the metabolic properties of the human brain from the brains of other mammals?

The autoradiographic technique for calculating quantitative rCMR was developed nearly thirty years ago for use in laboratory animals. By labeling an analog of the glucose molecule (deoxyglucose) with a radioactive tracer, uptake can be measured autoradiographically and used to calculate the regional cerebral metabolic rate of glucose (rCMRgl) (Sokoloff et al., 1977). The term "autoradiography" refers to the fact that the radiation source is internal to the tissue, rather than an external source such as conventional X-ray or CT. At first, autoradiography required a controlled injection of radiotracer followed by immediate sacrifice of the animal. The brain was removed, sectioned, and placed on radiographic film for optical quantification. More recently, the use of positron emitting atoms as radiotracers has allowed for *in vivo*, non-invasive measurements of rCMRgl by tomographic methods (Phelps et al., 1986), referred to as Positron Emission Tomography, or PET scanning.

There has been only one study to date (Blin et al., 1991) that explicitly examines inter-specific rCMR, but only between humans and rats. Fortunately, a rapid accumulation of experimental data on rCMR in the neurological literature includes several mammalian species. This thesis includes a major consolidation of

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published metabolic data, allowing greater sample sizes both for species and individuals. I have collected and pooled regional metabolic rates for six major subdivisions of the brain: the neocortex, the basal ganglia, the thalamus, the hippocampus, the cerebellum, and the white matter. These six regions constitute the major mass of the brain. Depending on the region and anesthetic regimen, rCMR data are available for nine mammalian species: mouse, rat, gerbil, rabbit, cat, squirrel-monkey, macaque, baboon, and human. Additional data on whole brain metabolism and blood flow are available for the dog, pig, sheep, goat, and horse. Since metabolic data are measured in mass-specific units (moles of glucose per 100 grams of tissue per minute), estimation of absolute regional metabolism requires normalization by the volume of the brain region. Therefore, I have also calculated regional volume estimates from MRI scans and stereotaxic atlases for those species in which rCMR data are available.

This thesis is the first broad-scale comparative analysis of baseline regional brain metabolism among mammals. The purpose is to describe variation in rCMR among different species of mammals in order to examine the scaling relationships of regional metabolic rate and brain size, and to test whether there is a consistent pattern of regional metabolism that characterizes the mammalian brain in general. By the term "scaling" I refer to the allometry of rCMRgl, or its proportional relationship with such size variables as body mass, brain volume, and region volume. Characterizing these allometric trends is, on the one hand, a method for understanding the fundamental biological scaling factors of brain metabolism, since estimates of allometric coefficients describing rCMRgl can be compared with other known scaling

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factors (such as the three-quarters scaling of brain size and body size). On the other hand, allometric trends are also the starting point for interspecific comparisons.

To maximize the number of species and individuals for these analyses, it was necessary to pool data from studies that differ in several important procedural details. Data for some species were collected only in conscious individuals and others only in the anesthetized state. Some species have been measured using only the terminal autoradiographic technique, while humans have been measured only with PET. Therefore, a key issue in this dissertation is the examination of error and variance in measurements of rCMRgl due to experimental disparities. My analysis will first explore intra-specific effects such as anesthesia, measurement technique, circulatory physiology, age, and sex on rCMRgl.

It is important to emphasize that much is still unknown about the basic processes of energy utilization in the brain. But the accumulation of data used to better understand brain metabolism has progressed to the point that comparative neurobiology can now transcend pure anatomy and begin to critically examine the function and physiology of the brain across species.

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2. Literature Review

While CT and MRI provide structural information about the brain, physiological processes are measured very differently. Because of its obvious advantage of non-invasive, *in vivo* measurement, PET is used widely in the bio-medical arena. However, much of the data analyzed here were measured by methods that preceded PET – in particular the Kety-Schmidt technique for measuring whole brain metabolism, and the Sokoloff method for calculating regional metabolism. All three methods (Kety-Schmidt, PET, Sokoloff) are new to anthropology. The following section will provide background on each of these three methods. A list of abbreviations used throughout this text is provided in Table 2.1.

Table 2.1. Abbreviations used in the text.

[^{14}C]DG	2-Deoxy-D-[^{14}C]glucose
ATP	Adenosine Triphosphate
BMR	Basal Metabolic Rate
CBF	Cerebral Blood Flow
CMR	Cerebral Metabolic Rate
CT	Computed Tomography
DG	Deoxyglucose
FDG	[^{18}F]2-fluoro-2-deoxy-D-glucose
fMRI	Functional Magnetic Resonance Imaging
HR	Heart Rate
MABP	Mean Arterial Blood Pressure
MRI	Magnetic Resonance Imaging
OGI	Oxygen/Glucose Index
paCO2	Partial Pressure of Carbon Dioxide
paO2	Partial Pressure of Oxygen
PET	Positron Emission Tomography
rCBF	Regional Cerebral Blood Flow
rCMR	Regional Cerebral Metabolic Rate
rCMRgl	Regional Cerebral Metabolic Rate of Glucose
rCMRO2	Regional Cerebral Metabolic Rate of Oxygen
RMA	Reduced Major Axis
ROI	Region of Interest
TCA	Tricarboxylic Acid Cycle

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I begin this background section by first describing the metabolic pathway in the brain, focusing on those portions most pertinent for this analysis. Then, I describe the Kety-Schmidt method for measuring whole-brain metabolism, since this provides a starting point for understanding the fundamentals of tracer kinetics. (All three methods are derived from principles of tracer kinetics – the mathematical modeling of a compartmented system that describes the concentrations and movements of reactants and products in and between compartments.) This is followed with a description of the Sokoloff method, and then PET, which is a non-invasive extension of the Sokoloff method. Then, I review the allometric scaling of various metabolic and structural brain variables including whole brain metabolism, neuron density, and white matter volume. This is followed by a review of the functional anatomy of the six major subdivisions that are examined in this thesis and their mosaic evolution across mammals. Finally, I review a variety of energetic implications for human brain evolution.

2.1 The Metabolic Pathway in the Brain

ATP synthesis begins with the anaerobic process of glycolysis, during which one mole of glucose is catalyzed into two moles of pyruvate (Figure 2.1). The net metabolic production of glycolysis is two moles of ATP and two moles of the electron carrier NADH. In the presence of oxygen, the two moles of pyruvate can be oxidized through the tricarboxylic acid (TCA) cycle to produce a total of 38 moles of ATP (including the two moles produced during glycolysis) (Figure 2.2). The complete pathway (glycolysis and TCA) can thus be represented as

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$$\text{Glucose} + 6O_2 \rightarrow 6CO_2 + 6H_2O \qquad \text{Eq. 1}$$

which expresses the fact that the oxidation of 1 mole of glucose requires 6 moles of O_2. Under aerobic conditions, the oxidation of one mole of glucose produces 38 moles ATP. For the purposes of measuring rCMRgl, the most important reaction in the metabolic pathway is the hexokinase-catalyzed phosphorylation of glucose into glucose-6-phosphate (highlighted in Figure 2.1). It is the rate of this reaction that can be measured using the Sokoloff model to calculate the local rate of glucose metabolism.

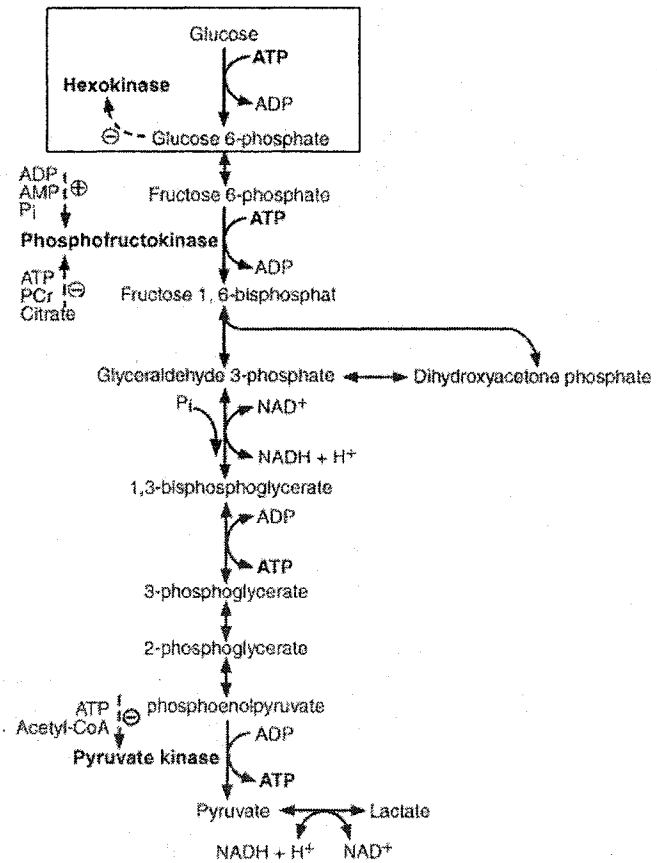


Figure 2.1. Glycolysis, the anaerobic pathway whereby glucose is catalyzed to pyruvate, produces 2 moles of ATP. Highlighted is the hexokinase catalyzed transformation of glucose to glucose-6-phosphate.

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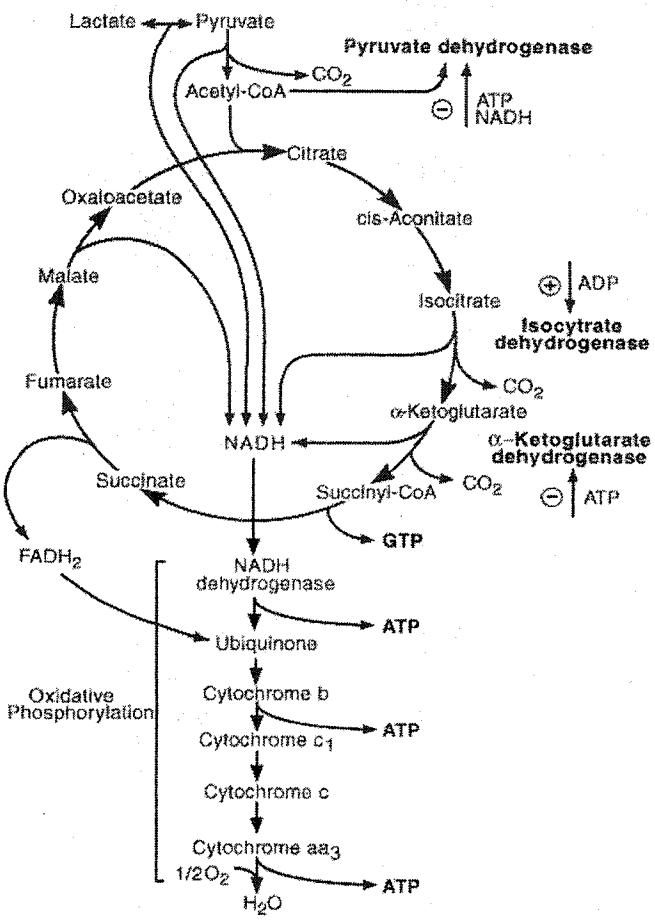


Figure 2.2. The tricarboxylic acid (TCA) cycle, whereby two moles of pyruvate are oxidized to produce 36 moles of ATP

Equation 1 can also be rearranged to describe the ratio of oxygen to glucose that enters the metabolic pathway, and thus gauge whether the system is operating under aerobic or anaerobic conditions. Cohen et. al (1964) created the oxygen/glucose index (OGI) to describe this ratio by using the function:

$$OGI = \frac{(a-v)_{O2}}{6*(a-v)_{gl}} * 100 \qquad \text{Eq. 2}$$

where the numerator denotes the arterio-venous difference in oxygen concentration and the denominator denotes the arterio-venous difference in glucose concentration multiplied by a factor of six to account for the stoichiometry described in Equation 1.

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An OGI close to 100 percent indicates that all glucose extracted from the arterial blood is completely oxidized to CO_2 and water. In this case, the rate of cerebral metabolism can be quantified by measuring either the cerebral metabolic of glucose (CMRgl) or the cerebral metabolic rate of oxygen ($CMRO_2$).

It has long been known brain blood flow, oxygen consumption, and glucose consumption are highly-coupled in the resting state (Siesjö, 1978). But, there is strong evidence that the immediate energy needs of neuronal activation are met, not by oxidative metabolism, but by anaerobic glycolysis (Fox and Raichle, 1986; Fox et al., 1988). This is perhaps not so surprising since the rate of ATP production during glycolysis is approximately one hundred times faster than the (more efficient) process of aerobic oxidative phosphorylation, so it can quickly accommodate spikes in cellular energy demand (Magistretti, 2003). Still, it came as a surprise when Fox and Raichle (1986) compared resting metabolism with that immediately following somatosensory and visual stimulation, and found that glucose uptake (and local cerebral blood flow) immediately jumped by 30-50%, while oxygen uptake increased by only about 5%. It appeared that, as in the case of muscle exertion, oxidative phosphorylation alone cannot deliver enough ATP to fuel neuronal activation, and so glycolysis must be used to supplement the difference.

This focal "uncoupling" between oxygen and glucose utilization is short-lived, and the tissues quickly return to an OGI close to 100%. But, it is worthwhile to note that the focal uncoupling of blood flow and oxygen uptake during neuronal activation forms the basis for functional magnetic resonance imaging (fMRI). During fMRI, hemoglobin acts as a paramagnetic contrast agent. The local uncoupling associated

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with activation causes a change in the local ratio of blood oxyhemoglobin and deoxyhemoglobin, which, in turn, alters the magnetic field detected by MRI. Although changes in the "blood oxygen level dependent" signal (called the BOLD signal) cannot yet be converted to absolute metabolic rates, fMRI has the advantage of very rapid signal acquisition so that activation changes can be monitored in real time and in a completely non-invasive manner.

In actuality, glucose uptake is almost always in slight excess of oxygen uptake. This is due to several factors (Magistretti, 2003). First, glucose is an essential component of many lipid and protein macromolecules present in neural cells, so a certain amount of glucose taken up from the cerebral circulation is necessary for production of these molecules. Second, glucose is also necessary for the synthesis of neurotransmitters, particularly glutamate, γ-aminobutyric acid (GABA), and acetylcholine. Third, in certain circumstances, glucose metabolism may proceed only through the glycolytic pathway that produces small amounts of ATP without any oxidation.

2.2 Whole Brain Metabolism: The Kety-Schmidt Method

In a biochemical process such as metabolism in which there is continual turnover of reactants and products, the process is said to be in a steady state if the concentrations of reactants and products remain constant over time. Although this is never completely accurate for any biological system, the assumption of steady state metabolism in the brain is valid if, over a particular period of time, levels of blood

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glucose and oxygen are constant, blood flow is constant, and there is no stimulation or inhibition of the metabolic pathway that would induce a change in metabolic rate.

Oxygen and glucose are delivered to the brain primarily via the internal carotid artery and drained primarily via the venous sinuses and through the jugular bulb. As they pass through the cerebral circulation, a fraction of both oxygen and glucose are carried from the arterial plasma, across the blood brain barrier, and into a precursor pool where they may enter the metabolic pathway, or pass back out into the circulation.

The Kety-Schmidt method is based on a derivation of the law of conservation of matter known as the Fick principle, which states that the amount of an inert substance taken up by a unit of tissue is equal to the difference in the amount delivered to the tissue by the arterial system and that removed from the tissue by the venous drainage (Fick, 1870; Hoff and Scott, 1948). If the metabolic pathway is assumed to be in a steady state, then the net rate of glucose metabolism in brain tissue is equal to the cerebral blood flow (volume per unit time) multiplied by the difference in glucose concentration between arterial plasma sampled from the internal carotid and that from the jugular or cerebral venous sinus:

$$CMR = CBF(C_a - C_v) \qquad \text{Eq. 3}$$

where CMR is the cerebral metabolic rate, CBF is the cerebral blood flow, and C_a and C_v are the concentrations of glucose in arterial and venous blood respectively.

Of the two terms on the right side of Equation 3, the arteriovenous difference in glucose concentration is the easier to measure. Standard enzymatic techniques have long been available to quantitatively assay glucose concentration in whole blood

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or plasma (Nelson, 1944; Saifer and Gerstenfeld, 1958). By withdrawing arterial and venous blood samples at regular intervals over the course of an experiment and assaying these samples for glucose, the *time course* of arteriovenous concentration difference can be calculated quite easily.

Measuring blood flow, without the use of mechanical probes, is a more difficult process. The watershed moment in the study of brain physiology came when Seymour Kety and Carl Schmidt (1945) devised a method for calculating cerebral blood flow by using a chemically inert tracer substance. Kety and Schmidt originally used low concentrations of nitrous oxide as their tracer, in part because it could be conveniently administered by inhalation. But, they noted that it does not matter what particular substance is used as a tracer so long as it has several properties (Kety and Schmidt, 1945, 1948): 1) it diffuses freely and rapidly across the blood-brain barrier; 2) it lends itself to accurate measurement in the blood; 3) it is physiologically inert; and 4) it can be administered in concentrations low enough that its presence does not alter the steady state.

The Kety-Schmidt method is based on the fact that, while cerebral metabolism is assumed to be in a steady state, the concentration of the tracer substance is not in a steady state. When a tracer is administered evenly and continuously over a period of time, the arterial concentration will rise rapidly immediately after the onset of administration and then level off asymptotically as the tracer reaches saturation. The venous concentration will exhibit a similar course, but its concentration *will lag behind* the arterial concentration, requiring a certain amount of time before the arterial and venous concentrations reach saturation equilibrium. An example of

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arterial and venous concentration curves in a typical Kety-Schmidt saturation experiment is presented in Figure 2.3. Kety and Schmidt demonstrated that the lag between the two curves (mathematically, the area between the curves) is proportional to the rate of blood flow.

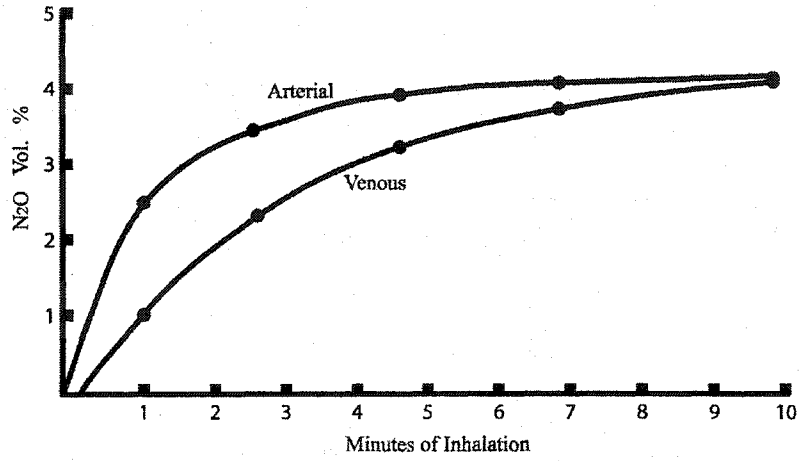


Figure 2.3. Typical blood curves using the Kety-Schmidt nitrous-oxide method. Adapted from Kety and Schmidt, 1945.

The operational equation for the Kety-Schmidt technique is expressed as:

$$CBF = \frac{100Q_t}{(A-V)t} \qquad \text{Eq. 4}$$

where Q_t represents the quantity of tracer (nitrous oxide) taken up by the brain from the beginning of inhalation to time t, and A and V represent the arterial and venous concentrations of nitrous oxide in the blood *expressed as percent of saturation volume*. As is evident from Figure 2.3, the arteriovenous difference is not constant over the course of the experiment. Therefore, values must be integrated over the course of the experiment, and Equation 4 becomes

$$CBF = \frac{100Q_t}{\int (A-V)dt} \qquad \text{Eq. 5}$$

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where the denominator now represents the area between the arterial and venous curves.

For the numerator, the total volume of nitrous oxide taken up by the brain during the experiment can be calculated by simply assuming that the mean nitrous oxide tension in the brain at saturation (which is equal to the amount taken up by the brain) is equal to its tension in the venous drainage at saturation. Although this assumption is only approximate during the first few minutes of inhalation, Kety and Schmidt demonstrated that it becomes increasingly valid over time as equilibrium comes to completion, generally after about 15 minutes of inhalation. In this case,

$$Q_t = \lambda V_t \qquad \text{Eq. 6}$$

where V_t represents the nitrous oxide content of the venous blood at saturation, and λ is the ratio of solubilities of nitrous oxide in brain and blood (also called the partition coefficient), which is approximately equal to 1. Substituting for Equation 5, Kety and Schmidt obtained their final operational equation:

$$CBF = \frac{100\, V_t}{\int_0^T (A - V)dt} \qquad \text{Eq. 7}$$

A typical Kety-Schmidt experiment proceeds as follows. Two catheters are introduced: one into an artery (usually the femoral artery, but often the internal carotid), and the other through a point in the venous system where blood is representative only of that which has circulated through the brain. For whole brain measurements, this generally means the jugular bulb or confluence of sinuses. However, some researchers will introduce the venous catheter as high as the superior sagittal sinus, where samples will be known to represent blood drained primarily from

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the cortex. The method has been described as follows: In animals, all surgery is performed under anesthesia, after which the incisions are sutured and the catheters are stabilized to allow the animal to recover. In some cases, the catheters are arranged in a mechanism that allows the animal to move freely in its cage. In other experiments, a plaster cast is placed around the lower torso to restrict movement. In either case, the animal is usually allowed to recover fully from anesthesia before the infusion of tracer begins. During tracer injection, regular samples are taken from both catheters to compute the time courses of tracer concentration, which are inserted into the operational equation (Equation 5). Since the method already requires regular arterial and venous blood sampling, these same samples can easily be assayed for glucose concentration, allowing for the calculation of whole-brain glucose metabolic rate (Equation 1).

Other than the incorporation of radioactive gases, such as ^{133}Xe or ^{85}Kr, to allow for more precise blood assay, the only significant modification of the Kety-Schmidt technique has been to measure the arterial and venous time courses during desaturation rather than saturation. In this so-called desaturation mode, continuous inhalation of the tracer is allowed to proceed until saturation equilibrium is complete. Tracer administration is then ceased, and regular arterial and venous blood samples are drawn as the tracer clears or "washes out" of the brain. The principle of the calculation remains the same as in the saturation mode, only now the integrated arteriovenous concentration is represented by the area between the desaturation curves, as illustrated in Figure 2.4.

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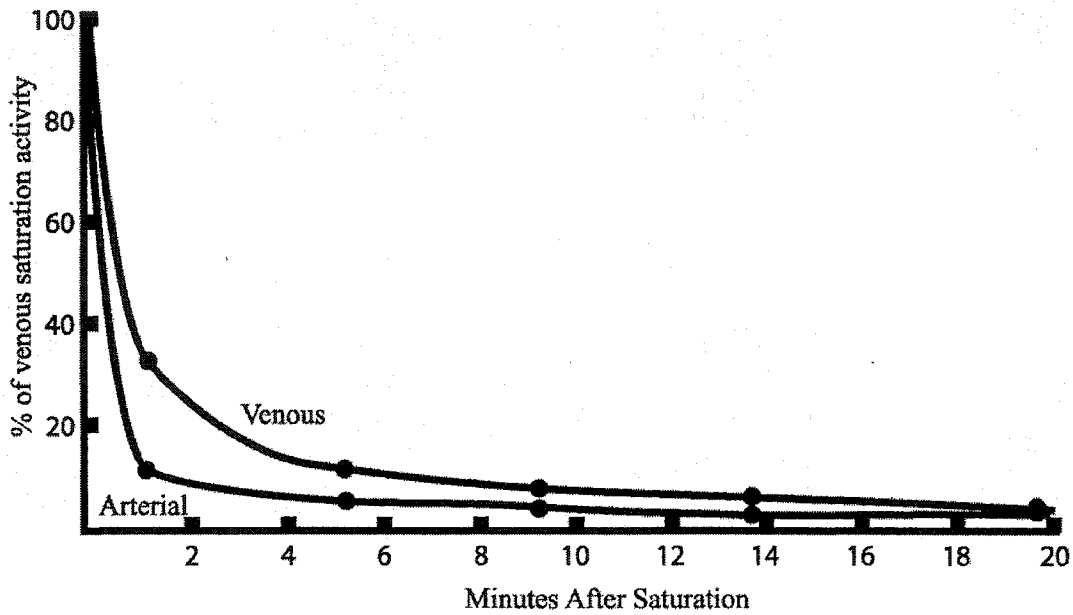


Figure 2.4. A typical blood curve for the Kety-Schmidt desaturation method using 133Xe as a tracer. Adapted from Gjedde et al., 1975.

Despite the passage of more than half a century, the Kety-Schmidt method remains the reference for quantitative determination of whole-brain blood flow. But the task of calculating the local rate of metabolism in particular regions of the brain is more complex. Sokoloff and his colleagues (1977) devised an ingenious method for measuring rCMRgl by using a radiolabeled analog of glucose.

2.3 Regional Metabolism: The Methods of Sokoloff and Ter-Pogossian

In the case of whole brain metabolism, the theoretical model (Equation 5) implies a relatively simple system based on exchange between two compartments: one compartment characterizing the brain circulation and the other characterizing the brain tissue. The task of calculating rCMRgl was solved by Louis Sokoloff and his

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colleagues (1977), who developed a three-compartment model that characterized the state, not just of glucose, but also of a particular analog of glucose: deoxyglucose (DG). DG has several convenient biochemical features that cause it to become trapped in the brain and accumulate for a period of time before it is cleared. Sokoloff and his colleagues demonstrated that a radiolabeled analog of the glucose molecule, 2-Deoxy-D-[^{14}C]glucose ([^{14}C]DG), can be used to calculate the rate of glucose uptake autoradiographically. Their model is illustrated in Figure 2.5.

In the blood plasma, which is the first compartment, glucose and [^{14}C]DG are present in concentrations denoted C_p and C_p*, respectively. While in the plasma, they share and compete for a common carrier that transports them across the blood-brain barrier and into the brain tissue, where they form a metabolic precursor pool.

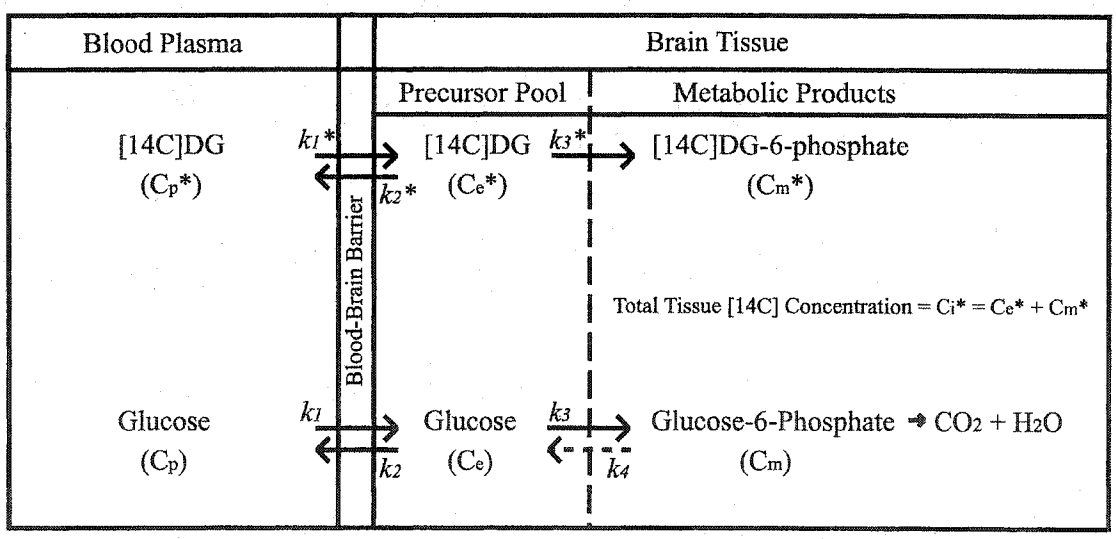


Figure 2.5. The three-compartment model of Sokoloff et al., 1977 for the measurement of rCMRgl.

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The metabolic precursor pool is the second compartment, with glucose and [^{14}C]DG concentrations denoted as C_e and C_e*). Here, inside the brain tissue, they compete again: either for transport back into the circulation, or for the metabolic enzyme hexokinase. Hexokinase catalyzes the first step of glycolysis (Figure 2.1, highlighted) during which glucose and [^{14}C]DG are phosphorylated to glucose-6-phophate and [^{14}C]DG-6-phosphate. The third compartment represents the concentrations of the normal and radiolabeled metabolic products (C_m and C_m^*).

At this point, the structural anomaly in the radiolabeled product, [^{14}C]DG-6-phosphate, is incompatible with the enzyme that converts glucose-6-phosphate to fructose-6-phosphate, which continues normally through the remainder of glycolysis and the TCA cycle to be metabolized to carbon dioxide and water. Since the hexokinase reaction that transforms [^{14}C]DG into [^{14}C]DG-6-phosphate is essentially irreversible (there is extremely little hydrolysis of glucose-6-phosphate back to glucose), and since [^{14}C]DG-6-phosphate does not interact with any other enzymes that are present, it becomes trapped in the cytoplasm and accumulates there.

The Sokoloff method for calculating rCMRgl (Reivich et al., 1969; Sokoloff et al., 1977) involves measuring 1) the time course of both glucose and [^{14}C]DG in the blood, and 2) the concentration of [^{14}C] in the brain tissue by autoradiography. In addition, the model requires that several constants are estimated independently. These constants include the rates at which glucose and [^{14}C]DG move from compartment to compartment (represented by the "kinetic constants," k values in Figure 2.5), and an "isotope correction factor" (lumped constant, see below) that accounts for the biophysical difference between glucose and [^{14}C]DG.

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The operational equation for the Sokoloff model is denoted as:

$$rCMRgl = \frac{\overbrace{C_i^*(t) - k_1^* e^{-(k_2^*+k_3^*)t} \int C_p^* e^{(k_2^*+k_3^*)} dt}^{\text{Amount of radioactive product, [}^{14}\text{C]DG-6-phosphate, formed in a given interval of time}}}{\underbrace{\left(\frac{\lambda V_M^* K_M}{\Phi V_M K_M^*}\right)}_{\text{The Lumped Constant}} \underbrace{\left[\int (C_p^*/C_p)dt - e^{-(k_2^*+k_3^*)t} \times \int (C_p^*/C_p) e^{(k_2^*+k_3^*)t} dt\right]}_{\text{The integrated specific activity of the precursor pool}}}$$

It is a menacing-looking equation, but actually it represents a simple ratio that characterizes the rate of an enzyme-catalyzed reaction. The numerator of the equation is the total amount of radioactive product, [^{14}C]DG-6-phosphate, formed in the tissue over a given interval of time. Note that the presence of ^{14}C can be detected by autoradiography, but it is not possible to distinguish [^{14}C]DG from [^{14}C]DG-6-phosphate in the radiograph. The total tissue concentration of ^{14}C, denoted C_i*, is equal to the combined concentrations of [^{14}C]DG and [^{14}C]DG-6-phosphate:

$$C_i^* = C_e^* + C_m^* \qquad \text{Eq. 9}$$

The numerator accounts for this problem by subtracting the amount of free unmetabolized [^{14}C]DG from the total tissue activity, assuming specific values of the kinetic constants. The denominator is composed of two terms. One term represents the complete time history of the arterial concentrations of [^{14}C]DG and glucose,

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called the "specific activity of the precursor pool." This term is multiplied by a correction factor called the "lumped constant."

The lumped constant is a series of ratios that correct for the difference in physiological behavior (transport across the blood-brain barrier, rate of phosphorylation, and distribution volumes) between normal glucose and radiolabeled deoxyglucose. It is composed of six separate constants. λ represents the ratio of distribution volumes for [^{14}C]DG and glucose in brain tissue. Φ is a constant between zero and one that describes the steady-state hydrolysis of glucose-6-phosphate back to free glucose and phosphate. As was stated earlier, the rate of this reaction in normal brain tissue is very low, so Φ is normally approximately equal to unity (no hydrolysis). V_M and V_M^* are the maximal velocities of phosphorylation for glucose and [^{14}C]DG, respectively. K_M and K_M^* are the concentrations of glucose and [^{14}C]DG at which V_M and V_M^* are 50% of maximum, i.e., half the maximal rate of phosphorylation. V_M and K_M are together known as the Michaelis-Menten constants (Michaelis and Menten, 1913).

A typical experiment using the Sokoloff method begins in much the same way as a Kety-Schmidt experiment. Catheters are positioned in the arterial and venous circulation while under surgical anesthesia. Animals are generally allowed to recover fully from anesthesia before the tracer is infused. A dose of [^{14}C]DG is then administered – one that is negligible in comparison to its K_M for hexokinase, but is high enough that its detectability is in the range for radiographic standards. Note that, unlike the Kety-Schmidt method, the Sokoloff method requires that the animal is sacrificed so that the brain can be sectioned and placed on radiographic film. After a

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prescribed amount of time (generally about 30-45 minutes), the animal is sacrificed and the brain is removed and frozen as quickly as possible. Thin-sections are placed on radiographic film, and its optical density is compared with calibrated standards. Combined with the blood curves for glucose and [^{14}C]DG, the mean glucose uptake rate of a particular brain region can be calculated using the operational equation.

The search for a method of *in vivo* autoradiography that could be safely applied to humans was solved by Ter-Pogossian and colleagues (Phelps et al., 1975; Ter-Pogossian and Powers, 1958). Their solution, leading to PET scanning, was an extension of the Sokoloff model that relied on the clever use of certain short-lived radionuclides that, when they decay, emit particles called positrons. A positron is the positively charged, anti-matter counterpart of an electron. The positron does not last long – it will quickly come into contact with a free electron and the two will annihilate each other in an event that produces energy. The key property of this annihilation event is that the resulting energy takes the form of two gamma rays that travel 180° from each other, directly away from the annihilation event (Figure 2.6).

The scanner consists of a ring of gamma-ray detectors that are wired together in "coincidence circuits." When any two gamma-ray detectors are triggered at exactly the same time, the scanner knows that an annihilation event occurred somewhere along a line between the two detectors. After recording hundreds of thousands of decay events, the local concentration of radiotracer can be reconstructed tomographically.

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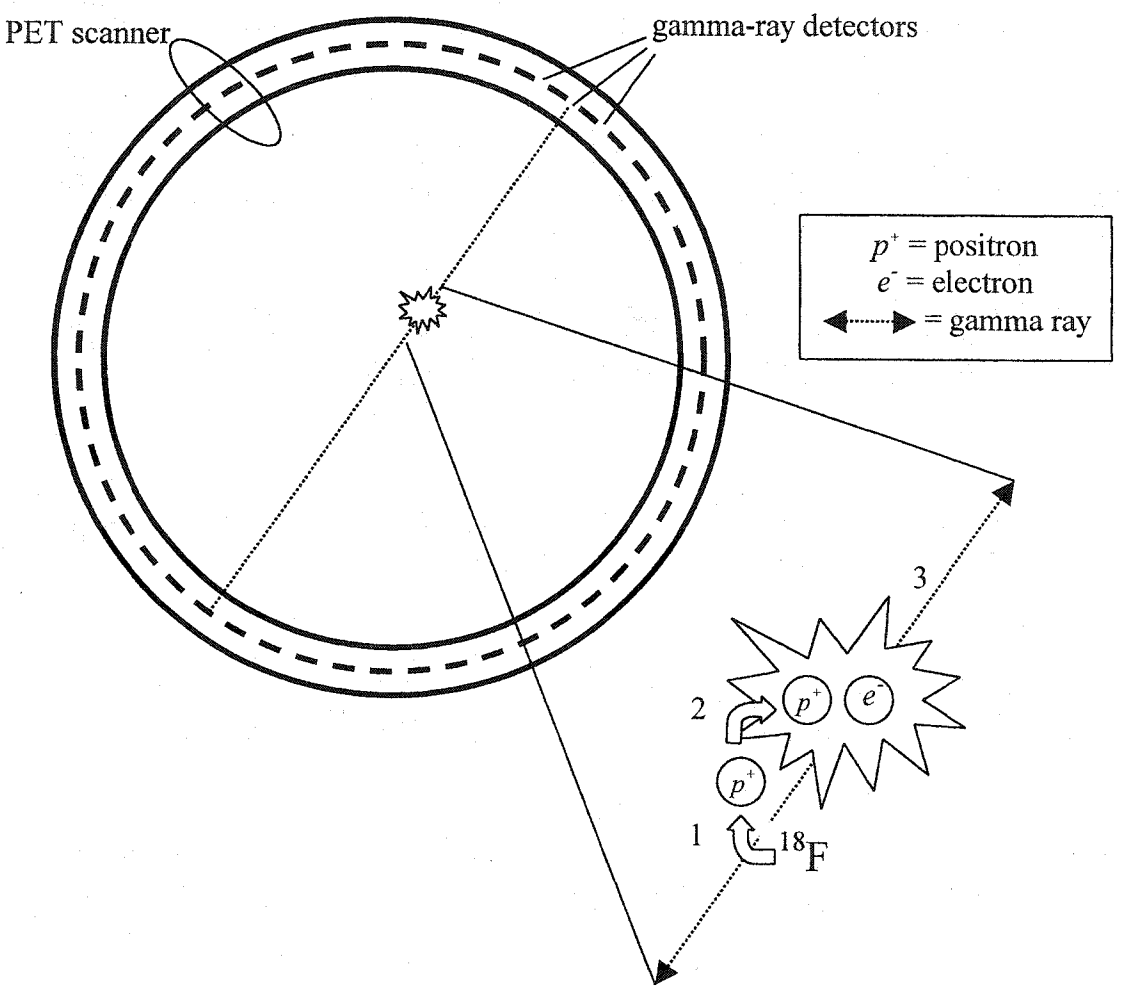


Figure 2.6. Diagram of the process of Positron Emission Tomography (PET). 1) ^{18}F decays and emits a positron. 2) The positron annihilates with a free electron. 3) The resulting energy takes the from of two gamma-rays that travel directly away from each other.

Most PET experiments that measure rCMRgl use the short-lived positron-emitter ^{18}F as a radiotracer. ^{18}F is attached to a deoxyglucose analog to form [^{18}F]2-fluoro-2-deoxy-D-glucose (FDG). FDG shares the same properties that made [^{14}C]DG so useful: it is taken up by the same carriers as normal glucose, but it only proceeds through the first step of glycolysis (the hexokinase reaction) before it becomes trapped in the cell and accumulates. Just as with [^{14}C]DG autoradiography, quantitative PET requires measurement of the blood curves both for glucose and the

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radiolabeled analog, in this case FDG. The rate of glucose metabolism is then calculated using an extension of the three-compartment Sokoloff model (Brooks, 1982; Huang et al., 1980).

Along with the refinement of equipment and mathematical algorithms to analyze PET data, many of the most exciting advances in PET imaging have come from new developments in radio-pharmaceuticals. Chemists have created a variety of positron emitting compounds that can be used to measure a multitude of brain features. In addition to metabolic analogs like FDG and O_2, positron emitters can be attached to a variety of receptor-binding ligands, allowing for measurement of receptor density and distribution. The versatility of PET promises new dimensions of exploration in anthropological, primatological, and comparative neurological research.

2.4 Brain Metabolism and Allometry

Allometry as a method of biometrics was initially developed by Huxley (1932) as a theory of growth, and has since been generalized to a variety of questions regarding size and proportionality in biological systems. The allometric approach to brain research first came to prominence with the question of brain-body size scaling. Early investigators (Brandt, 1867) first reported an allometric coefficient of two-thirds when log brain size was regressed on log body size, which was rationalized as a reflection of geometric scaling of surfaces versus volumes, and the brain's need to manage information between these two dimensions (Snell, 1891). Later, Jerison

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(1955) published an allometric coefficient of 0.73 for a large range of mammalian species, a value close to the currently accepted value of 0.75 (Martin, 1990).

This revision of brain:body allometry from two-thirds to three-quarter scaling was paralleled in studies of whole body basal metabolic rate (BMR) and body size. As was the case with brain:body allometry, early researchers speculated that negative allometry of BMR and body size was associated with geometric scaling of surfaces and volumes and should therefore equal two-thirds (Rubner, 1883). Subsequent research once again demonstrated that the empirically derived scaling coefficient of BMR and body size is approximately equal to 0.75 (Brody, 1945; Kleiber, 1932, 1947; Martin, 1981). New research suggests that three-quarters scaling of brain and body size (along with a host of other biological size variables) may be a result of fractal geometry in bifurcating systems (West et al., 1997), but this explanation remains disputed.

While this thesis is the first to examine scaling of regional brain metabolism, there is consensus among investigators that the metabolic rate of the entire brain scales negatively with brain and body size, and that this is most likely due to the reduction in cell density as organs become larger. This observation is not new: Nilsson and Siesjö (1976) commented that

> . . . CMRO2 of the rat brain is at least twice as high as that of the human brain. Furthermore, cortical CMRO2 in the rat seems to be twice as high as that in the dog. These differences probably reflect an inverse relationship between cerebral metabolic rate and body weight. This conclusion is in agreement with previous findings of differences in oxygen uptake by cortical tissue from various species *in vitro* (Elliott and Henderson, 1948; Tower and Young, 1973).

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Nilsson and Siesjö further suggested that differences in cerebral metabolic rate among species are a result of an inverse relationship between neuron density and brain volume. They concluded that "the relative decrease in neuron density in larger brains can readily explain the species differences in CMRO2."

Similarly, Mink et al., (1981) derived a regression equation of

$$\log y = 0.86 - 0.13 \log x$$

for five mammals, where x is brain weight and y is whole brain metabolic rate in cm^3 O_2/100g/min. Like Nilsson and Siesjö, Mink et al. also conclude that "the declining rates of metabolism in larger brains presumably reflect the lower packing densities of neuronal cell bodies."

Table 2.2 summarizes predicted and empirically-derived scaling exponents for a variety of structural brain variables. Of particular importance are the scaling coefficients for neuron density (–0.33) and white matter volume (1.33). Larger brains have relatively fewer neurons, but they also possess a relatively greater volume of white matter. As the number of neurons increases (absolutely) with increasing brain size, interconnectivity between neurons requires an exponentially greater amount of wiring (Zhang and Sejnowski, 2000).

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Table 2.2. Inter-specific scaling factors for cortical variables regressed on brain size.

Variable	Predicted Exponent	Empirically-Derived Exponent	Reference
Synapse density	0	0	(Abeles, 1991)
Synapse diameter*	0	0	(Abeles, 1991)
Neuron number	0.67	0.62	(Jerison, 1973)
		0.67	(Passingham, 1973)
Neuron density	-0.33	-0.31	(Prothero, 1997b)
		-0.28	(Prothero, 1997b)
		-0.28	(Tower, 1954)
		-0.32	(Tower, 1954)
Cortical thickness	0.11	0.09	(Prothero, 1997a)
		0.12	(Prothero, 1997a)
		0.13	(Hofman, 1991)
		0.20	(Hofman, 1989)
		0.08	(Prothero and Sundsten, 1984)
		0.17	(Jerison, 1982)
		0.11	(Rockel et al., 1980)
Soma radius	0.11	0.10	(Changizi, 2001)
White matter volume	1.33	1.32	(Allman, 1999)
		0.99	(Prothero, 1997b)
		1.28	(Hofman, 1991)
		1.37	(Hofman, 1989)
		1.31	(Frahm et al., 1982)

* Synaptic diameter is the average length of the observed dense synaptic membranes.

It is equally important to note that synaptic density is constant across species (Blinkov and Glezer, 1968; Tower, 1954; Tower and Elliott, 1952). Empirical estimates of cortical synaptic density in various mammals consistently average 7-8 x 10^8 synapses per $1mm^3$ of cortex (Cragg, 1967, 1975a, 1975b; Lennie, 2003; Schuz and Palm, 1989; Vrensen et al., 1977). Abeles (1991) remarks that

> the similarity between figures [on synaptic density] for different animals, different brain regions, and different investigators are amazing, particularly if compared with the large variations in neuronal densities. This suggests that the limiting factor for local connectivity is the packing density of the synapses. All cortices have synapses that are packed as densely as possible.

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Again, the consensus is that the "large variations in neuronal densities" among species are responsible for the negative allometry of brain metabolism regressed on brain size.

The previous results are all limited to examination of whole brain metabolism. There has been only one study to date that explicitly examines the pattern of regional brain metabolism across species. Blin et al. (1991) compared rCMRgl in 18 regions for rats and humans. Using regional metabolic rates as data points, they demonstrated a consistent pattern of rCMRgl between rats and humans ($r = 0.72$; $p < 0.001$). Their results indicated that, in both species, cerebral cortex, thalamic, and striatal regions metabolize at higher rates than deeper structures such as the hypothalamus, amygdala, hippocampus, substantia nigra, and cerebellum.

Although in this thesis I examine fewer regions than Blin (6 as opposed to 18), the inclusion here of rCMR data for up to nine species (as opposed to 2) allows for a quantitative cross-species comparison of regional brain metabolism. The lack of data for more primates, large-bodied anthropoids and small-bodied strepsirrhines in particular, is a weakness for the comparative interpretation of human regional metabolism. As data for additional species become available in the future, this analysis can be further refined.

2.5 Functional Morphology: The Brain Mosaic

The regions of the brain examined in this study are: the cerebral cortex, the thalamus, the basal ganglia, the hippocampus, the cerebellum, and the cerebral white matter. These subdivisions were chosen because they are readily distinguishable both

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in histological sections and in neuroimaging scans like CT and MRI. Together, these six subdivisions represent the bulk of the brain caudal to the brainstem and pons.

These subdivisions of the central nervous system derive from different portions of the embryonic neural plate and neural tube (Figure 2.7). The hindbrain, or rhombencephalon, includes the medulla oblongata, pons, and a specialized outgrowth called the cerebellum. The cerebellum is the largest part of the mammalian rhombencephalon. It is interesting to note that the cerebellum, which accounts for approximately 10% of brain weight in mammals, contains more neurons and synapses than the entire remainder of the brain (Houk and Mugnaini, 2003). Most of our knowledge about the cerebellum relates to its function as a regulator of the sensorimotor system (Houk and Mugnaini, 2003; Ito, 1984). In this capacity, the cerebellum is interconnected with many regions of the brain including the cortex, thalamus, and basal ganglia. The cerebellum appears to be primarily active in regulating motor neurons, both for executing movements and for motor learning (Raymond et al., 1996). Recent research suggests that the cerebellum

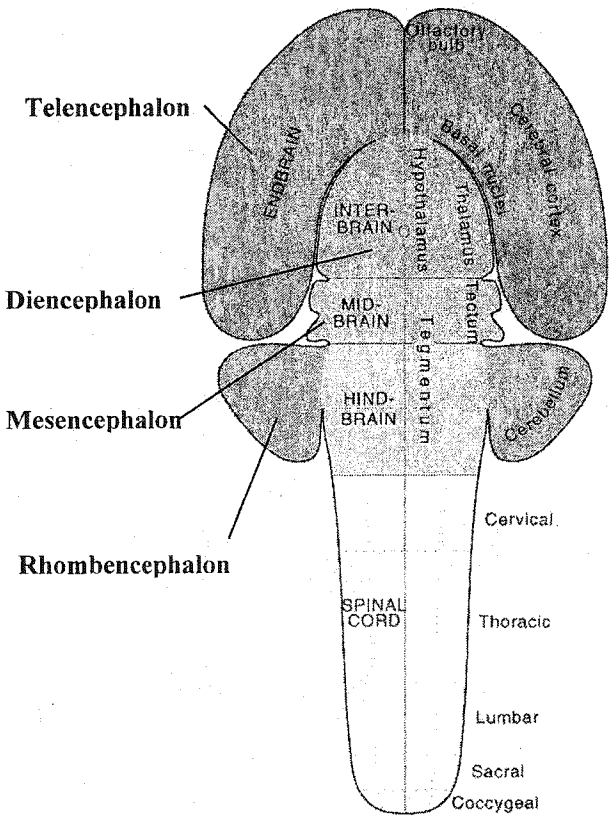


Figure 2.7. Illustration of the embryonic subdivisions of the brain.

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is also involved in higher cognitive processes, such as problem solving and motor planning (Thach, 1998).

The thalamus is a derivative of the diencephalon, or "inter-brain" (Figure 2.7). The thalamus is actually a diverse set of nuclei that function as the principal synaptic relay between the peripheral nervous system and the cortex (Jones, 1984). Various divisions of the thalamus relay all manner of motor and sensory modalities, including mechanoreception, thermoreception, nociception, and proprioception from the periphery, as well as sensory reception from the viscera. The thalamus also contains relay nuclei for visual signals (lateral geniculate nucleus) and auditory signals (medial geniculate nucleus).

The basal ganglia – a derivative of the endbrain (telencephalon) – also consists of several nuclei, the largest of which include the caudate, putamen, and globus pallidus. The basal ganglia have a diverse set of functions and interconnections, but their primary action is inhibition between the motor cortex and its targets (Mink, 1996, 2003). Pathologies of the basal ganglia are related to various movement disorders, including Parkinson's Disease, Huntington's Disease, and Tourette Syndrome (Watts and Koller, 1997).

The telencephalon also gives rise to the cerebral cortex. The phylogenetically oldest part of the cortex contains the hippocampus. The hippocampus is the dominant feature of the limbic system, which is associated with emotion and memory (MacLean, 1952). The hippocampus has interconnections with the cingulate gyrus of the cortex, as well as with the olfactory bulbs and the amygdala – a region known to be involved in reward and conditioning.

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The phylogenetically newer part of the cortex, the neocortex, contains stratified layers of neurons that are organized into functionally discrete regions (Brodmann, 1909). Some regions, like the visual or sensorimotor cortices, are dedicated to discrete sensory modalities, while other areas, like the parietal and frontal cortices, operate as "association areas" between sensory modalities. Although the mammals in this analysis share a relatively similar cortical architecture, the problems of homology are very real (Campbell, 1982). In an effort to maintain homologous comparisons, I have pooled the published metabolic data from a broad range of cortical areas into major anatomic regions: frontal, sensorimotor, parietal, temporal, occipital, and cingulate cortices.

The final subdivision examined here is the white matter. White matter consists almost exclusively of myelinated axons that provide interconnections between brain regions. White matter is the literal "wiring" of the brain, and it is therefore present in all parts of the central nervous system. For this analysis, however, metabolic data for white matter refers only to the cerebral white matter – namely the corpus callosum, internal capsule, and centrum semiovale.

Comparative morphologic research suggests that the human brain is not simply a scaled-up version of the nonhuman primate brain, though the exact picture of so-called mosaic brain evolution depends largely on the numbers and kinds of species that are compared. The largest dataset on comparative brain size comes from Stephan and colleagues who, in the 1980s, measured regional brain volumes in a variety of insectivores, bats, and primates (Baron et al., 1983, 1988, 1987, 1990; Frahm et al., 1984a, 1984b, 1982; Stephan et al., 1982, 1988, 1981, 1987). Factor

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analysis of their dataset indicates that, after brain size, the olfactory bulb and associated limbic structures account for most of the interspecific variance (Finlay and Darlington, 1995). In fact, Gould (1975) demonstrated that brain size and olfactory/limbic size could discriminate simians, prosimians, and insectivores with near perfect accuracy. As one moves from insectivores to primates, the size of the olfactory bulb and associated limbic structures are relatively reduced, while the neocortex is relatively expanded (Stephan et al., 1981). There is also a large degree of correlated evolution among brain divisions (Barton and Harvey, 2000) that may be related to their embryonic order of development (Finlay et al., 2001).

Among primates, there are important differences in the relative size of brain structures that appear to distinguish the human brain from all others. First, the human cerebral cortex is larger than predicted by nonhuman primate allometry (Deacon, 1988; Rilling and Insel, 1999), but there is still debate as to which parts of the cortex are expanded and which are reduced. There is a long tradition arguing that the human frontal lobe is larger than would be expected for a primate of our body size (Deacon, 1997), but new research using MRI has indicated that the human frontal lobe is not disproportionately large (Semendeferi and Damasio, 2000). New evidence is also indicating that the human temporal lobes are disproportionately large in relation to brain size (Rilling and Seligman, 2000; Semendeferi and Damasio, 2000) – a feature that could be related to the expansion of language cortices. Holloway has argued that the human visual cortex is smaller than would be expected, and that this feature is even evident on endocasts of fossil australopithecines (Holloway, 1979, 1981, 1985). Finally, aside from relative size differences in the cortex, the human cerebellum

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appears to be smaller than expected for an anthropoid of our brain size (Rilling and Insel, 1998; Semendeferi and Damasio, 2000).

2.6 Energetic Implications for Human Brain Evolution

The human brain constitutes approximately 2% of body mass, but it receives 15% of cardiac output, is responsible for 20% of total body oxygen utilization, and 25% of total body glucose utilization (Leonard and Robertson, 1994; Mink et al., 1981). In infants, where relative brain size is especially large, cerebral metabolism is responsible for approximately 40% of the entire energy budget (Holliday, 1971; Holliday et al., 1967). Many anthropologists suggest that the human brain's high energy demand has a direct relationship to various anatomical, physiological, and genetic features unique to humans or anthropoids. A few examples will provide the reader with an idea of the breadth of metabolic implications for human brain evolution.

As for the co-evolution of brain size with post-cranial anatomy, Aiello and Wheeler (1995) argued that, since human BMR is no greater than expected for an animal of our body mass, the energy demands of encephalization must have necessitated a trade-off with some other metabolically "expensive" organ. They hypothesized that the adoption of a high quality diet relaxed the need for a large gastro-intestinal tract, and thus allowed for more energy to be shunted to the brain in a kind of zero-sum game. I recently tested their hypothesis using a group of large-brained freshwater fish, and found evidence for reduction of the GI tract in the large-brained mormyrid, *Gnathonemus petersii* (Kaufman, 2003).

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A large "metabolically-expensive" brain also radiates more heat than smaller brains. This observation led Falk and colleagues (Falk, 1986; Falk and Conroy, 1983) to suggest that morphological changes in the occipital venous sinus system of modern human and fossil hominin skulls may be related to heat dispersion.

Molecular biologists and geneticists have also been interested in the unique energy demands of the anthropoid brain. Modifications of the metabolic pathway over the course of anthropoid evolution may offer greater aerobic efficiency for brain metabolism. One such modification is a shift in the proportion of lactate dehydrogenase isozymes in primate brain tissue from type M (muscle type, best suited for anaerobic metabolism and dominant in the strepsirrhine brain) to type H (heart type, best suited for aerobic metabolism and dominant in the anthropoid brain) (Goodman et al., 1969; Koen and Goodman, 1969; Syner et al., 1966). Several research groups have also identified accelerated evolution in cytochrome c oxidase (COX) subunits during the early evolution of catarrhines (Baba et al., 1981; Wu et al., 1997). The COX enzyme is a final component in the TCA cycle and plays a vital role in aerobic tissues. The accelerated evolution of this component is thought to be related to the expansion of the neocortex. Finally, anthropoid primates may also exhibit a modification of the binding properties of fetal hemoglobin that could be related to extended prenatal brain development (Goodman, 1999).

Common to each of these examples is the idea that brain metabolism has had a significant influence on human evolution. Since most theories about brain evolution treat the brain as a uniform energy-consumer, this thesis is intended to provide a more detailed view of mammalian brain metabolism at the level of specific regions. A

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better understanding of the basic principles of pattern and scaling in regional brain metabolism is an important step toward a more sophisticated and nuanced view of human brain evolution.

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3. Methods

The data for this thesis are derived from three types of sources. Most of the metabolic data are derived from a major literature review. To this I have added metabolic data from FDG PET scans on humans and baboons performed at the Mallinckrodt Institute of Radiology. The third data category consists of regional brain volume estimates across species, collected by measuring serial brain sections from MRI scans and from stereotaxic atlases. These volume estimates are necessary to convert mass-specific metabolic data into absolute regional metabolism.

3.1 Published Data

This thesis incorporates neurological data from more than 350 studies published over the last five decades, beginning with the earliest Kety-Schmidt experiments and proceeding to the modern methods of PET and Sokoloff autoradiography. The neurological literature containing these data is continually growing. This thesis includes data published up to and including December 2003. Most of the studies included here were designed to test and measure the effects of various abnormal physiological states (e.g., hypoxia or hypercapnia), neurological trauma (e.g., cerebral artery occlusion and stroke), or the basis of various neuropathologies (e.g., Parkinsonism, autism, etc.). In all cases, studies were selected for inclusion only if baseline control values were published, and only control values are included in this dataset.

Although I focus on glucose metabolic rate, data on the metabolic rate of oxygen and of cerebral blood flow were also collected for comparative purposes.

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Most studies reported metabolic values in units of micromoles (glucose or oxygen) per 100 grams of tissue per minute. However, many researchers reported their results in units of milligrams (glucose or oxygen) per 100g per minute. In these cases, milligrams were converted to micromoles by dividing the reported value by the appropriate molecular weight (180.162 for glucose; 31.9988 for O2) and multiplying by 1000. In the case of oxygen metabolism, some researchers reported values in units of milliliters of oxygen per 100 grams per minute. In these cases, milliliters were converted to micromoles by applying Boyle's Law assuming standard temperature and pressure. Under these conditions (the Ideal Gas Law), one mole of O2 is equal to 22.4 liters, so values reported in milliliters were divided by 22.4 and multiplied by 1000 to obtain micromoles.

In most studies, particularly in the high-resolution Sokoloff autoradiography experiments, regional metabolism is reported for specific components of individual subdivisions. For example, metabolic rate in the basal ganglia is often separately reported for the caudate, putamen, and globus pallidus. Oftentimes, even smaller components are reported, such as the head, body, and tail of the caudate nucleus. In these cases, the separate component values were averaged to obtain a mean metabolic rate for the particular subdivisions analyzed here: the cortex, thalamus, hippocampus, basal ganglia, cerebellum, and white matter. If values were reported separately for the left and right hemispheres, the average was computed and included in the dataset.

In addition to blood flow and metabolic variables, physiological variables describing blood chemistry and vital signs were included in the dataset whenever available. These variables include: heart rate (HR), mean arterial blood pressure

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(MABP), the partial pressure of oxygen (paO2), the partial pressure of carbon dioxide (paCO2), blood pH, and plasma glucose concentration. Mean body mass and age of the study population were also collected whenever they were reported. It is important to note that virtually none of the human studies report body mass statistics, and virtually none of the animal studies report age statistics.

Approximately half of the studies reported variance in the form of standard error of the mean. In these cases, standard error was multiplied by the square root of *n* to obtain the standard deviation of the population. For each species, standard deviations were pooled by squaring the values to obtain variance, computing a weighted average of the variance, and then taking the square root of the final value. Additionally, the standard deviations among mean population values were also computed. These two standard deviations describe different types of variance. While the pooled standard deviations represent the average variance within study populations, the standard deviation of mean values represents the variance among study populations. Coefficients of variation were also computed, including the unbiased method of Sokal and Rohlf (1995).

3.2 Mallinckrodt PET Studies

The protocol for human and baboon ^{18}FDG PET scans performed at the Mallinckrodt Institute of Radiology have been described in detail elsewhere (Mintun et al., 1984; Powers et al., 1995). Human subjects were all normal controls measured in the conscious state, while baboon subjects were pre-operative controls measured under N2O sedation. Quantitative images of rCMRgl (Figure 3.1) were obtained by

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applying the 4-compartment autoradiographic model of Brooks (1982) on a pixel by pixel basis. Rate constants for humans were those given by Brooks (1982), using a lumped constant of 0.54. Baboon rate constants were those given by Kennedy et al. (1978) using a lumped constant of 0.34. A low-pass Butterworth filter was applied to both human and baboon scans to smooth the image. To test the sensitivity of the data

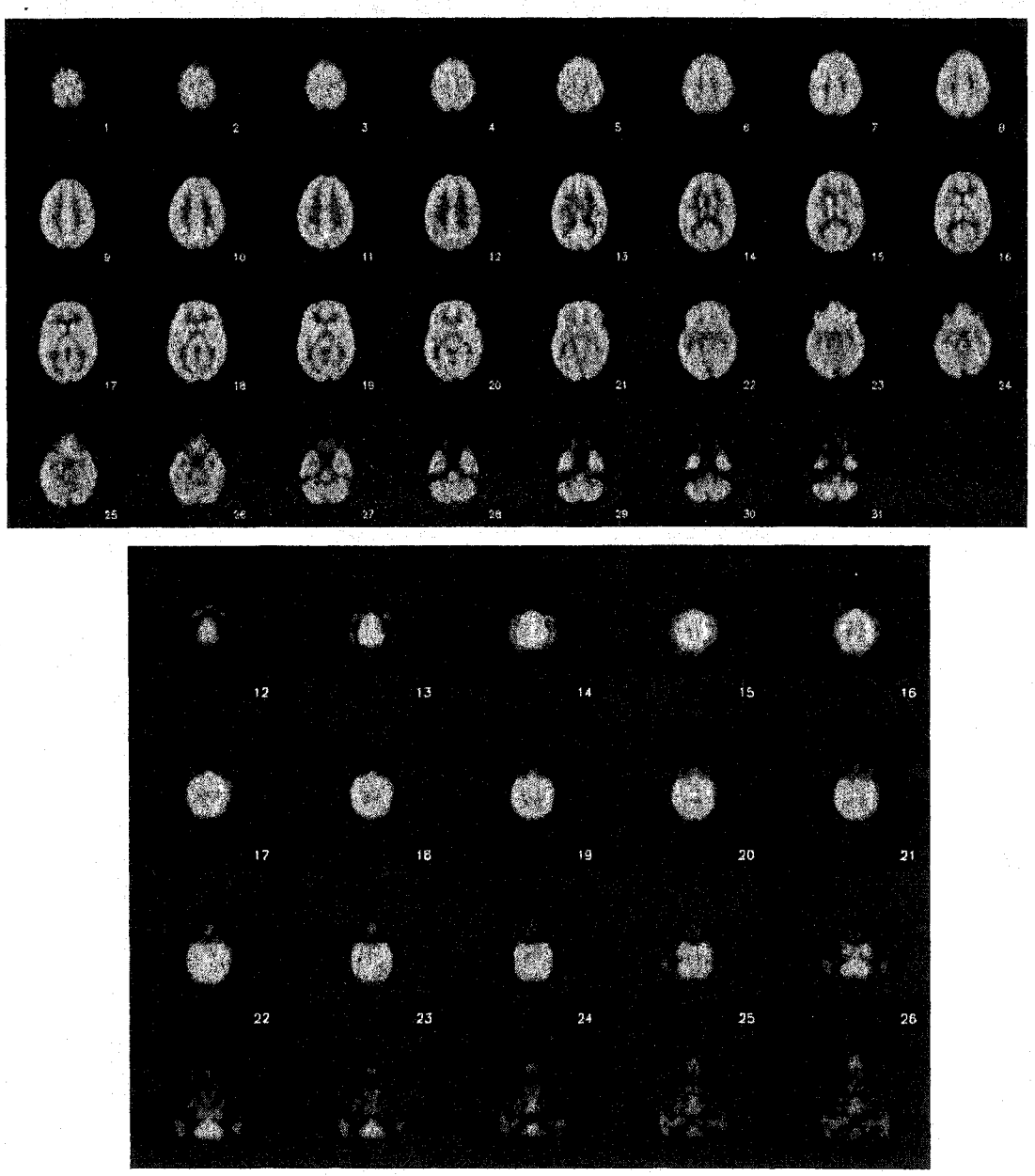


Figure 3.1. PET scans of human (top) and baboon (bottom) performed at the Mallinckrodt Institute of Radiology, Washington University.

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to variation in rate constants, I processed the baboon PET data using three sets of kinetic constants: those for humans (Brooks, 1982), macaques (Kennedy et al., 1978), and rats (Sokoloff et al., 1977). The use of rat constants yielded values lower than expected, but the human and macaque constants produced similar results. The PET data was quite robust to variation in kinetic constants.

Regions of interest (ROI) were defined with the assistance of magnetic-resonance images registered by hand, with no warping. Since a statistical correction for partial-volume averaging was not possible for these scans, rCMR ROI boundaries were defined 2-3 mm within the anatomical boundaries of the region so as to lessen the probability of partial-volume effects. White matter was sampled from the centrum-semiovale, making sure to keep the ROI away from gray matter and ventricular spaces.

3.3 Volume Estimation

Whole brain and regional volume estimates were calculated by applying the Cavalieri principle to serial brain sections (Baddeley et al., 1986). The Cavalieri principle states that the volume of an object can be estimated by first calculating the planar surface area of a region on individual sections. Since both section thickness and inter-slice distance are known, the volume of a region can then be calculated by summing the individual volumes measured from the serial sections. This requires identifying the presence of a subdivision on each slice of a scan or atlas, tracing its boundary, calculating the planar area multiplied by the inter-slice distance, and then summing these section volumes.

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Human and baboon regional volumes were estimated from magnetic resonance scans, while stereotaxic brain atlases were used for all other species (Table 3.1). In total, more than 600 sections were measured for this study. The basic procedure was identical both for MRI scans and stereotaxic atlases. For each MRI or atlas, 30-80 slices of known position (e.g. known inter-slice distance) were selected throughout the entire length of the brain. ROIs were traced for each of the six subdivisions whenever present in the section. Tracings of MRI sections were performed manually using the ANALYZE software package (Mayo Clinic, Rochester) (Figure 3.2). Atlas plates were digitally scanned using an Agfa DuoScan T2000XL, capable of scanning at 4000dpi optical resolution. These images were then imported into the software package ImageJ (NIH, Bethesda) where regions of interest were defined for each subdivision on each slice in which it appears.

Table 3.1. Stereotaxic atlases used for volume estimation.

Species	# of slices*	reference
Mouse	37	(Franklin and Paxinos, 1997)
Mouse	46	(Williams, 2004)
Rat	80	(Pellegrino et al., 1979)
Rat	39	(Paxinos and Watson, 1998)
Gerbil	57	(Loskota et al., 1974)
Rabbit	30	(Girgis and Shih-Chang, 1981)
Cat	62	(Snider and Niemer, 1961)
Rhesus macaque	76	(Paxinos et al., 1999)
Rhesus macaque	52	(Snider and Lee, 1961)

* number of slices analyzed. In some cases, this is fewer than the number of slices provided in the atlas.

For MRI scans, the scale is contained in the scan data, and ANALYZE will quickly return the planar area of any ROI. For atlas images, scale was calibrated to pixel size by referring to a known distance on the stereotaxic grid. After each ROI was traced, planar area was entered into an Excel spreadsheet.

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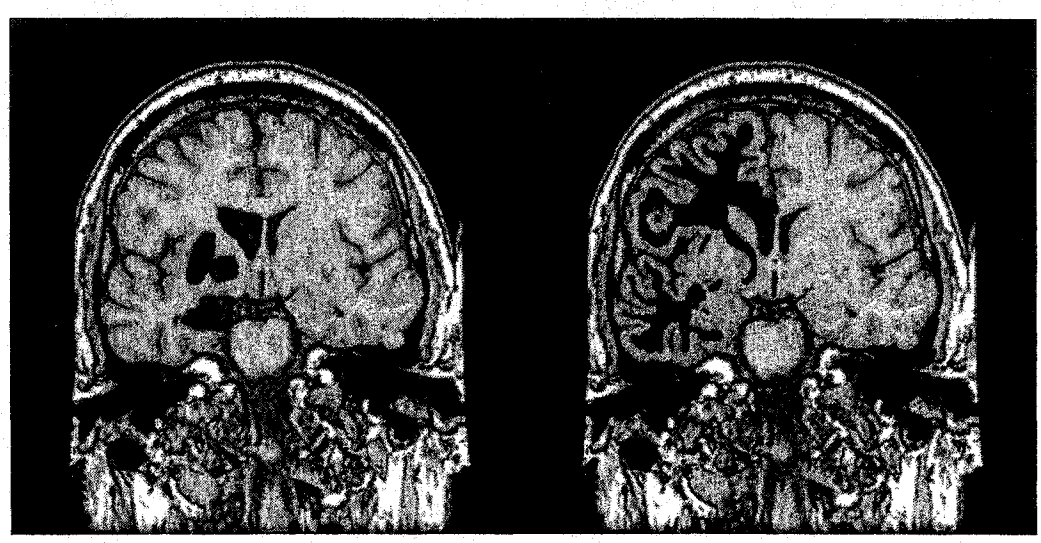


Figure 3.2. Regions of interest as delineated on human MRI slices. Left: sub-cortical gray. Right: white matter.

Planar area for each ROI was then multiplied the by the inter-slice distance to its next adjacent slice. The final slice was multiplied by the inter-slice distance from the previous section. These slice volumes were then summed to obtain regional volume estimates. In the case of bi-lateral regions, the final volume was doubled.

Atlases provide an exceptional method for estimating regional volumes, since most atlases include labeled, diagrammatic tracings of each histological section, in addition to photographic plates. To ensure accurate anatomical definitions, diagrammatic tracings were used for volume estimation whenever possible, since boundaries were clearly delineated and identified by the authors (Figure 3.3). For these diagrammatic illustrations, ImageJ contains a feature whereby a region of interest can be defined by having the computer automatically trace the interior of a solid boundary. Prior to automatic tracing, the image was manually thresholded from gray-scale to a black-and-white line drawing. In this manner, the surface area of each subdivision on any particular section could be calculated consistently and automatically with very little error.

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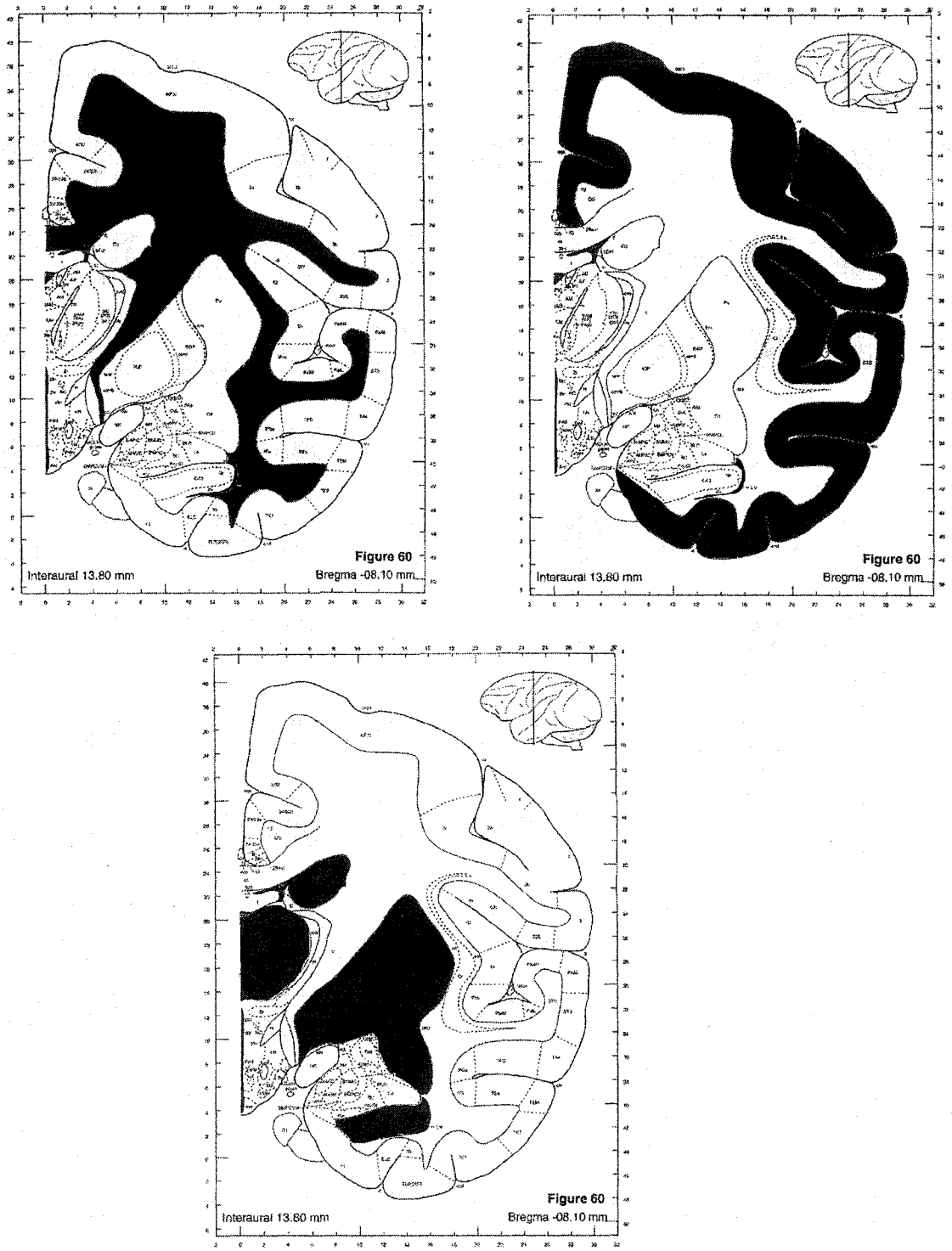


Figure 3.3. Regions of interest as delineated on a macaque stereotaxic atlas slice. Above left: white matter. Above right: cortex. Below: sub-cortical gray.

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Most of the atlases were created using rapid-freezing methods that suffer relatively little shrinkage or distortion. In cases where brains were sectioned after chemical fixation, a shrinkage factor of 30% was assumed (Stephan et al., 1981) and the volume estimates were adjusted accordingly.

In cases where only photographic plates or MRI sections were available, repeatability tests were performed to estimate the error involved in manually tracing regions. These tests involved manually tracing a variety of regions at least five times on different occasions. The error variance (variance among the five intrarater values) was then divided by the total variance in intraspecific region size. The results indicate that structures which have relatively regular shapes, like the thalamus, basal ganglia, and cerebellum, incur relatively little error in tracing: in humans, the ratio of error variance to total variance was less than 3%, while the value in rats was approximately 5%. Complex shapes, such as the cortical ribbon or white matter, are more difficult to trace and consequently have greater error variance among the replicates: the ratio of error variance to total variance for these regions was between 5% and 9%.

The final volume estimates were compared with independent reports of subdivision volumes from a variety of sources (Bramlett and Dietrich, 2002; Cahn et al., 2002; Fox and Wilczynski, 1986; Harvey and Napper, 1991; Herbert et al., 2003; Hofman, 1988; Latimer, 1956; Mega et al., 2002; Scahill et al., 2003; Schlenska, 1974; Seidman et al., 1999; Szabo et al., 2003; West, 1990; Whalley and Wardlaw, 2001). There is a significant amount of variation in brain structure volume within species, though this variance is far less than that between species. Intra-specific

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variation in this case is present and unavoidable. However, given the range of brain sizes in this study and the fact that all volume estimates are log-transformed, intra-specific error in volume estimates should have a relatively minimal effect on the results.

3.4 Statistical Analysis

Regression equations and confidence intervals were calculated using three different methods: *n*-weighted least-squares regression of tip (species) data, non-weighted reduced major axis regression of tip data, and least-squares regression of phylogenetically-independent contrasts. Analysis of tip data is not a "non-phylogenetic" method; it simply assumes a "star" phylogeny in which every species is equidistant from an assumed common ancestor (Felsenstein, 1985). However, the "true" phylogeny reflects the fact that species are differentially related to each other, and therefore do not constitute independent data points in comparative analyses. Felsenstein (1985) proposed a solution to this problem whereby phylogenetic history could be taken into account during comparative analyses. He hypothesized that the differences between species should be independent of phylogeny, and proposed a method of independent contrasts that incorporates an *a priori* cladogram to reduce *n* species data to *n*-1 nodal differences.

The *a priori* phylogeny used here is derived from several sources (Kay et al., 1997; Martin, 1990; Purvis, 1995; Springer and de Jong, 2001; Springer et al., 2003) (Figure 3.4). Although the original phylogeny included estimations of "real" branch lengths, the analyses here are performed using transformed branch lengths. This was

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necessary because one of the assumptions of the independent contrasts method is that contrasts have equal expected variances. To test this assumption, I plotted the absolute values of the contrasts against their standard deviations (the square-root of the sum of corrected branch lengths) and computed their correlation coefficients (Garland et al., 1992; Purvis and Rambaut, 1995). After trying several branch length transformations, I found that Pagel's arbitrary method (Pagel, 1992) provided the best

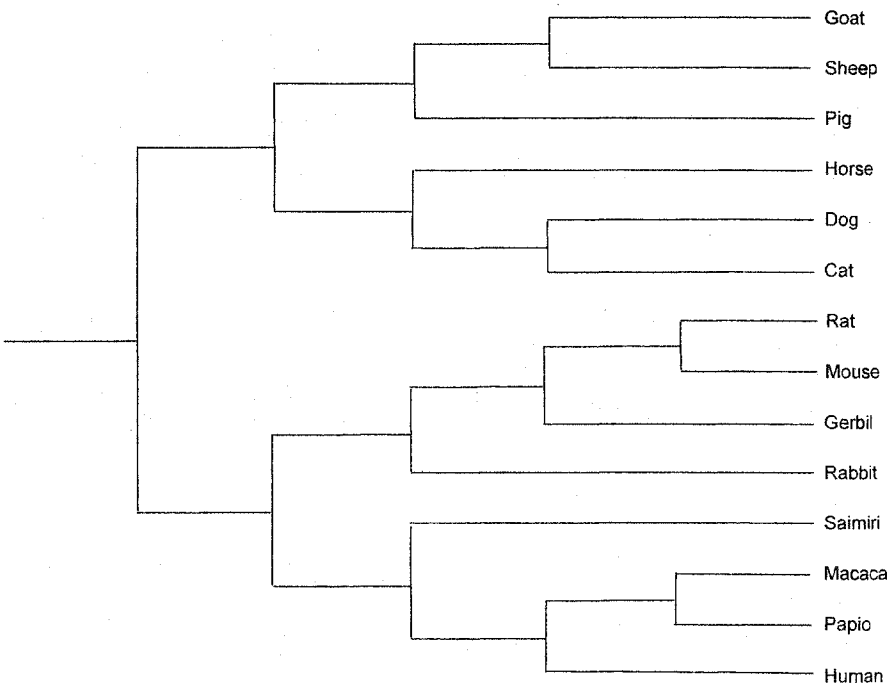


Figure 3.4. Phylogeny for calculating independent contrasts; derived from (Kay et al., 1997; Martin, 1990; Purvis, 1995; Springer and de Jong, 2001; Springer et al., 2003). Branch lengths transformed by Pagel's arbitrary method.

fit for this assumption. Pagel's method sets all branch lengths equal to one with the condition that tips are contemporaneous.

To test for phylogenetic independence in the tip data, and to test whether the contrast data produce values unrelated to the phylogeny, I applied Abouheif's Test for Serial Independence (TFSI) on both the tip data and the contrast data (Abouheif,

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1999). The TFSI is designed to detect phylogenetic autocorrelation among observations. It does not assume a particular model of evolutionary change (such as Brownian motion) and does not require information about branch lengths.

N-weighted least-squares regressions were calculated using SPSS version 10.0. Reduced major axis (RMA) regressions were calculated using the RMA software package (Bohonak, A.J.; San Diego State University). Independent contrasts were calculated using the PDTREE module of the Phenotypic Diversity Analysis Programs (Garland et al., 1993). This version also supports the computation of confidence intervals based on the method of Garland and Ives (2000).

Species means for body mass, brain mass, and basal metabolic rate were compiled either from the atlas/MRI studies, or from published values (Crile and Quiring, 1940; Genoud, 2002; Langer, 2002; Mink et al., 1981; Smith and Jungers, 1997; White and Seymour, 2003).

There are several important statistical properties of this dataset that should be emphasized. First, the raw data represent average values for study populations, not individuals. Sample sizes of study populations vary widely both within and between species. The squirrel monkey, *Saimiri*, is represented by a single study of four anesthetized individuals. Conversely, more than one hundred studies of CMRgl in the rat have been incorporated into the dataset. Sample sizes among the studies on rats vary from four individuals to forty individuals. Therefore, the pooled values for rCMRgl in each species represent the average of study populations, weighted by sample size.

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Second, since the raw data represent study populations, there are two variance statistics for each species: the weighted average of variances *within* study populations, and the variance *among* the sample-size-weighted population values. Comparison of these two variances provides an important barometer of measurement reproducibility within species. Results from a single laboratory tended to be more similar than those from other laboratories, but there was little systematic variation within specific modalities or protocols. For example, there was no correlation between rCMR and the year the data were published.

Third, there is a significant amount of error variance in rCMRgl since the population data that make up the species means were collected under non-identical control conditions. Certain control conditions, such as physiological descriptors like blood pressure and gas tensions, have been recorded in most studies, and their contribution to error variance can be estimated. Other sources of error are not so easily quantified. For example, regional sampling between populations is not stereotaxically precise since two published values of metabolism in the temporal cortex will not have been sampled from exactly the same portions of cortex.

Furthermore, the calculation of rCMRgl is derived from theoretical models of tracer kinetics that, though they have been experimentally validated, are still simplified models that introduce error. Finally, the detection and localization of radiotracers entails errors inherent to the measuring systems. PET, for example, has an intrinsic error component due to the short distance (approximately 2 mm) between the point of positron emission and the point of the annihilation event that is detected by the PET cameras. Partial-volume averaging is another source of error intrinsic to

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PET, since any region of interest that is sampled will be "contaminated" to some degree by surrounding tissues or spaces. All of these factors will be examined in detail in order to characterize the methodological issues involved in comparing regional glucose metabolic rate among different studies and among different species.

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4. Results

The total sample is summarized below for conscious subjects (Table 4.1) and anesthetized subjects (Table 4.2). Population data from individual experiments, arranged by brain structure, are presented in Appendix Tables A1-A14. Sample size, anesthetic regimen, and measurement method accompany the metabolic and blood flow statistics for each reference. Note that, for any given species, the majority of data from different regions were collected from the same populations.

Table 4.1. Summary of sample sizes for conscious subjects: # of studies (# of individuals)
G = CMRgl, O = CMRO2, F = CBF

Genus	Whole Brain			Cortex			Thalamus			Basal Ganglia		
	G	O	F	G	O	F	G	O	F	G	O	F
Homo	67 (823)	37 (358)	35 (335)	49 (650)	13 (169)	10 (125)	33 (406)	9 (102)	8 (114)	32 (372)	11 (133)	4 (52)
Macaca	15 (72)	4 (22)	4 (22)	10 (46)		1 (6)	10 (44)			9 (40)	1 (6)	1 (6)
Canis	1 (8)	1 (8)	1 (12)									
Felis	1 (5)			1 (11)			2 (11)			2 (11)		
Lepus	1 (4)	1 (38)	1 (38)	1 (5)			1 (5)			1 (5)		
Meriones	3 (13)		1 (10)	3 (13)		2 (7)	2 (8)		1 (2)	3 (13)		2 (7)
Rattus	56 (524)	7 (106)	7 (106)	47 (347)	1 (6)	3 (32)	42 (321)		4 (26)	40 (294)		2 (12)
Mus	12 (80)		1 (5)	8 (43)			5 (29)			5 (30)		
Equus			2 (18)									
Ovis		3 (18)	4 (30)									
Capra		2 (23)	1 (17)									
Sus					1 (20)	1 (6)			1 (6)			1 (6)

	Hippocampus			Cerebellum			White Matter		
	G	O	F	G	O	F	G	O	F
Homo	8 (106)	1 (14)	2 (40)	22 (299)	8 (92)	5 (73)	13 (158)	10 (146)	11 (172)
Macaca	7 (31)		1 (6)	6 (25)		1 (6)	6 (24)		
Canis									
Felis	2 (11)			1 (4)			1 (4)		
Lepus	1 (5)						1 (5)		
Meriones	1 (6)		1 (2)	1 (6)		1 (10)			1 (2)
Rattus	42 (314)		5 (30)	28 (220)		2 (14)	34 (252)		1 (4)
Mus	6 (49)			4 (25)			5 (28)		
Equus									
Ovis									
Capra									
Sus						1 (6)			1 (6)

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Table 4.2. Summary of sample sizes for anesthetized subjects: # of studies (# of individuals)
G = CMRgl, O = CMRO2, F = CBF

Genus	Whole Brain			Cortex			Thalamus			Basal Ganglia		
	G	O	F	G	O	F	G	O	F	G	O	F
Homo	1 (10)	3 (32)	3 (32)					1 (8)	1 (8)		1 (8)	
Papio	10 (69)	17 (134)	17 (134)	6 (39)	2 (16)	2 (16)	3 (20)	1 (4)		2 (10)		
Macaca	6 (41)	3 (40)	3 (40)	3 (7)	1 (4)	2 (8)	1 (4)		1 (4)	4 (11)		
Saimiri	1 (4)			1 (4)		1 (4)	1 (4)		1 (4)	1 (4)		1 (4)
Canis	5 (49)	35 (311)	33 (288)									
Felis	5 (48)	9 (103)	8 (96)	1 (6)			1 (6)			1 (6)		
Lepus		4 (35)	4 (35)			3 (28)						
Meriones	2 (7)			1 (3)		2 (11)	1 (3)		1 (4)	1 (3)		1 (4)
Rattus	7 (48)	26 (242)	26 (242)	6 (41)	2 (12)	4 (29)	1 (21)		1 (6)	3 (25)		
Mus												
Equus												
Ovis												
Capra												
Sus		5 (39)	6 (57)		3 (23)	3 (16)			2 (12)			2 (12)

	Hippocampus			Cerebellum			White Matter		
	G	O	F	G	O	F	G	O	F
Homo					1 (8)	1 (8)			
Papio	3 (20)			2 (10)			1 (2)		
Macaca				2 (3)				1 (4)	1 (4)
Saimiri	1 (4)		1 (4)	1 (4)		1 (4)	1 (4)		1 (4)
Canis									
Felis				1 (6)			1 (6)		
Lepus			1 (12)						
Meriones	2 (7)		2 (11)	1 (3)		1 (7)			
Rattus	2 (15)			1 (10)					
Mus									
Equus									
Ovis									
Capra									
Sus						3 (16)			3 (16)

There was high correlation among gray matter metabolism . Cortical CMRgl was highly correlated with thalamic CMRgl (Humans: $R = 0.79$; $n = 31$; $p < 0.001$; Rats: $R = 0.75$; $n = 42$; $p < 0.001$) and basal ganglia CMRgl (Humans: $R = 0.85$; $n = 29$; $p < 0.001$; Rats: $R = 0.70$; $n = 40$; $p < 0.001$). The highest correlation was between human thalamic CMRgl and basal ganglia CMRgl ($R = 0.95$; $n = 28$; $p < 0.001$). White matter metabolism did not generally correlate with other regions.

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Intra-specific Variation

4.1 Anesthesia

It was hypothesized that the primary source of intra-specific variation in this dataset would be due to anesthetic regimen, categorized as "conscious," "N2O sedated," and "other." The latter category includes a variety of general anesthetics administered with or without N2O.

Nested ANOVAs for each region, in which the independent factors are defined as "species" and "anesthetic regimen within species," consistently show that the nested factor explains a significant portion of the total variance in virtually all regions, regardless of whether the analysis is weighted by sample size. Tables 4.3 & 4.4 provide examples of these results. The ratios of sums of squares between "species" and "anesthetic regimen within species" are generally between 4:1 and 8:1, meaning that variation between genera is far greater than variation among anesthetic regimens within genera.

Table 4.3. Nested ANOVA (unweighted) of Cortical CMRgl in which the independent factors are "species" and "anesthetic regimen within species."

	Sum of Squares	df	Mean Square	F	Sig.
Model	119350.97	14	8525.07	48.36	<0.001
Intercept	114811.35	1	114811.35	651.27	<0.001
Species	46848.46	8	5856.06	33.22	<0.001
Anesth. within Species	6494.70	6	1082.45	6.14	<0.001
Error	22388.52	127	176.29		
Total	775155.16	142			

R Squared = .842 (Adjusted R Squared = .825)

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Table 4.4. Nested ANOVA (weighted) of Hippocampus CMRgl in which the independent factors are "species" and "anesthetic regimen within species."

Source	Sum of Squares	df	Mean Square	F	Sig.
Model	153588.95	13	11814.53	131.14	<0.001
Intercept	241648.78	1	241648.78	2682.29	<0.001
Species	82019.00	8	10252.37	113.80	<0.001
Anesth. within Species	11009.71	5	2201.94	24.44	<0.001
Error	52612.90	584	90.09		
Total	2058900.60	598			

R Squared = .745 (Adjusted R Squared = .739)

The anesthetic effect can be examined in greater detail by performing independent-samples t-tests for each species and each possible region. There are 50 possible comparisons in which conscious whole brain and regional cerebral metabolic rates can be compared with anesthetized values ("N2O" and "other"). Table 4.5 presents the results of these comparisons, with statistics for the mean difference between conscious and anesthetized subjects (positive difference indicates conscious>anesthetized) The results indicate a consistent depression of cerebral metabolism and blood flow in the anesthetized state – 39 of 50 possible comparisons have *p*-values (2-tailed) less than 0.001. Of those 39 comparisons, 32 indicate decreases in metabolism or blood flow in the anesthetized state, while 7 indicate increases.

Table 4.5. Comparison of rCMR and rCBF in conscious versus anesthetized subjects.

df* = unweighted degrees of freedom; Mean Difference = anesthetized minus conscious.

		t	df	df*	Sig. (2-tailed)	% Change	Mean Difference	Std. Error of the Difference
Homo	Whole Brain CMRgl	-5.10	224	21	<0.001	30.48	-10.10	1.98
	Whole Brain CMRO2	-9.88	388	38	<0.001	21.73	-33.77	3.42
	Whole Brain CBF	-7.18	365	36	<0.001	17.66	-9.41	1.31
	Thalamus CMRO2	-0.98	108	8	0.328	8.32	-13.45	13.69
	Thalamus CBF	-6.18	120	7	<0.001	26.37	-14.28	2.31
	Basal Ganglia CMRO2	-0.48	139	10	0.634	3.23	-5.04	10.56

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	Cerebellum CMRO2	-5.17	98	7	<0.001	34.06	-61.34	11.86
	Cerebellum CBF	-4.11	79	4	<0.001	37.97	-21.64	5.26
Macaca	Whole Brain CMRgl	0.74	111	19	0.461	3.53	1.31	1.77
	Whole Brain CMRO2	-4.50	60	5	<0.001	24.70	-53.10	11.81
	Whole Brain CBF	-5.91	60	5	<0.001	34.12	-22.97	3.89
	Cortex CMRgl	-2.38	51	11	0.021	17.35	-8.67	3.65
	Cortex CBF	-1.25	12	1	0.235	12.74	-6.75	5.40
	Thalamus CMRgl	-3.63	46	9	0.001	20.79	-9.32	2.57
	Basal Ganglia CMRgl	-1.68	49	11	0.099	7.85	-3.48	2.07
	Cerebellum CMRgl	4.74	26	6	<0.001	29.69	11.88	2.51
Canis	Whole Brain CMRgl	-10.28	55	4	<0.001	33.70	-15.00	1.46
	Whole Brain CMRO2	-0.93	317	34	0.351	8.82	-18.11	19.40
	Whole Brain CBF	4.33	298	32	<0.001	54.44	36.80	8.50
Sus	Thalamus CBF	-8.73	16	1	<0.001	31.51	-24.20	2.77
	Cerebellum CBF	-2.32	20	2	0.031	20.69	-18.38	7.93
	Cortex CBF	-5.53	20	2	<0.001	42.66	-46.16	8.35
	Cortex CMRO2	-86.06	16	1	<0.001	51.11	-174.77	2.03
	Basal Ganglia CBF	-35.65	16	1	<0.001	17.30	-17.75	0.50
	White Matter CBF	-3.92	20	2	0.001	21.58	-7.25	1.85
Lepus	Whole Brain CMRO2	-24.83	71	3	<0.001	30.12	-61.87	2.49
	Whole Brain CBF	7.22	71	3	<0.001	23.11	14.43	2.00
Felis	Whole Brain CMRgl	-0.99	50	4	0.328	7.42	-2.33	2.36
	Cortex CMRgl	-4.01	15	1	0.001	38.62	-25.67	6.40
	Thalamus CMRgl	-8.18	15	1	<0.001	62.09	-46.84	5.73
	Basal Ganglia CMRgl	-9.16	15	1	<0.001	33.24	-18.57	2.03
Meriones	Whole Brain CMRgl	0.67	18	3	0.511	4.90	3.53	5.27
	Cortex CMRgl	4.63	14	2	<0.001	28.61	29.40	6.35
	Cortex CBF	4.79	16	2	<0.001	23.68	31.97	6.67
	Thalamus CMRgl	14.22	9	1	<0.001	28.64	30.65	2.16
	Basal Ganglia CMRgl	0.59	14	2	0.565	8.13	6.62	11.24
	Basal Ganglia CBF	1.58	9	1	0.147	21.20	26.71	16.86
	Hippocampus CMRgl	-1.27	11	1	0.230	14.31	-9.87	7.77
	Hippocampus CBF	-4.39	11	1	0.001	18.92	-20.62	4.70
Rattus	Whole Brain CMRgl	-0.65	570	61	0.516	2.01	-1.37	2.12
	Whole Brain CMRO2	-3.65	345	31	<0.001	8.52	-36.59	10.03
	Whole Brain CBF	4.59	346	31	<0.001	10.04	11.51	2.51
	Cortex CMRgl	-9.97	386	51	<0.001	23.86	-23.69	2.38
	Cortex CMRO2	-4.32	41	2	<0.001	20.10	-93.34	21.60
	Cortex CBF	-2.18	59	5	0.034	12.24	-17.82	8.19
	Thalamus CMRgl	-3.47	340	43	0.001	12.35	-11.11	3.20
	Thalamus CBF	-10.90	30	41	<0.001	45.04	-58.52	5.37
	Hippocampus CMRgl	-3.47	327	42	0.001	14.19	-9.32	2.68
	Basal Ganglia CMRgl	-1.25	317	41	0.211	3.69	-2.67	2.13
	Cerebellum CMRgl	-4.14	228	27	<0.001	28.90	-15.85	3.83

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When the mean differences in Table 4.5 are converted to a percentage of the average conscious value for the species, the average percent reduction due to anesthesia (across regions and species) is 35% for glucose metabolism (min. 19%, max. 62%), 29% for oxygen metabolism (min. 9%, max. 51%), and 29% for blood flow (min. 12%, max. 49%). These anesthesia-induced depressions appear to be unbiased across species, except for the gerbil, *Meriones*, in which 3 of the 4 statistically-significant comparisons surprisingly indicate an increase in blood flow and metabolism.

When conscious subjects are compared to those receiving only N2O, there is still a general pattern of depression of blood flow and metabolism, but the results are less regionally consistent (Table 4.6). For example, the basal ganglia are unaffected in *Meriones* and *Rattus*, but are significantly reduced in *Felis*. Whole-brain blood flow is reduced by 12% in *Homo* and by 54% in *Macaca mulatta*, but is increased by 32% in *Canis* and by a modest 6% in *Rattus*. Whole-brain and cortical metabolism and blood flow appear to be most sensitive to the effects of N2O. The average percent reduction (across regions and species) in CMRgl is 32% (min. 6%, max. 62%). For oxygen uptake, the average percent reduction is 28% (min. 10%, max. 55%), and for blood flow the average percent reduction is 33% (min. 12%, max. 54%).

Table 4.6. Comparison of rCMR and rCBF in conscious versus those given N2O only.
df* = unweighted degrees of freedom; Mean Difference = anesthetized minus conscious.

		t	df	df*	Sig. (2-tailed)	% Change	Mean Difference	Std. Error of the Difference
Homo	Whole Brain CMRO2	3.40	381	37	0.001	11.09	-17.26	5.07
	Whole Brain CBF	3.59	358	35	<0.001	13.26	-7.03	1.96
Macaca	Whole Brain CMRgl	1.28	89	14	0.204	8.69	-3.12	2.44
	Whole Brain CMRO2	2.90	39	3	0.006	23.15	-49.75	17.18

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	Whole Brain CBF	4.80	39	3	<0.001	39.70	-26.73	5.57
Felis	Whole Brain CMRgl	0.72	26	1	0.477	3.57	-1.12	1.55
	Cortex CMRgl	4.01	15	1	0.001	38.62	-25.67	6.40
	Thalamus CMRgl	8.18	15	1	<0.001	62.09	-46.84	5.73
	Basal Ganglia CMRgl	9.16	15	1	<0.001	33.24	-18.57	2.03
Meriones	Whole Brain CMRgl	-0.08	14	2	0.936	0.95	0.66	8.09
	Cortex CMRgl	-4.63	14	2	<0.001	28.61	29.40	6.35
	Thalamus CMRgl	-14.22	9	1	<0.001	28.64	30.65	2.16
	Basal Ganglia CMRgl	-0.59	14	2	0.565	8.13	6.62	11.24
Rattus	Whole Brain CMRgl	-1.38	557	59	0.170	5.08	3.65	2.65
	Whole Brain CMRO2	2.13	232	18	0.034	5.06	-21.74	10.21
	Whole Brain CBF	-3.16	226	17	0.002	4.99	5.41	1.71
	Cortex CMRgl	7.26	380	50	<0.001	20.29	-20.09	2.77
	Cortex CMRO2	4.32	41	2	<0.001	20.10	-93.34	21.60
	Cortex CBF	0.62	53	4	0.539	3.23	-4.71	7.61
	Thalamus CMRgl	0.62	339	43	0.534	2.57	-2.31	3.70
	Basal Ganglia CMRgl	1.21	322	42	0.227	3.54	-2.56	2.12
	Hippocampus CMRgl	3.38	332	43	0.001	13.93	-9.12	2.69
	Cerebellum CMRgl	4.14	228	27	<0.001	28.90	-15.85	3.83

There are also significant differences between subjects receiving only N2O and those receiving N2O plus another anesthetic (Table 4.7). Under this regimen, both glucose and oxygen metabolism are significantly reduced in most species and region. Whole-brain blood flow is unaffected in *Homo*, *Canis,* and *Felis*, but shows a 32% decrease in *Papio* and a 15% increase in *M. mulatta*.

Table 4.7. Comparison of rCMR and rCBF in those receiving only N2O versus those receiving N20 plus another anesthetic.

df* = unweighted degrees of freedom; Mean Difference = other anesthetic minus N2O only.

		t	df	df*	Sig. (2-tailed)	% Change	Mean Difference	Std. Error of the Difference
Genus								
Homo	Whole Brain CMRO2	6.31	30	1	<0.001	20.36	-28.18	4.46
	Whole Brain CBF	1.43	30	1	0.164	7.89	-3.63	2.55
Papio	Whole Brain CMRgl	1.55	67	8	0.126	31.89	-8.12	5.23
	Whole Brain CMRO2	2.13	132	15	0.035	7.36	-10.64	4.99
	Whole Brain CBF	11.14	132	15	<0.001	7.36	-19.42	1.74
	Cortex CMRgl	2.36	37	4	0.024	26.24	-12.86	5.45
	Thalamus CMRgl	26.88	18	1	<0.001	4.36	-1.99	0.07
	Hippocampus CMRgl	2.71	18	1	0.014	4.55	-1.51	0.56
Macaca	Whole Brain CMRgl	-7.69	39	4	<0.001	20.12	8.25	1.07
	Whole Brain CMRO2	4.05	38	1	<0.001	3.86	-6.37	1.57
	Whole Brain CBF	-33.82	38	1	<0.001	14.99	7.16	0.21

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Canis	Whole Brain CMRO2	1.77	296	32	0.078	14.47	-31.66	17.89
	Whole Brain CBF	-1.27	273	30	0.207	17.49	12.08	9.55
Sus	Whole Brain CMRO2	-5.52	37	3	<0.001	41.57	73.08	13.23
	Whole Brain CBF	-3.99	55	4	<0.001	23.23	10.29	2.58
Felis	Whole Brain CMRgl	1.84	46	3	0.072	8.00	-2.42	1.32
	Whole Brain CMRO2	2.54	101	7	0.013	18.61	-36.54	14.39
	Whole Brain CBF	1.30	94	6	0.197	9.99	-5.10	3.92
Rattus	Whole Brain CMRgl	3.26	46	5	0.002	17.99	-12.92	3.96
	Whole Brain CMRO2	2.20	229	23	0.029	6.79	-27.67	12.60
	Whole Brain CBF	-4.94	230	23	<0.001	12.54	15.55	3.15
	Cortex CMRgl	3.25	39	4	0.002	15.73	-12.42	3.82
	Cortex CBF	5.66	27	2	<0.001	44.98	-63.37	11.21
	Thalamus CMRgl	9.18	19	1	<0.001	34.05	-29.73	3.24

The pattern of anesthetic effect is one in which N2O alone results in a less severe physiologic effect than when N2O is administered in combination with a general anesthetic, as illustrated in Figures 4.1 and 4.2. However, the anesthetic effect is not altogether consistent. In some cases, particularly for cerebral blood flow, there may be no statistical difference among anesthetic regimens (Figure 4.3). In other cases, particularly for subcortical gray regions, measurements performed under N2O alone are not statistically different from measurements performed in the conscious state, but measurements performed under general anesthesia show a clear depression (Figure 4.4).

For the purposes of this thesis, the principal result regarding anesthesia is that the effect is certainly strong enough that it must be taken into consideration when comparing cerebral physiology across species. For this reason, all of the following analyses are performed separately for conscious and anesthetized subjects, and most of the results will focus on subjects measured in the conscious state.

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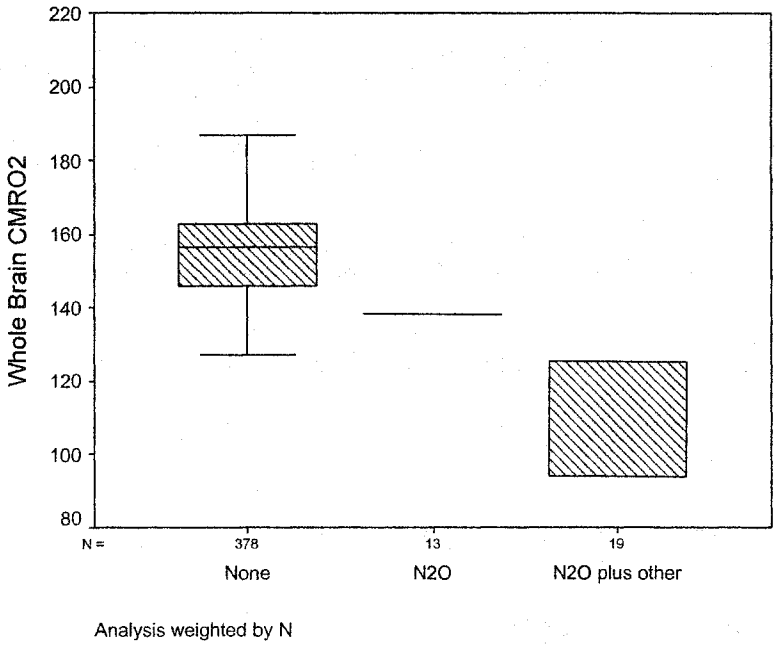


Figure 4.1. Human Whole Brain CMRO2 compared by anesthetic regimen.

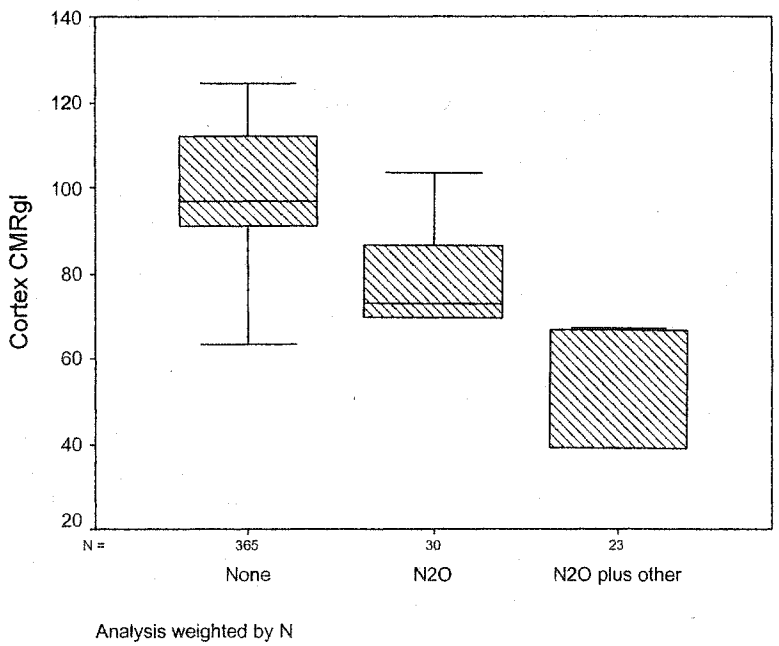


Figure 4.2. Rat Cortex CMRgl compared by anesthetic regimen.

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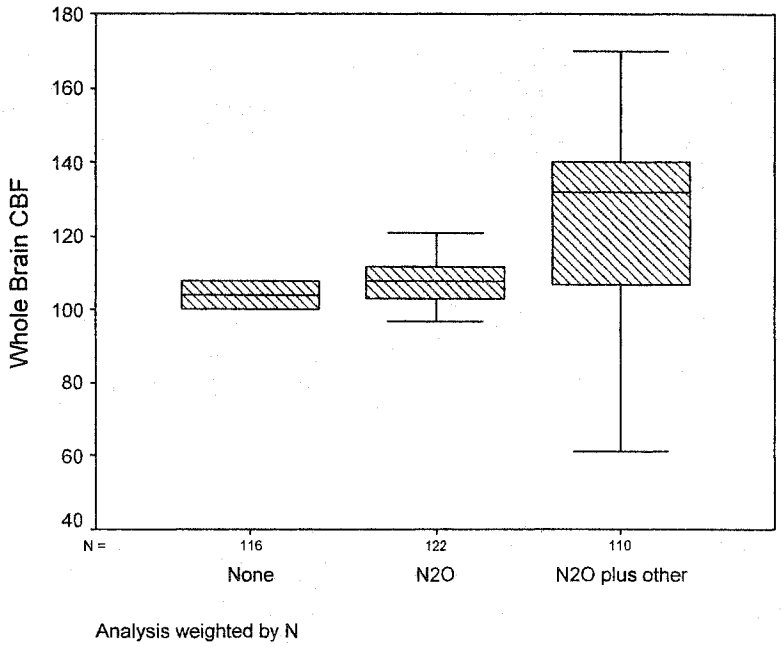


Figure 4.3. Rat Whole Brain CBF compared by anesthetic regimen.

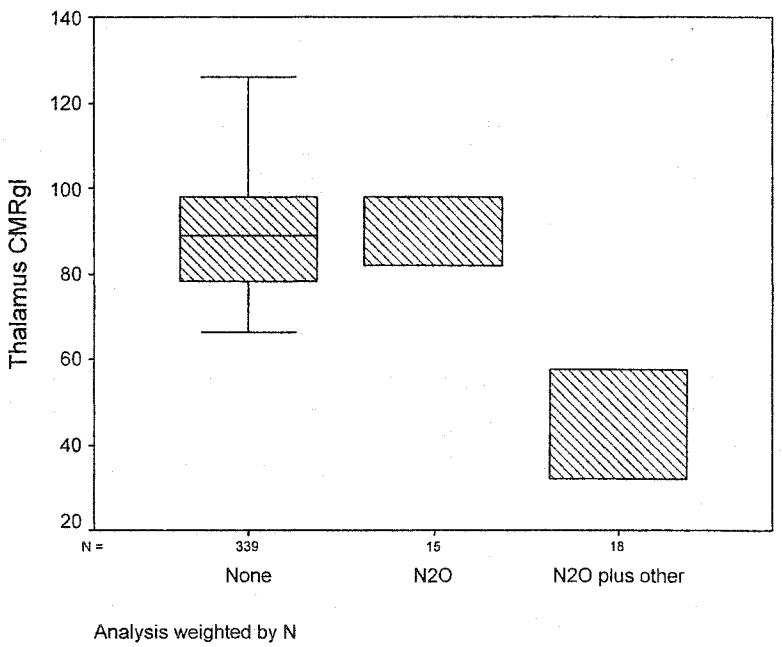


Figure 4.4. Rat Thalamus CMRgl compared by anesthetic regimen.

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4.2 Measurement Technique

A second likely source of variation in the data set will be that associated with measurement technique: the Kety-Schmidt method, the Sokoloff method, and PET. A key methodological goal for this particular analysis is to assess the degree of error bias introduced by partial-volume averaging in PET, a phenomenon which is virtually non-existent in terminal autoradiography. If partial-volume averaging were to affect CMRgl, it would tend to artificially deflate high values and inflate low values in PET experiments. For example, partial-volume averaging would cause cortical CMR to be artificially low and white matter CMR to be artificially high.

Of 12 possible independent samples *t*-tests in which PET values can be compared with Sokoloff-derived values, 7 are statistically distinct based on measurement technique (Table 4.8). Particularly high *t*-values are present for cortical CMRgl, which is lower in PET experiments (Figure 4.5). Unfortunately, it was not possible to test for differences in white matter CMRgl between PET and Sokoloff-derived values. However, the indication of lower values for PET-derived cortical rCMRgl fits the pattern one would expect from partial-volume averaging. For the purposes of inter-specific comparisons in this thesis, the indication of partial-volume effects means that human PET values must be interpreted cautiously: cortical rCMRgl may be artificially low, while white matter rCMRgl may be artificially high.

With regard to measurements of whole brain metabolism, there also appears to be a measurement effect regarding the Kety-Schmidt technique and PET (Table 4.9), though the effect on CBF is opposite that on CMR. For whole brain CMRgl, PET yields lower values than the Kety-Schmidt technique among anesthetized baboons

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and conscious humans. Conversely, whole brain CBF values are higher in PET experiments than Kety-Schmidt experiments among conscious humans as well as among anesthetized baboons and rats.

Table 4.8. Independent Samples t-test: PET vs. Sokoloff

Conscious subjects. Mean Difference = Sokoloff minus PET.

		t	df	df*	Sig. (2-tailed)	% Change	Mean Difference	Std. Error of the Difference
Macaca	Whole Brain CMRgl	0.76	48	9	0.454	5.97	2.11	2.79
	Cortex CMRgl	16.03	44	8	<0.001	32.45	17.72	1.11
	Thalamus CMRgl	3.51	42	8	0.001	15.22	6.97	1.99
	Hippocampus CMRgl	1.57	29	5	0.127	8.52	3.00	1.91
	Basal Ganglia CMRgl	1.71	38	7	0.095	8.99	4.04	2.36
	Cerebellum CMRgl	-0.86	23	4	0.401	4.51	-1.30	1.52
Rattus	Whole Brain CMRgl	-3.73	388	47	<0.001	16.56	-12.57	3.37
	Cortex CMRgl	7.61	345	45	<0.001	30.21	30.32	3.98
	Thalamus CMRgl	3.15	319	40	0.002	14.30	12.93	4.10
	Basal Ganglia CMRgl	-2.83	292	38	0.005	10.12	-8.11	2.87

Anesthetized subjects. Mean Difference = Sokoloff minus PET.

		t	df	df*	Sig. (2-tailed)	% Change	Mean Difference	Std. Error of the Difference
Rattus	Whole Brain CMRgl	2.90	46	5	0.006	18.66	13.11	4.52
	Cortex CMRgl	2.03	39	4	0.049	14.65	11.28	5.55

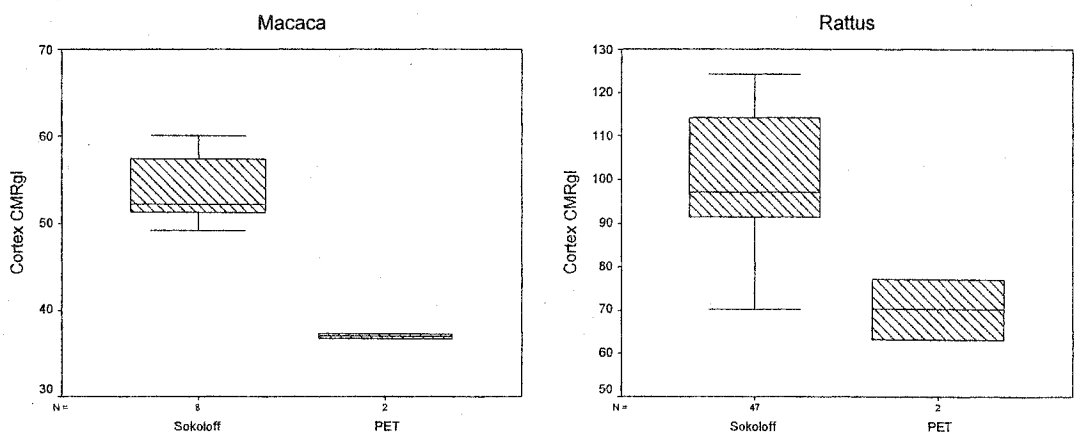


Figure 4.5. The effect of measurement technique on rCMRgl values. PET-derived cortical rCMRgl values are less than Sokoloff-derived values.

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Table 4.9. Independent Samples t-test: Kety-Schmidt vs. PET

Conscious subjects. Mean Difference = PET minus Kety-Schmidt.

		t	df	df*	Sig. (2-tailed)	% Change	Mean Difference	Std. Error of the Difference
Homo	Whole Brain CMRgl	-5.86	814	64	0.000	13.00	-4.37	0.75
	Whole Brain CMRO2	-0.90	349	34	0.368	1.23	-1.94	2.16
	Whole Brain CBF	9.70	326	32	0.000	13.73	7.61	0.78
Macaca	Whole Brain CMRgl	1.27	32	4	0.214	13.74	5.28	4.17
Rattus	Whole Brain CMRgl	0.89	104	6	0.376	4.10	3.25	3.65

Anesthetized subjects. Mean Difference = PET minus Kety-Schmidt.

		t	df	df*	Sig. (2-tailed)	% Change	Mean Difference	Std. Error of the Difference
Papio	Whole Brain CMRgl	-26.24	50	7	0.000	40.49	-15.68	0.60
	Whole Brain CMRO2	3.32	92	12	0.001	11.05	16.35	4.92
	Whole Brain CBF	13.13	92	12	0.000	39.23	21.02	1.60
	Cortex CMRgl	-1.25	37	4	0.219	11.67	-4.37	3.50
Macaca	Whole Brain CMRgl	-1.10	39	4	0.276	5.38	-2.08	1.89
Canis	Whole Brain CMRO2	1.68	241	25	0.095	19.12	37.78	22.53
	Whole Brain CBF	2.47	228	23	0.014	44.03	31.39	12.73
Sus	Whole Brain CMRO2	1.62	25	1	0.118	3.67	7.11	4.39
	Whole Brain CBF	17.72	25	1	0.000	28.27	15.86	0.90
Rattus	Whole Brain CBF	6.08	240	24	0.000	47.50	55.18	9.07

In summary, the likelihood of a partial-volume effect means that regional comparisons involving PET- and Sokoloff-derived values need to be considered with care. In addition, although differences concerning the Kety-Schmidt method are less consistent, the magnitude of the effect size once again commands a degree of caution when comparing Kety-Schmidt-derived Whole Brain values with PET- or Sokoloff-derived values.

4.3 Circulatory Physiology

A third potential source of intra-specific variation is the physiological state of the circulatory system: heart rate, mean arterial blood pressure, blood O2 tension,

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blood CO2 tension, and blood pH. These physiological properties often co-vary – blood pressure and heart rate for example. But of particular interest are the potential relationships between CMR/CBF and circulatory variables that could account for some portion of intra-specific variation. Correlation analysis among all regions within each species entails a large number of multiple comparisons and yields numerous statistically-significant correlation coefficients, particularly when the analysis is weighted by sample size. A more robust test is weighted least squares regression. The results of this procedure indicate that blood gas tensions in particular have a statistically-significant effect on cerebral metabolism, and this effect is most detectable in the rat where large sample sizes provide greater power.

Table 4.10 presents a list of *only* statistically-significant ($\alpha = 0.05$) regression coefficients in which the total degrees of freedom are greater than or equal to five. It is important to note that under these restrictions ($p < 0.05$; and $df > 4$), the rat is the only species to yield significant regression results. However, the presence and number of these significant correlations confirms the notion that metabolic and blood flow data are sensitive to circulatory physiology, and that the effect size can be sizeable. To help interpret effect size, Table 4.11 provides descriptive statistics of the circulatory variables in conscious rats. In the case of carbon dioxide tension, where there is a consistent negative correlation with regional metabolism, a one-standard-deviation rise in CO2 tension will produce a 14% reduction in whole brain CMRgl (Figure 4.6, top). A two-standard-deviation rise in blood pH will produce a 14% increase in thalamic CMRgl (Figure 4.6, bottom). Clearly, these are meaningful

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correlations that account for a significant proportion of intra-specific variation in cerebral blood flow and metabolism.

Table 4.10. Statistically significant results from weighted least-squares regression of CMR/CBF on circulatory variables in conscious rats.

		Beta	Sig.	df
HR				
	Cortex CMRgl	0.11	0.012	11
	Basal Ganglia CMRgl	0.11	0.017	10
MABP				
	Whole Brain CBF	1.10	0.014	5
	White CMRgl	-0.33	0.033	26
paO2				
	Cortex CMRgl	-0.53	0.004	25
	Thalamus CMRgl	-0.43	0.031	22
	Hippocampus CMRgl	-0.46	0.015	23
	Basal Ganglia CMRgl	-0.37	0.050	22
paCO2				
	Whole Brain CMRgl	-3.70	0.019	7
	Whole Brain CMRO2	-21.60	0.023	5
	Thalamus CMRgl	-1.70	0.037	22
	Cerebellum CMRgl	-2.30	0.018	16
	White CMRgl	-1.45	0.009	20
pH				
	Thalamus CMRgl	301.00	0.002	20

Table 4.11. Descriptive statistics of physiological variables in the conscious rat.

	N	Range	Min.	Max.	Mean	SD
Heart Rate	15	141.00	346.00	487.00	410.53	37.51
MABP	39	47.80	90.20	138.00	117.26	9.52
paO2	32	82.24	75.76	158.00	95.20	20.23
paCO2	34	11.95	30.75	42.70	37.98	2.81
pH	29	0.11	7.36	7.47	7.42	0.02

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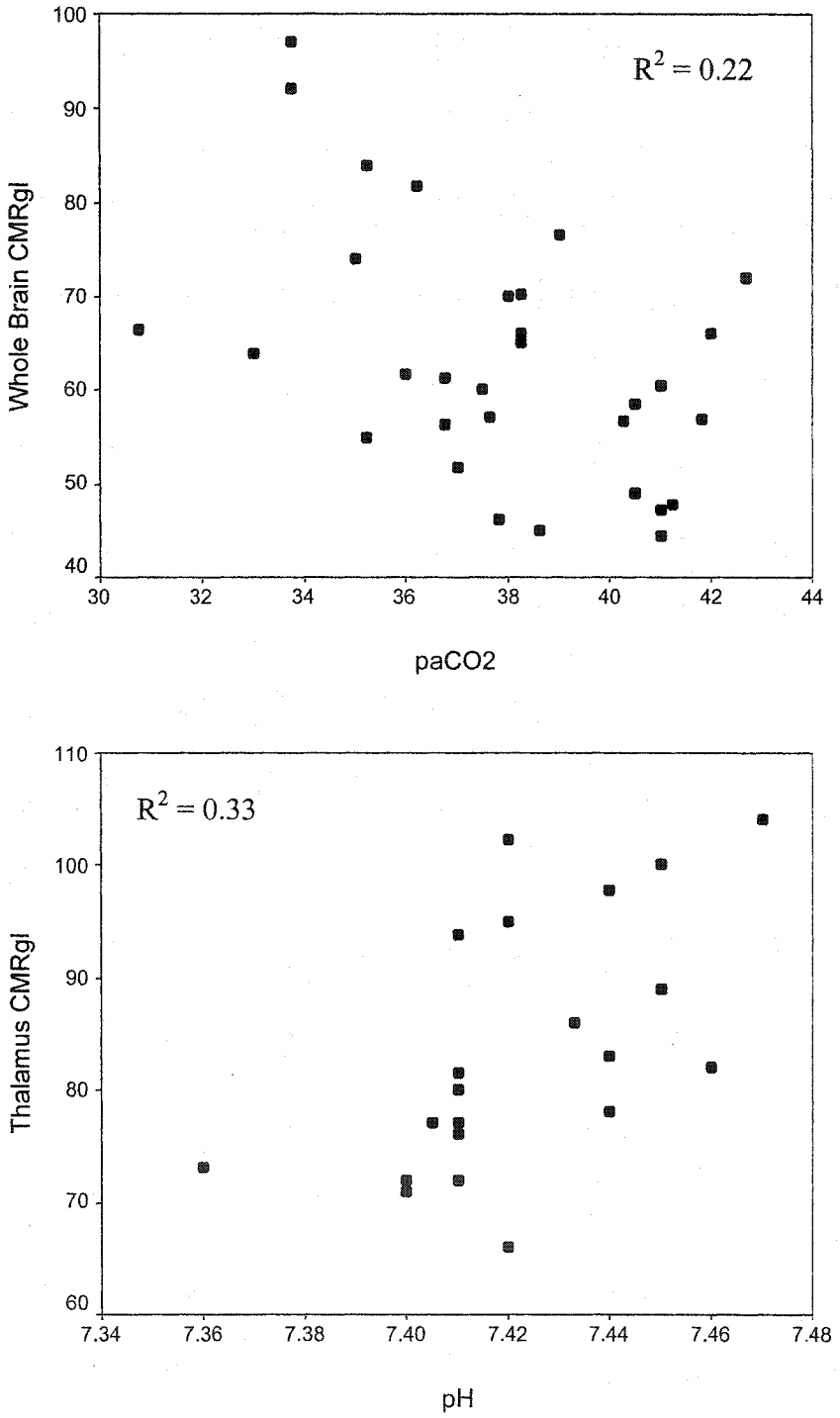


Figure 4.6. Top: bi-variate plot of whole brain CMRgl and carbon dioxide tension in conscious rats. Bottom: bi-variate plot of thalamic CMRgl and pH in conscious rats.

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With regard to estimating species means, however, these correlations are relatively benign. In the first place, variation in circulatory physiology is random, normal, and common to all species. Second, laboratory conditions allow for a high degree of control over physiological variables. Carbon dioxide tension, for example, is monitored very closely and can be manipulated during ventilation to maintain normal values. Finally, species means for CMR and CBF are intrinsically weighted towards the mean values among inter-study circulatory variables. For these reasons, I believe it is unnecessary to statistically adjust population values of CMR/CBF according to circulatory physiology when estimating species means. It is understood that the data are subject to the effects of circulatory physiology, and that this is a source of random error when extrapolating species distributions.

4.4 Body Mass, Fasting, Restraint, and Age

Most of the animal studies provide sample statistics for body mass (in contrast, almost none of the human studies include body mass statistics). The most complete series of data is that for the rat, where mean body mass is normally distributed (skewness = 0.01; kurtosis = -1.57) around a mean of 330 grams with a standard deviation of 60 grams (n = 60, conscious). Of the sixty studies, 58 studies measured males exclusively and only two measured females exclusively. Keeping in mind the unbalanced sample sizes, there is no evidence for sexual dimorphism in this sample ($p = 0.67$; df = 58). In fact, the mean female body mass is slightly larger than the mean male body mass (340g in males; 370g in females). With sexual dimorphism

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ruled out, weighted univariate least-squares regression was performed using CMR/CBF dependent variables and log body mass as the independent variable. Neither in whole brain CMR/CBF nor in any regional CMR/CBF variable did the regression yield a significant result.

The effect of pre-experimental fasting on CMRgl can also be examined in conscious rats. Fasting helps ensure an accurate and stable blood glucose level for deriving metabolic rate, and has become a standard practice in most present-day human PET studies. Fasting periods vary in the rat studies from 12 hours to 4 hours. Analysis of variance indicates that CMRgl values are generally higher in non-fasted rats than in fasted rats (Figure 4.7). The effect appears stronger in the thalamus and white matter than in other regions, although all regions (except for whole-brain measurements) reach significant *p*-values when weighted by sample size. However, this does not necessarily mean that regional metabolic rate is actually higher in non-fasted rats. Eating will elevate plasma glucose concentration, which is a component of the operational equations for all measurement methods. Elevated plasma glucose

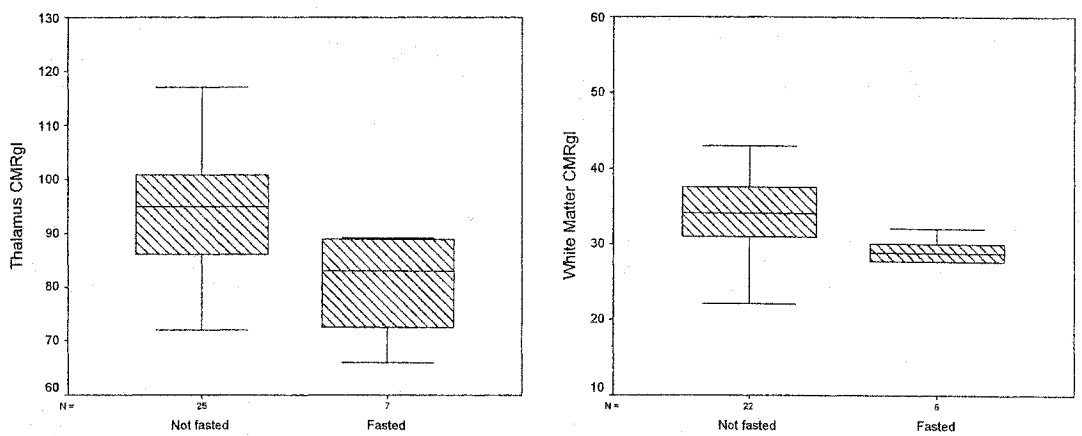


Figure 4.7. The effect of fasting on CMRgl in conscious rats.

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levels would lead to artificially high rCMRgl values, which could explain the data seen here.

The effect of restraint can also be examined in rats. In this case, comparison of mean values of 14 regions from studies using restrained rats versus studies using free-moving rats does not yield any significant results.

Among humans, there are sufficient data to test for a relationship between age and CMR/CBF. The results of correlation analysis and weighted least-squares regression do not provide convincing support for any relationship between age and CMR/CBF in the studies included here. Only one of 21 tests is significant (whole-brain blood flow: $r = -0.51$; $df = 28$; $p = 0.005$), but since we may expect 1 in 20 results to be a false positive, this can probably be dismissed as Type 1 error.

Inter-specific variation

Mean values for rCMRgl in conscious subjects are presented in Appendix Table A15. As described in the previous chapter, there are two measures of species variance: "unweighted standard deviation" measures the variation *among* study populations, while "weighted standard deviation" measures the average standard deviation *within* study populations. In general, these two statistics are consistently similar among regions and among species, indicating strong measurement reproducibility. Coefficients of variation exhibit a broad range of values, typically ranging from 15%-30%, but are also consistent across regions.

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4.5 Allometry

Allometric equations are calculated for the whole brain and each of the six subdivisions by regressing CMRgl values from conscious subjects on three independent variables: log body mass, log brain mass, and log basal metabolic rate (BMR). Each analysis is performed using *n*-weighted least-squares regression of tip data, reduced major axis regression of tip data, and least-squares regression of contrast data. As stated in the previous chapter, branch lengths for independent contrasts were transformed using Pagel's method (Pagel, 1992).

The results of Abouheif's Test For Serial Independence (TFSI) reveal that there is a significant amount of phylogenetic signal in the data (Table 4.12). In this test, phylogenetic independence is indicated by a *p*-value for the average of C-statistics that is close to 0.5. With the exception of the cerebellum, the *p*-values for the average of C-statistics for tip data are all less than 0.05, indicating phylogenetic correlation. When the TFSI is applied to contrast data, results are improved though the phylogenetic signal is not completely alleviated. In particular, the contrasts for thalamic CMRgl do not appear to be adequately controlling for phylogeny. I tested

Table 4.12. Test for Serial Independence, and diagnostic correlation coefficients for absolute values of contrasts and their standard deviations.

	TFSI *p*-value for average C-stat (tips)	TFSI *p*-value for average C-stat (contrasts)	Brain Mass diagnostic *R*	CMRgl diagnostic *R*
Whole Brain	0.008	0.425	0.01*	0.10*
Cortex	0.018	0.157	0.17	0.01
Thalamus	0.030	0.043	0.17	0.26
Hippocampus	0.028	0.143	0.17*	0.01*
Basal Ganglia	0.032	0.128	0.17	0.12
Cerebellum	0.155	0.335	0.13	0.06
White Matter	0.027	0.320	0.29	0.25

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several transformations of both data and branch lengths, but none provided better results.

Correlation coefficients for the absolute values of contrasts and their standard deviations are within acceptable limits (Table 4.12). However, this test reveals an important feature of the data set: in all regions, the contrast for the node separating the primate clade from the rodent/lagomorph clade is significantly higher than any other contrast on the tree. This effect is especially strong for whole brain CMRgl and hippocampus CMRgl (asterisked *R* values in Table 4.12). Figure 4.8 provides an example of this effect in the case of whole brain CMRgl regressed on log brain mass. The 28.99 contrast is an outlier. Nunn and Barton (Nunn and Barton, 2000) note that an outlier contrast typically represents a grade shift between clades. In this case, the data indicate a grade shift between primates and rodents/lagomorphs. The direction of the grade shift indicates higher CMRgl in the rodent/lagomorph clade (see below).

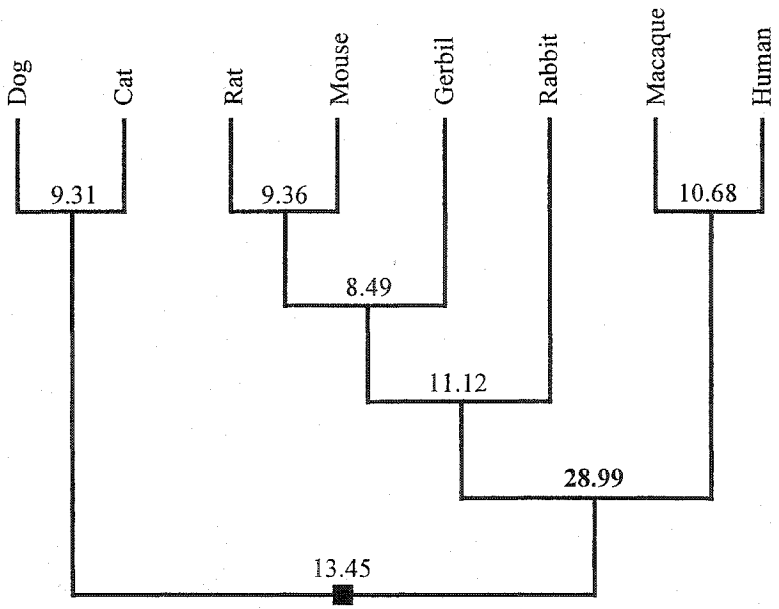


Figure 4.8. Cladogram indicating phylogenetic independent contrasts for whole brain CMRgl. Note that the node separating the primate clade from the rodent/lagomorph clade has the highest contrast.

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The limited number of species restricts the options for dealing with the grade shift statistically. Contrast data already represent n-1 species data, and removing another contrast reduces the sample yet again. Moreover, the only clade that contains more than two species is the rodent/lagomorph clade. This means that the human-macaque "slope" and the cat-dog "slope" are each based on an n of 2, which is not adequate for estimating a regression line. I did conduct regressions both with and without the outlying contrast.

Slopes and their 95% confidence intervals obtained from regressing CMRgl on log body mass, log brain mass, and log basal metabolic rate are presented in Figures 4.9 – 4.11. Brain regions are abbreviated as follows: WB = whole brain, CX = cortex, TH = thalamus, HI = hippocampus, BG = basal ganglia, CB = cerebellum, and WM = white matter. Correlation coefficients are provided below each of the error bars.

The pattern of slopes is consistent for body mass, brain mass, and BMR (mirror imaged), but of the three independent variables log brain mass yields the strongest correlation coefficients. As expected, the n-weighted least-squares regressions yield the highest correlations and tightest confidence intervals. Correlations are lower when reduced major axis regression is used, and lower still for independent contrasts. Of the various regions, the cerebellar data consistently generate the lowest correlations with size/BMR, while the hippocampus and basal ganglia consistently produce high correlations, even in RMA and independent contrast regressions. Note that the slopes for gray matter regions (cortex, thalamus, hippocampus, and basal ganglia) are consistently negative.

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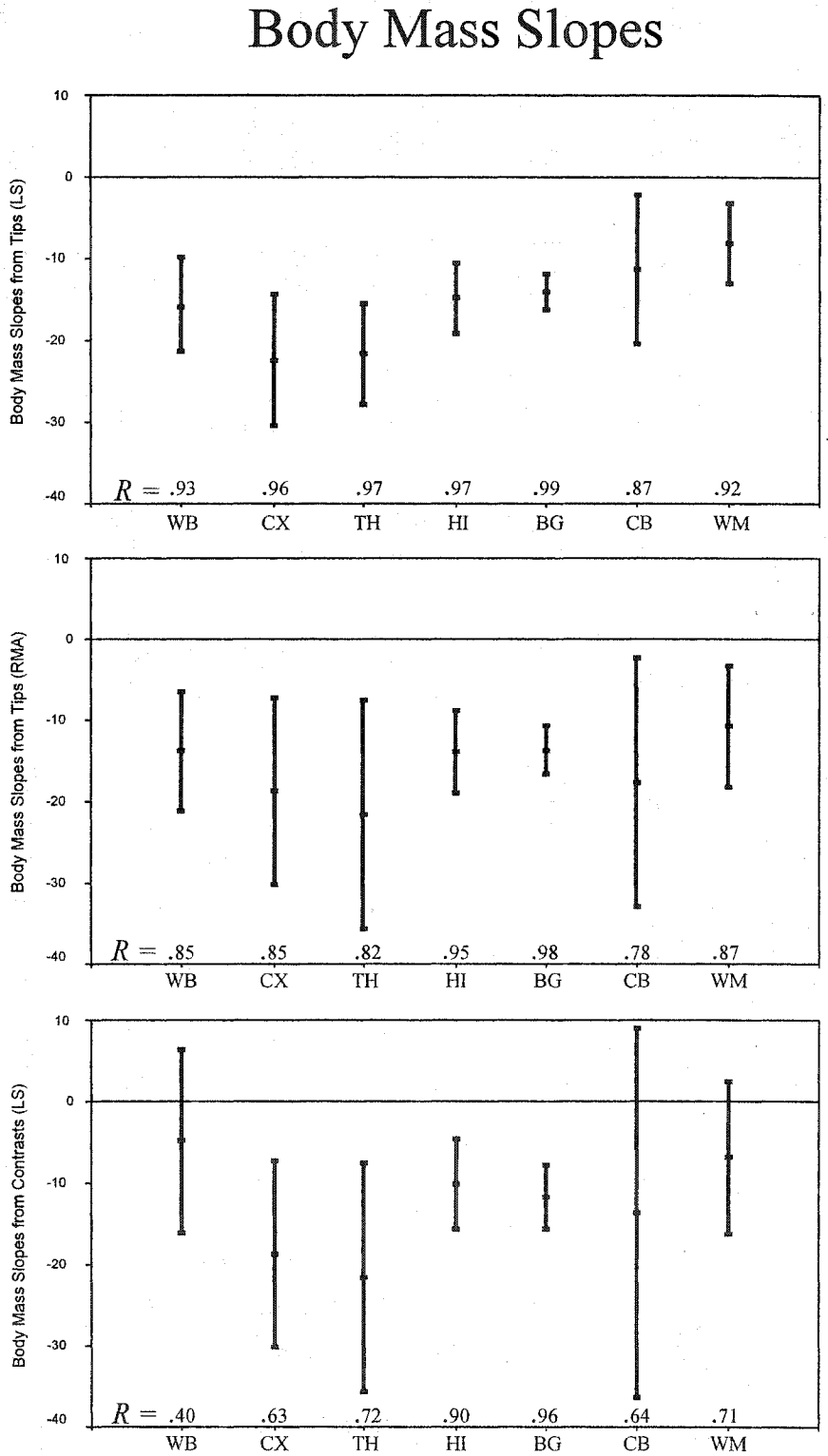


Figure 4.9. Comparison of CMRgl / log body mass slopes (± 95% C.I.) for the whole brain and six subdivisions computed using weighted least-squares regression of tip data (top), reduced major axis regression of tip data (middle), and least-squares regression of contrasts (bottom). Correlation coefficients are provided at the bottom. See text for abbreviations.

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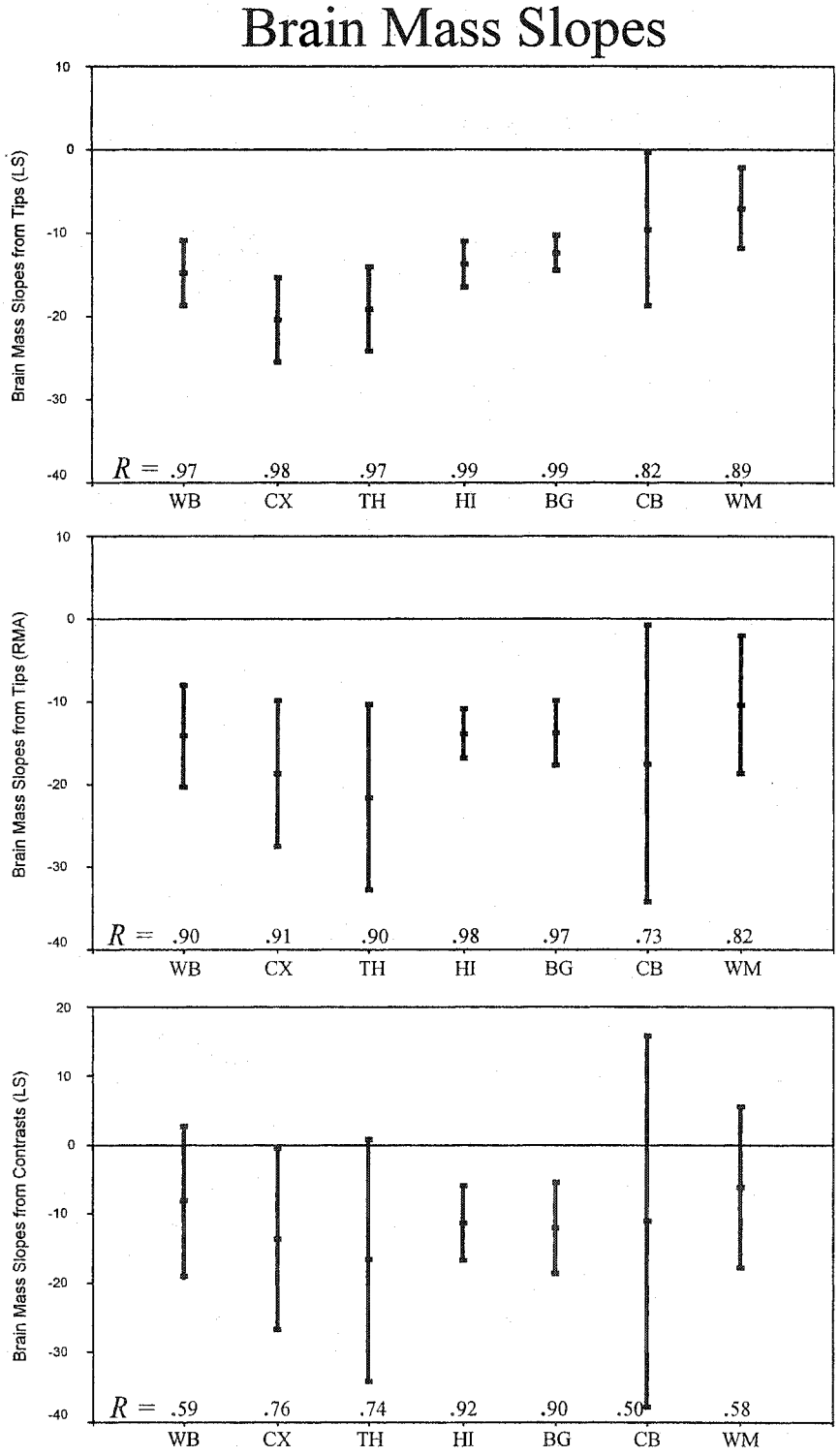


Figure 4.10. Comparison of CMRgl / log brain mass slopes (± 95% C.I.) for the whole brain and six subdivisions computed using weighted least-squares regression of tip data (top), reduced major axis regression of tip data (middle), and least-squares regression of contrasts (bottom). Correlation coefficients are provided at the bottom.

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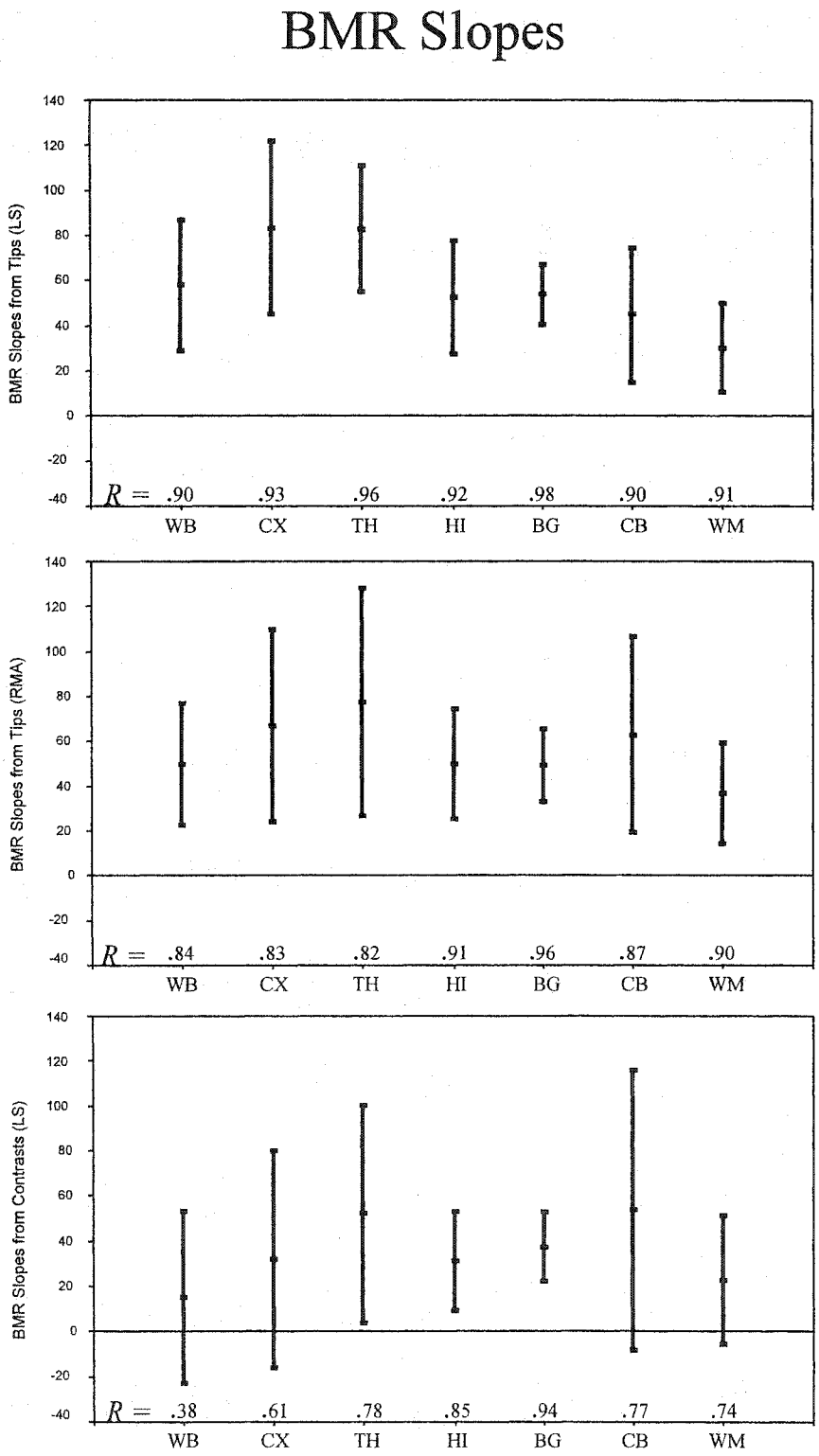


Figure 4.11. Comparison of CMRgl / log BMR slopes (± 95% C.I.) for the whole brain and six subdivisions computed using weighted least-squares regression of tip data (top), reduced major axis regression of tip data (middle), and least-squares regression of contrasts (bottom). Correlation coefficients are provided at the bottom.

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The results indicate that mass specific CMRgl decreases with increasing brain size. However, there are conflicting results as to the homogeneity of slopes. Least-squares regression of tip data suggests that the cortex and thalamus exhibit steeper slopes than other regions. Slopes for the hippocampus and basal ganglia are intermediate. The slope for white matter is consistently shallow.

On the other hand, reduced major axis and independent contrast slopes are more homogeneous. In both cases, the pattern of mean slopes resembles that of *n*-weighted least-squares, but the confidence intervals are much larger. Although the slopes are directionally consistent (negative for body and brain size, positive for BMR), they are not statistically different from each other when analyzed using independent contrasts. Confidence intervals become much larger when RMA and independent contrasts are used, which contributes to the homogeneity of slopes. When the outlier contrast is removed from the independent contrast regression, the slopes are more homogenous and closer to zero. However, it is interesting to note that the hippocampus and basal ganglia retain relatively tight confidence intervals even from independent contrasts regression.

Analyses of the intercepts from the allometric regressions indicate that regional CMRgl varies in magnitude across species. Figure 4.12 illustrates the regression intercepts and their confidence intervals for each of the three line-fitting methods. These results indicate that the cortex and thalamus metabolize at higher rates than the brain as a whole. The hippocampus and basal ganglia again exhibit intermediate intercepts, while white matter displays the lowest intercept. This pattern

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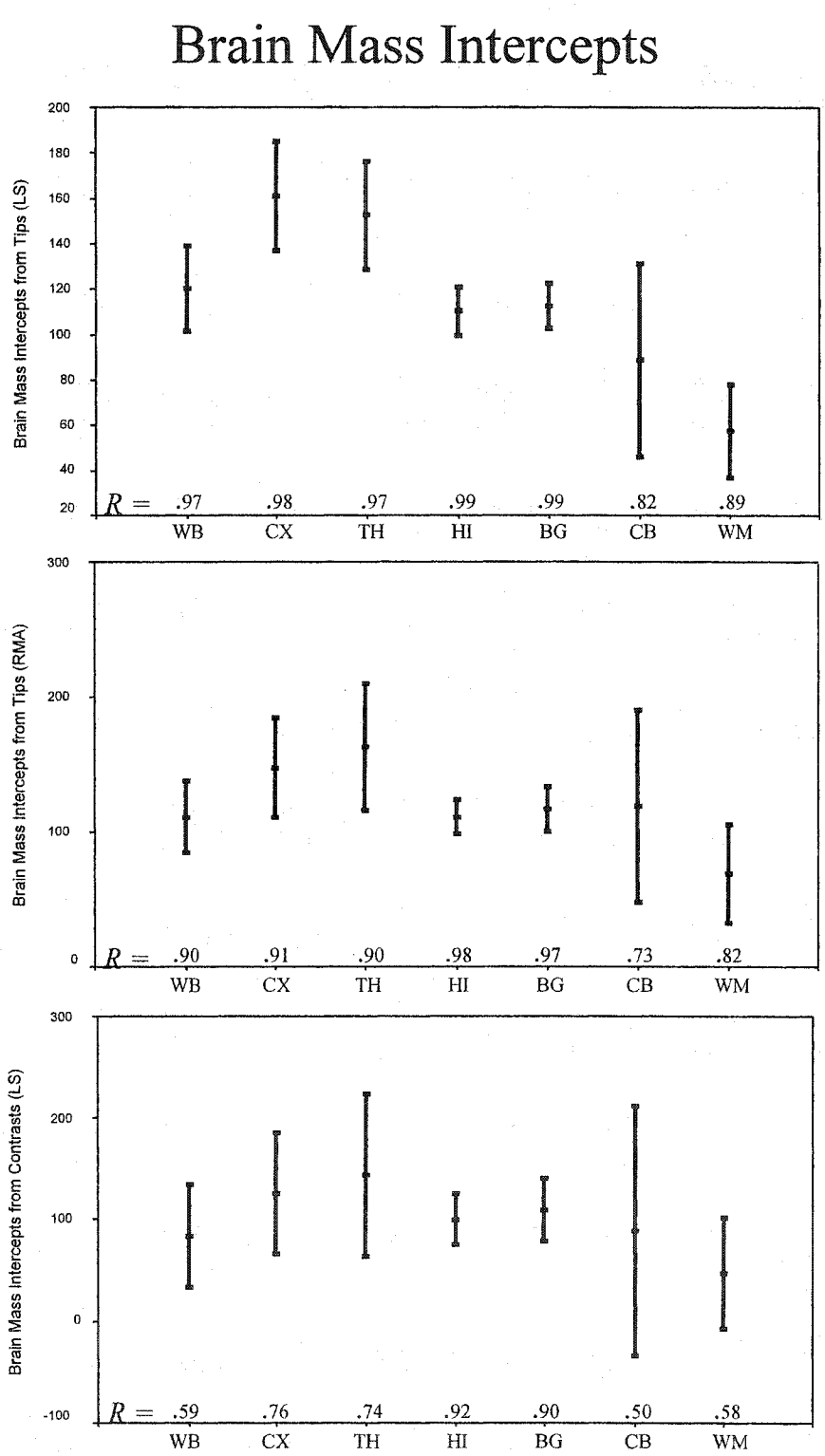


Figure 4.12. Comparison of CMRgl / log brain mass intercepts (± 95% C.I.) for the whole brain and six subdivisions computed using weighted least-squares regression of tip data (top), reduced major axis regression of tip data (middle), and least-squares regression of contrasts (bottom). Correlation coefficients are provided at the bottom.

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is consistent for each of the three regression methods, though the differences are less pronounced with reduced major axis and independent contrast regressions than with *n*-weighted least squares of tip data.

The comparison of regression lines is examined in greater detail in Figures 4.13a-g. In these graphs, CMRgl is plotted against log brain mass separately for each region and for the brain as a whole. Regression lines are those computed from independent contrasts mapped back into raw data space. Although I show both conscious and anesthetized individuals, the regressions are calculated for conscious individuals only. For each subdivision, the regression line describing conscious whole brain metabolism is provided as a reference.

Perhaps the most important feature of these graphs is that they illustrate the probable effects of partial-volume averaging. The human metabolic data, all collected with PET, are much more homogeneous than in other species. This would be expected if partial-volume averaging is occurring. If we assume that, due to partial-volume averaging, the metabolic rates for the cortex and thalamus in humans are actually underestimates, than the slopes for these regions are probably not different from the whole brain slope. In the same way, if we assume that white matter rates in humans are being overestimated because of partial-volume averaging, then the white matter slope is again probably no different than the whole brain slope. Partial-volume effects may therefore distort the slopes, indicating heterogeneity when in fact the slopes are homogenous.

At the same time, the partial-volume effects actually corroborate the magnitude differences previously indicated by the intercept values. If we assume

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homogenous slopes then the human values for cortical and thalamic CMRgl appear too low, and the value for white matter appears too high. In this case, the pattern of magnitude differences among regions remains the same: the cortex and thalamus metabolize at high rates, the hippocampus and basal ganglia metabolize at intermediate rates, and white matter metabolizes at a relatively low rate.

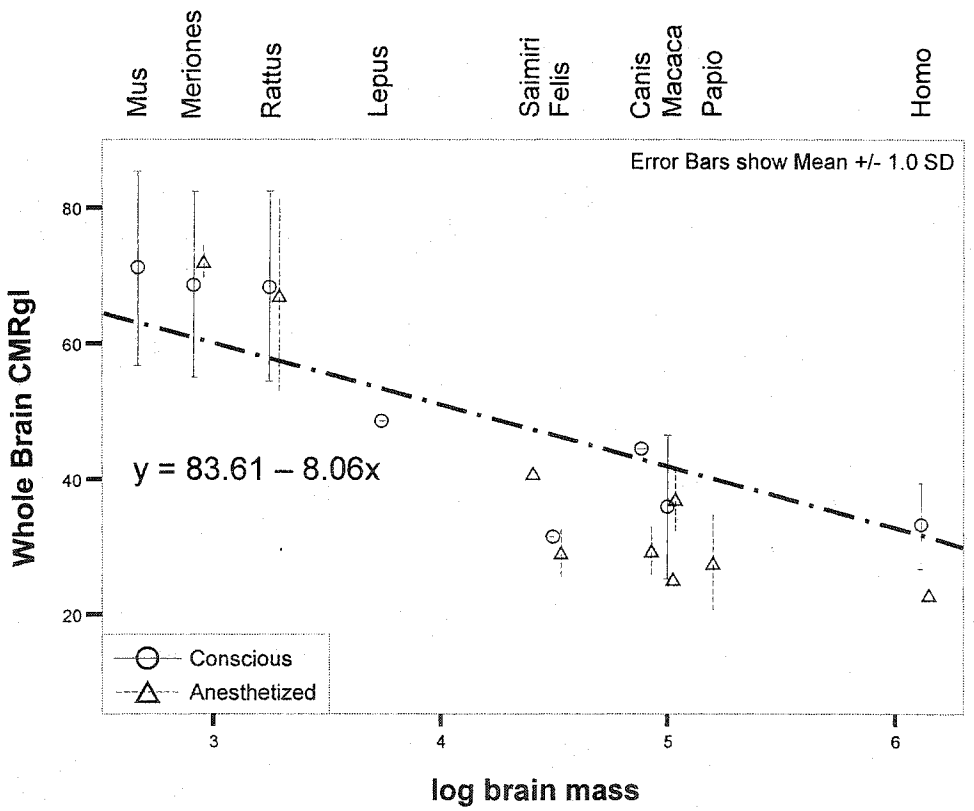


Figure 4.13a. CMRgl plotted against log brain mass. Best fit lines represent independent contrasts regression for conscious individuals only. The line for whole brain CMRgl — · – is provided as a reference for other regions ——.

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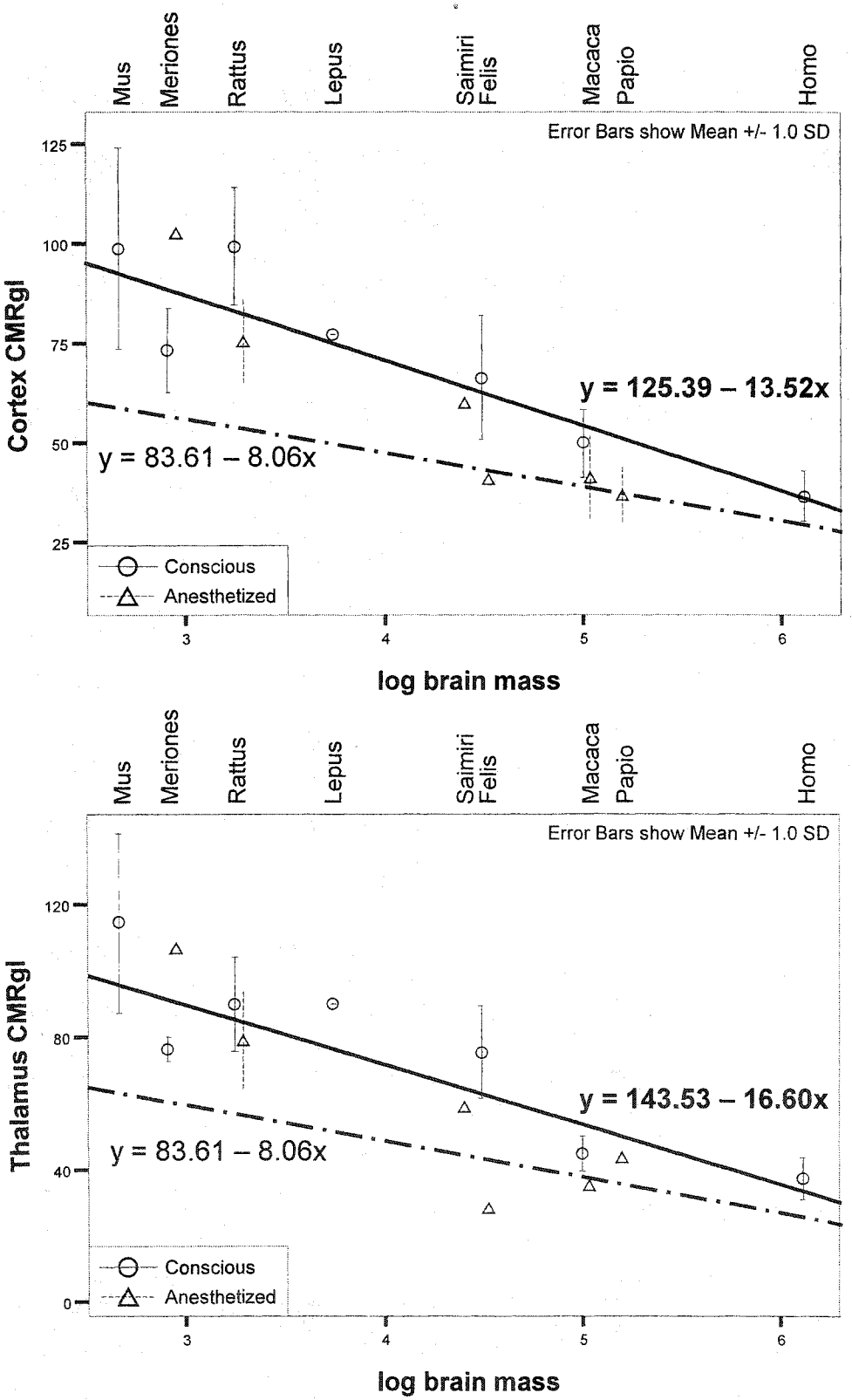


Figure 4.13b,c. CMRgl plotted against log brain mass. Best fit lines represent independent contrasts regression for conscious individuals only. The line for whole brain CMRgl — · – is provided as a reference for other regions ——.

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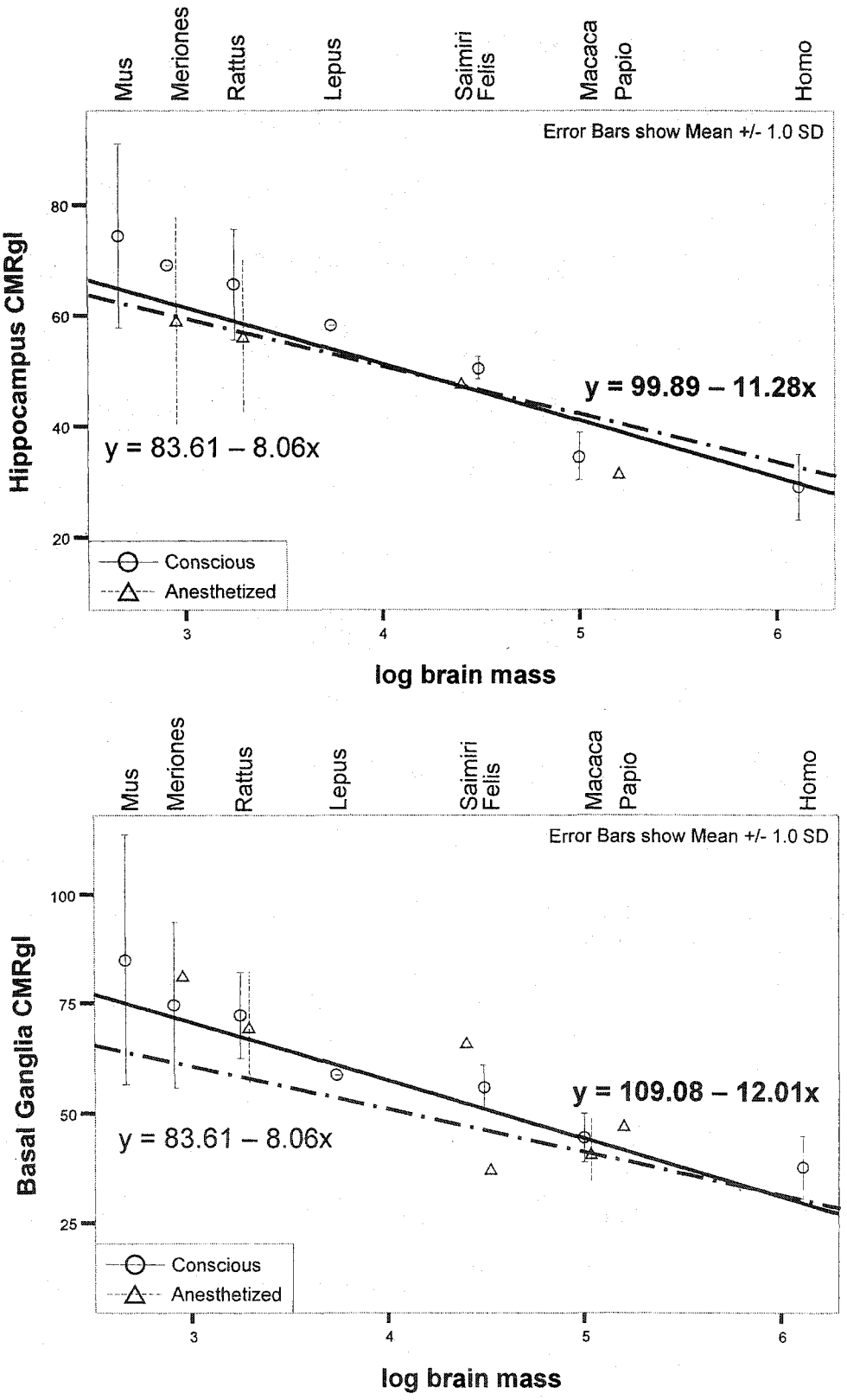


Figure 4.13d,e. CMRgl plotted against log brain mass. Best fit lines represent independent contrasts regression for conscious individuals only. The line for whole brain CMRgl — · – is provided as a reference for other regions ——.

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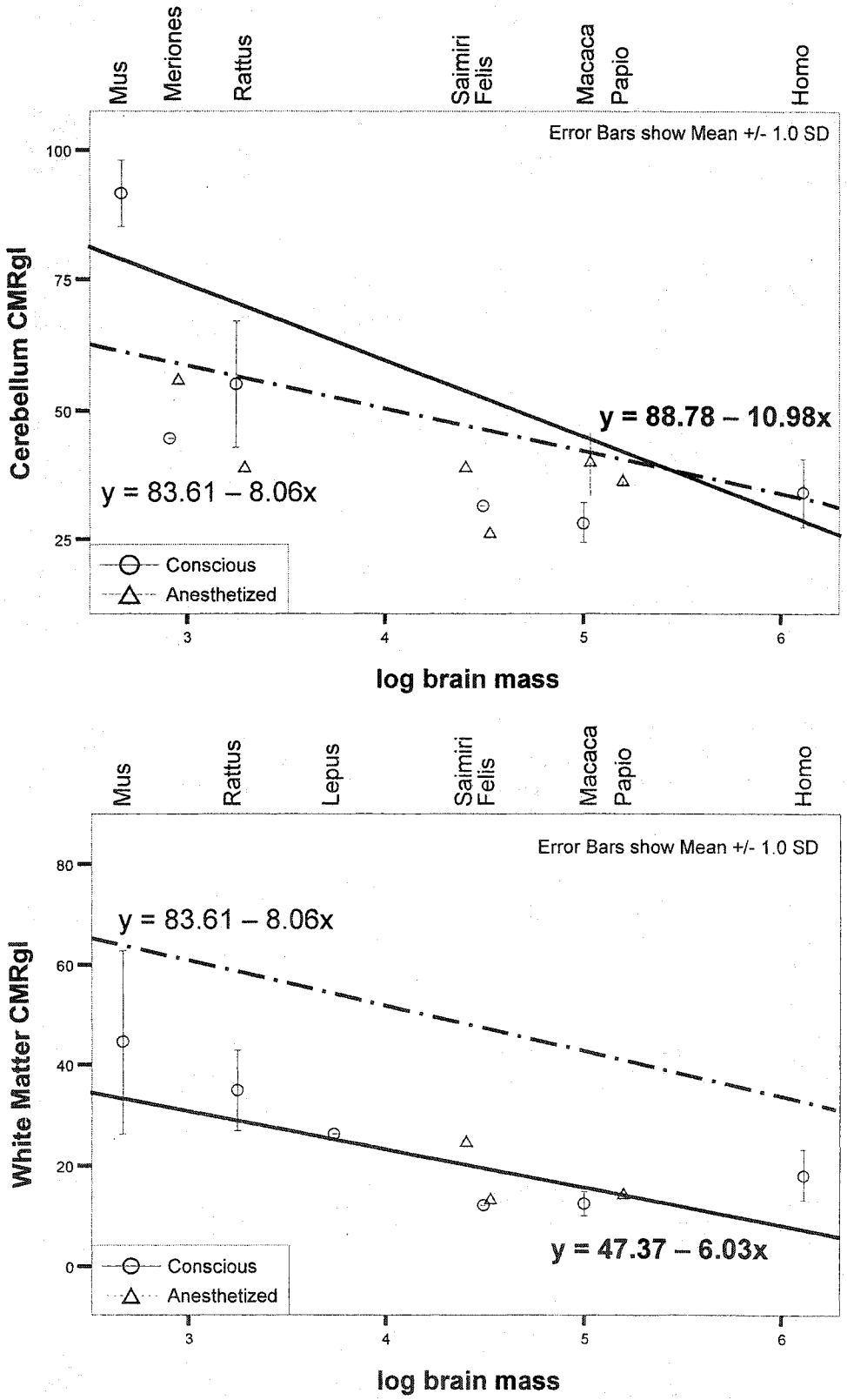


Figure 4.13f,g. CMRgl plotted against log brain mass. Best fit lines represent independent contrasts regression for conscious individuals only. The line for whole brain CMRgl — · – is provided as a reference for other regions ——.

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The slope for the cerebellum is strongly influenced by the mouse value, which appears to be an outlier. When the mouse data are removed from the regression the slope is indistinguishable from the whole brain reference slope.

Regressions of whole brain CMRO2 and CBF on log brain volume show the same negative allometric relationship (Figures 4.14 & 4.15). Whole brain blood flow is particularly interesting since this test allows for the greatest number of species, including animals that are larger in body mass than humans, such as *Equus*. The "middle" of the regression is fairly odd, with *Lepus*, *Felis,* and *Canis* displaying relatively low blood flow values. In *Rattus* and *Lepus*, the anesthetized values are actually higher than the conscious values. Blood flow appears to be a more complex factor than glucose metabolic rate.

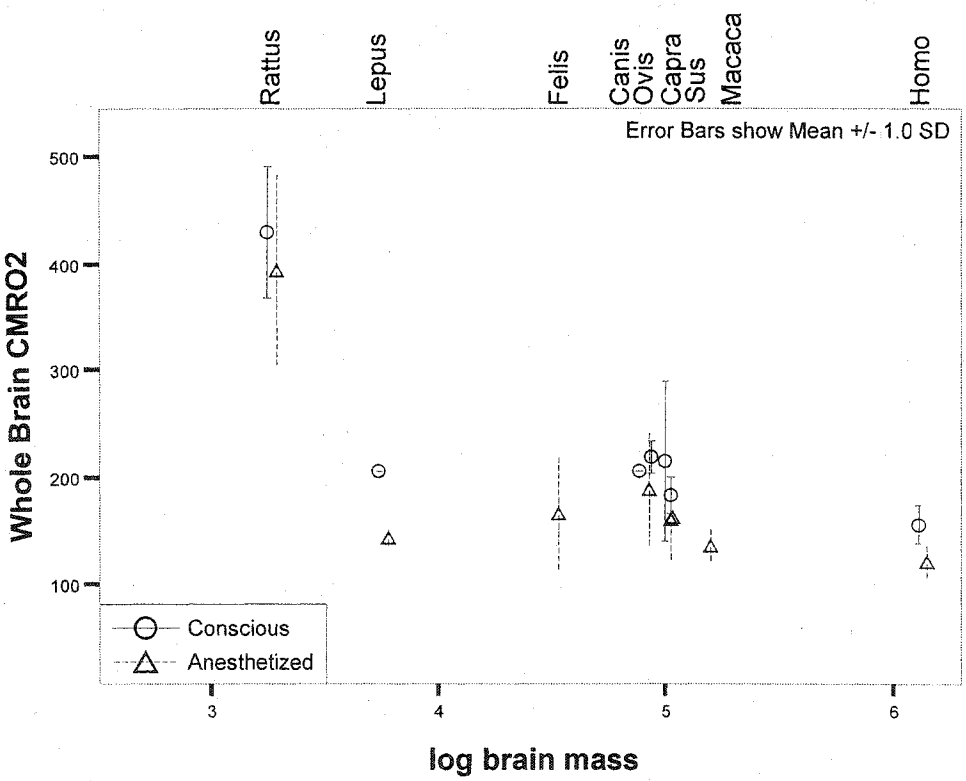


Figure 4.14. Whole Brain CMRO2 (umol/100g/min) regressed on log brain mass.

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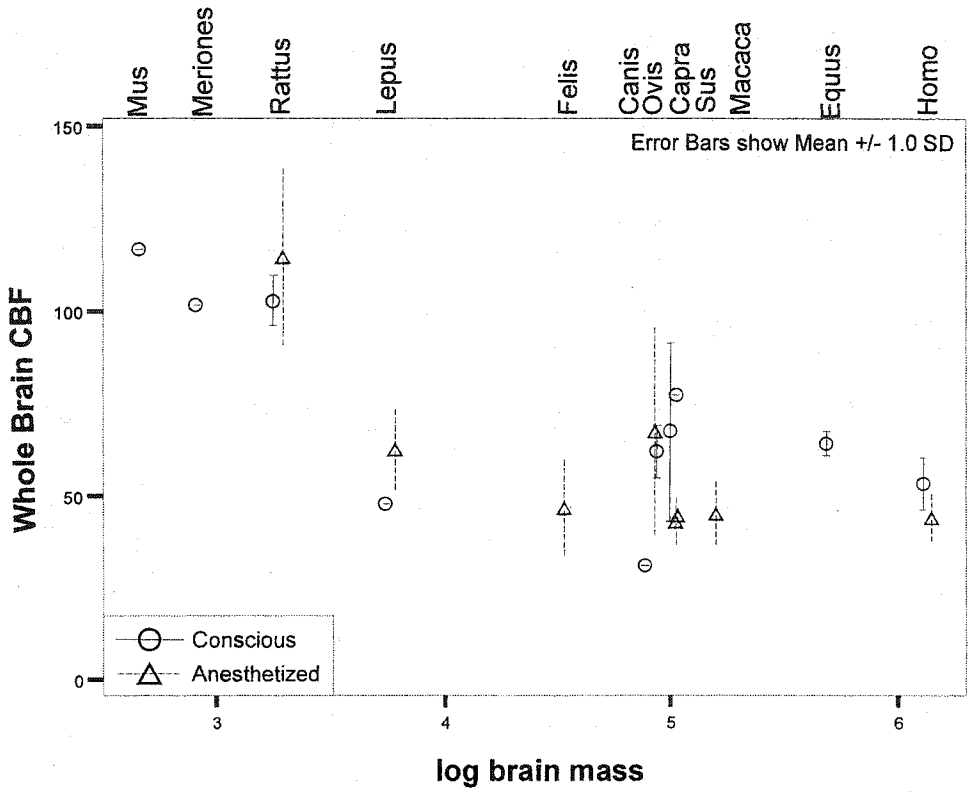


Figure 4.15. Whole Brain CBF (ml/100g/min) regressed on log brain mass.

When CMRgl values are log-transformed, conventional log-log allometry once again suggests homogeneity of slopes with "grade shifts" between regions. Figure 4.16 illustrates the slopes and their 95% confidence intervals for log CMRgl regressed on log brain mass. All three line-fitting methods yield consistent negative scaling, though the independent contrast confidence intervals for whole brain, cerebellum, and white matter overlap zero. The mean slopes range from –0.08 to –0.14, with an average of –0.12.

Figure 4.17 illustrates the intercepts from the same regressions (log CMRgl on log brain mass). The pattern is the same as that for the untransformed metabolic data: cortex and thalamus metabolize at high rates across all species, hippocampus and basal ganglia exhibit intermediate rates, and cerebellum and white matter (though highly variable) metabolize at low rates.

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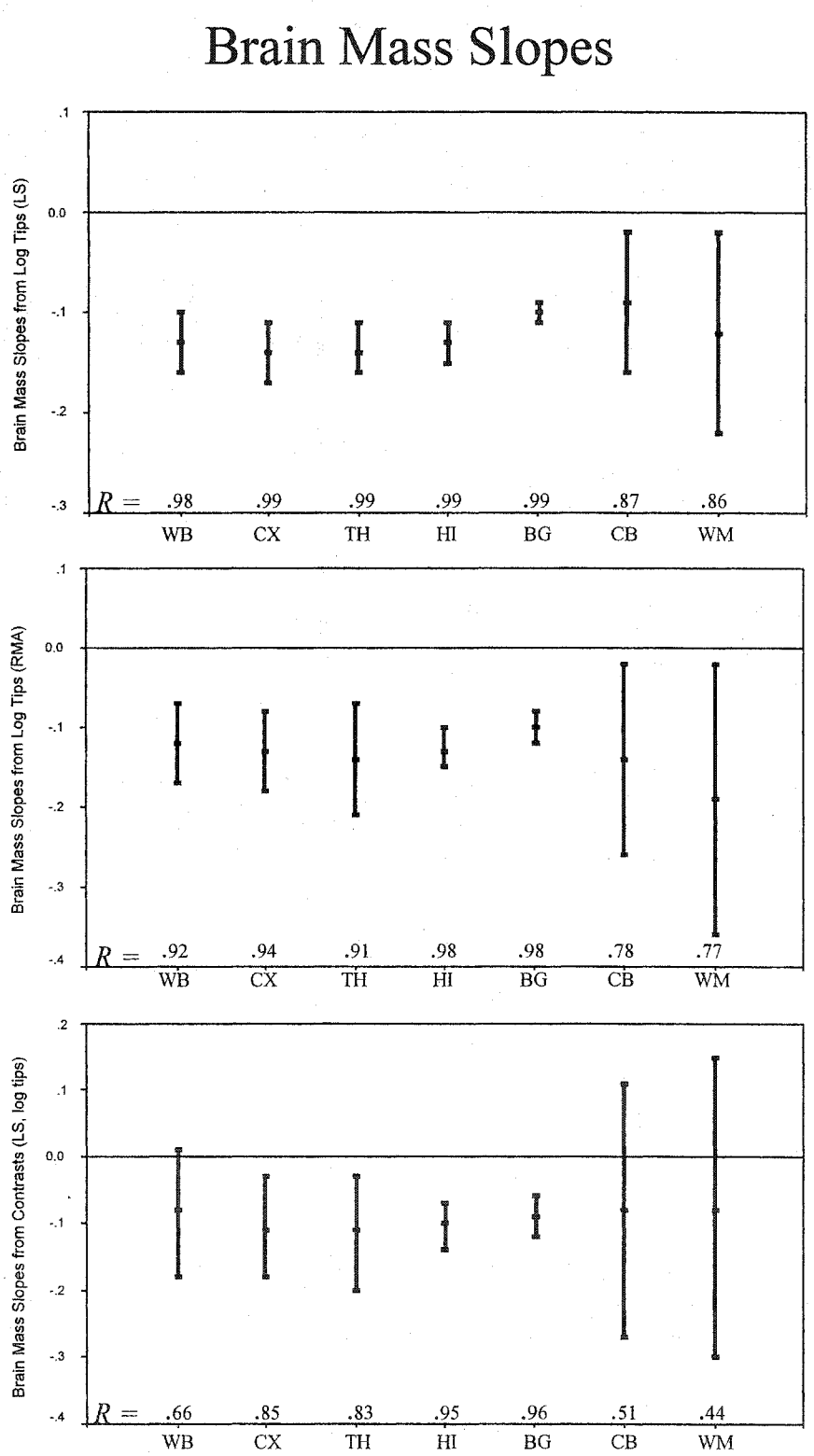


Figure 4.16. Comparison of log CMRgl / log brain mass slopes (± 95% C.I.) for the whole brain and six subdivisions computed using weighted least-squares regression of tip data (top), reduced major axis regression of tip data (middle), and least-squares regression of contrasts (bottom). Correlation coefficients are provided at the bottom.

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Brain Mass Intercepts

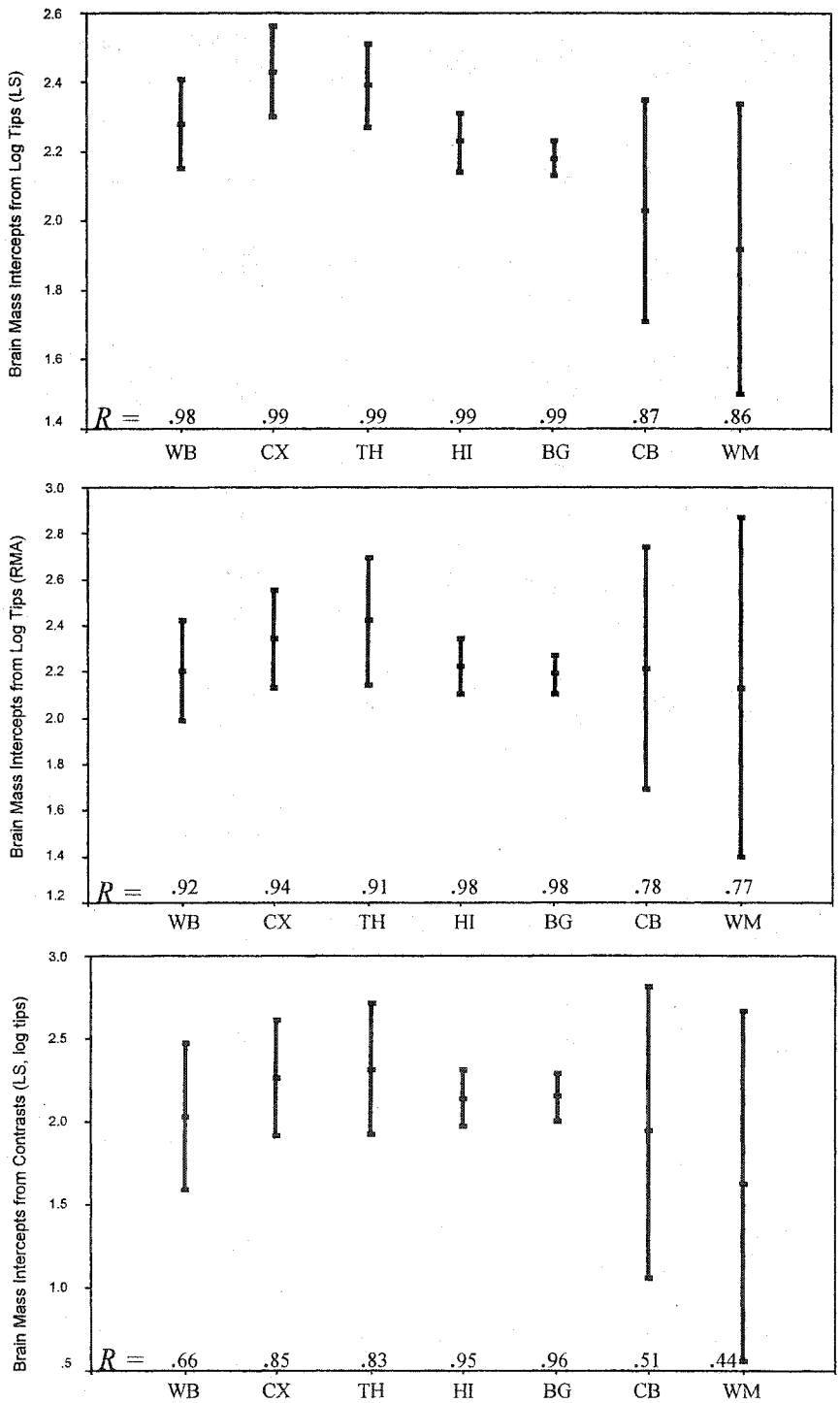


Figure 4.17. Comparison of log CMRgl / log brain mass intercepts (± 95% C.I.) for the whole brain and six subdivisions computed using weighted least-squares regression of tip data (top), reduced major axis regression of tip data (middle), and least-squares regression of contrasts (bottom). Correlation coefficients are provided at the bottom.

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Analysis of residual values for *Homo* depends on whether the best-fit lines are computed with or without humans in the sample. When the best-fit lines are computed with humans in the sample, the residuals are not significant – humans fall within the confidence intervals for the regression line in all regions. When the best-fit lines are computed without humans in the sample, humans fall within the prediction intervals only for whole brain metabolism. In every other region for which the regression model is statistically significant, the human values for regional metabolism fall well above the best-fit line.

Several factors must be considered when interpreting residuals from regression equations in which humans are removed from the sample. First, many of these equations actually predict negative values for regional metabolism in a brain as large as that of humans. In other words, the prediction line crosses into negative territory when humans are not included in the sample, and negative metabolic rates are obviously not possible. A second, and related, issue is that since the human brain is the largest in the sample, by far, the confidence intervals for the prediction equations are extremely large when the best-fit line is extended beyond the regression sample to the level of human brain size. If it were possible to include in this sample brains larger than that of humans, it may then make sense to compute prediction equations without humans in the sample. But this is not the case.

Since the large human brain has high leverage when incorporated in the regression, it will have a larger effect on the position of the best-fit line than the brains of other animals in the sample. The fact that human residuals are not markedly

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large when the regression includes humans is therefore a fairly conservative result. Given the limitations of the sample, residuals derived from regressions in which humans are intentionally removed are highly suspect.

In sum, the results of this allometric analysis reveal that 1) there is consistent negative scaling of mass-specific metabolic rate with brain size and body size; 2) the metabolic slopes are likely homogeneous among regions, averaging –0.12 in log-log space; 3) there is a grade shift among regions such that the highest metabolic rates occur in the cortex and thalamus, and the lowest rates occur in the white matter, and; 4) human values for mass-specific brain metabolism are consistent with the inter-specific models and essentially "fall on the line."

4.6 Variation Within the Cortex

The most complete series for metabolic rates from individual regions of the cortex are available for humans, macaques, and rats. For comparative purposes, metabolic rates from individual cortical regions are normalized by the average total cortical metabolic rate for the species. Figure 4.18 illustrates the resulting metabolic ratios for seven cortical regions in the three species. It is immediately clear that auditory cortex metabolizes at a high rate in all three species. The magnitude of the difference is particularly extreme in macaques and rats, but even in humans (where partial-volume averaging is probably occurring) auditory cortex has the highest metabolic ratio.

The rat pattern is interesting in that, aside from auditory cortex, the ratios of the other six cortical areas are very similar. This is in contrast to the macaque where

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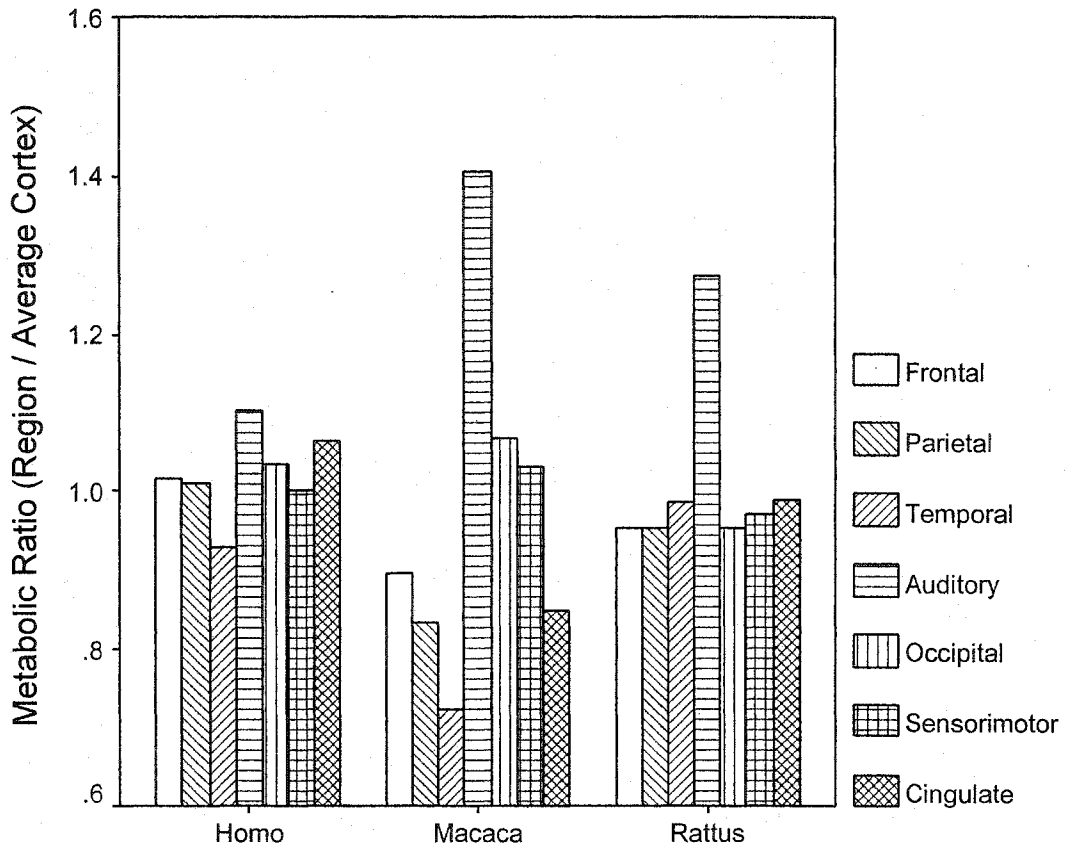


Figure 4.18. Comparison of cortical metabolism in humans, macaques, and rats. Bars represent the ratio of regional CMRgl to average cortical CMRgl.

cortical metabolism appears quite heterogeneous. In the macaque, occipital and sensorimotor areas retain relatively large metabolic ratios while frontal, parietal, temporal, and cingulate ratios are markedly less. Relative to the entire cortex, temporal metabolism is low both in macaques and humans. However, partial-volume effects limit the interpretation of the human data. The fact that cingulate metabolism appears high in humans could be the result of a high degree of gyrifcation in a region that is directly adjacent to its contralateral counterpart.

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4.7 Absolute Regional Metabolism

All of the results discussed thus far have been concerned with mass-specific metabolism: moles of glucose per unit volume of tissue per unit time. Deriving absolute regional metabolic rate (the amount of glucose taken up by an entire brain subdivision per unit time) requires knowledge of the volumes of brain subdivisions. The results of volumetric estimates from serial sections are presented in Table 4.13. These volume estimates are multiplied by the mass-specific metabolic rate of each corresponding region to obtain absolute regional metabolic rates.

Table 4.13. Volume of brain subdivisions (ml).

Genus	Body mass (g)	Brain vol.	Neocortex vol.	Hippocampus vol.	Thalamus vol.
Homo	65000	1335328	669286	7694	14841
Papio	14000	151000	67200	2520	3810
Macaca	8200	103484	45675	1383	2038
Lepus	2750	5700	2700	479	181
Felis	4500	31800	10500	920	869
Rattus	298	1837	637	110	81
Meriones	65	853	258	52	40
Mus	24	480	147	29	21

Genus	Body mass (g)	Brain vol.	Basal Ganglia vol.	Cerebellum vol.	White Matter vol.
Homo	65000	1335328	25353	142657	448282
Papio	14000	151000	5820	16500	38600
Macaca	8200	103484	4048	11219	26450
Lepus	2750	5700	214	618	600
Felis	4500	31800	507	3600	5180
Rattus	298	1837	101	196	86
Meriones	65	853	55	88	32
Mus	24	480	31	46	18

I have noted that partial-volume averaging may distort human CMRgl values, particularly in the cortex, thalamus, and white matter. To make sure that this error is not compounded when computing absolute metabolic rates, I ran all of the following tests both on unadjusted values and on values in which human cortical and thalamic CMRgl were increased by 30% and white matter decreased by 30%. A 30% effect

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size for partial-volume averaging is a very liberal estimate (see Discussion). However, in the case of this analysis, variation in regional volume far exceeds variation in regional CMRgl, and the results using partial-volume-adjusted values are not different from those obtained using unadjusted values. I present here the results obtained using unadjusted values.

Log-log regressions of regional metabolism on brain volume are illustrated in Figure 4.19, along with the corresponding regression coefficients and their confidence intervals in Table 4.14. These results confirm that whole brain metabolism scales negatively with brain volume (slope less than unity), but because of differential brain expansion, scaling relationships vary among regions. The scaling coefficient of the cerebral cortex is not significantly different from unity, indicating that cortical metabolism scales proportionally with brain size. The slope for the cerebellum is also not significantly different from unity. On the other hand, sub-cortical gray regions exhibit slopes well below unity, indicating negative allometry among these structures. White matter exhibits a slope significantly greater than unity, indicating positive allometry.

In relation to humans, there are two interesting patterns evident from the relationships in Figure 4.19. First, the hippocampus in particular makes up less of the brain's total energy consumption in macaques and humans than in other mammals. Second, the positive scaling of white matter actually reaches a point in humans such that, except for the cortex, the white matter requires more energy than any other region.

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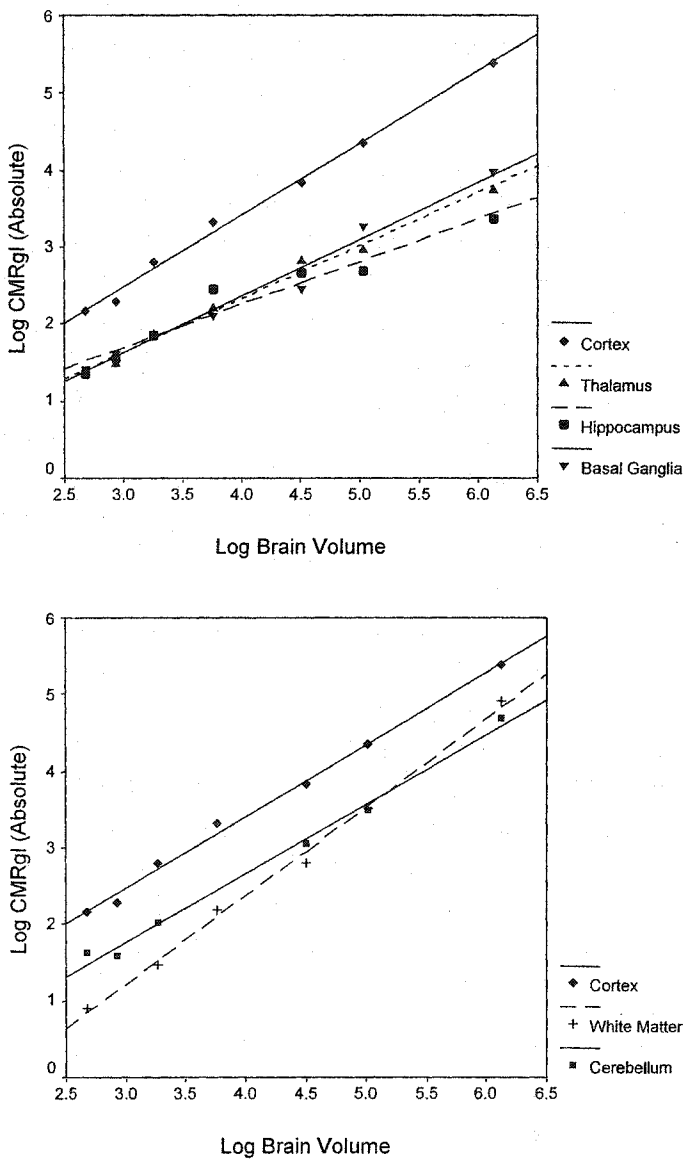


Figure 4.19. Log absolute metabolic rate regressed on log brain mass in conscious subjects.

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Table 4.14. Log absolute CMRgl regressed on log brain mass in conscious subjects.

	R	Intercept	95% C.I.		Slope	95% C.I.	
			Lower	Upper		Lower	Upper
Whole Brain CMRgl							
Least-squares	0.99	0.16	0.00	0.32	0.89	0.85	0.93
Reduced Major Axis	0.99	0.16	0.00	0.32	0.89	0.86	0.93
Independent Contrasts	0.99	0.15	-0.12	0.41	0.90	0.84	0.95
Cortex CMRgl							
Least-squares	0.99	-0.34	-0.68	-0.01	0.94	0.86	1.02
Reduced Major Axis	0.99	-0.35	-0.69	-0.02	0.94	0.86	1.02
Independent Contrasts	0.99	-0.34	-0.83	0.15	0.94	0.83	1.04
Thalamus CMRgl							
Least-squares	0.99	-0.45	-0.78	-0.12	0.69	0.61	0.77
Reduced Major Axis	0.99	-0.46	-0.79	-0.13	0.70	0.62	0.77
Independent Contrasts	0.99	-0.40	-0.80	0.01	0.69	0.60	0.77
Hippocampus CMRgl							
Least-squares	0.97	0.02	-0.68	0.72	0.56	0.39	0.72
Reduced Major Axis	0.94	-0.05	-0.75	0.64	0.58	0.41	0.75
Independent Contrasts	0.96	0.11	-0.76	0.98	0.55	0.36	0.73
Basal Ganglia CMRgl							
Least-squares	0.99	-0.61	-1.14	-0.07	0.74	0.61	0.87
Reduced Major Axis	0.98	-0.64	-1.17	-0.11	0.75	0.62	0.87
Independent Contrasts	0.99	-0.71	-1.25	-0.16	0.75	0.63	0.87
Cerebellum CMRgl							
Least-squares	0.99	-0.94	-1.40	-0.48	0.90	0.79	1.01
Reduced Major Axis	0.99	-0.95	-1.41	-0.49	0.91	0.80	1.01
Independent Contrasts	0.99	-0.97	-1.69	-0.26	0.91	0.75	1.06
White Matter CMRgl							
Least-squares	0.99	-2.25	-2.73	-1.77	1.16	1.05	1.27
Reduced Major Axis	0.99	-2.26	-2.74	-1.78	1.16	1.05	1.27
Independent Contrasts	0.99	-2.27	-2.79	-1.76	1.16	1.04	1.27

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The pattern is further illustrated in Figure 4.20, where percentage of total brain metabolism is graphed for each region in different species. In humans, macaques, cats, and rabbits, approximately 50-60% of total brain metabolism takes place in the cortex. Among gerbils, rats, and mice, the figure is approximately 35-45%. In humans and macaques, the subcortical gray structures account for far less of the brain's energy use than in other animals, while the cerebellum remains relatively stable in its energy demands. Finally, the increasing volume of white matter reaches a peak in humans: nearly 20% of total brain metabolism in humans occurs in the white matter, more than double the amount of the other mammals in the sample.

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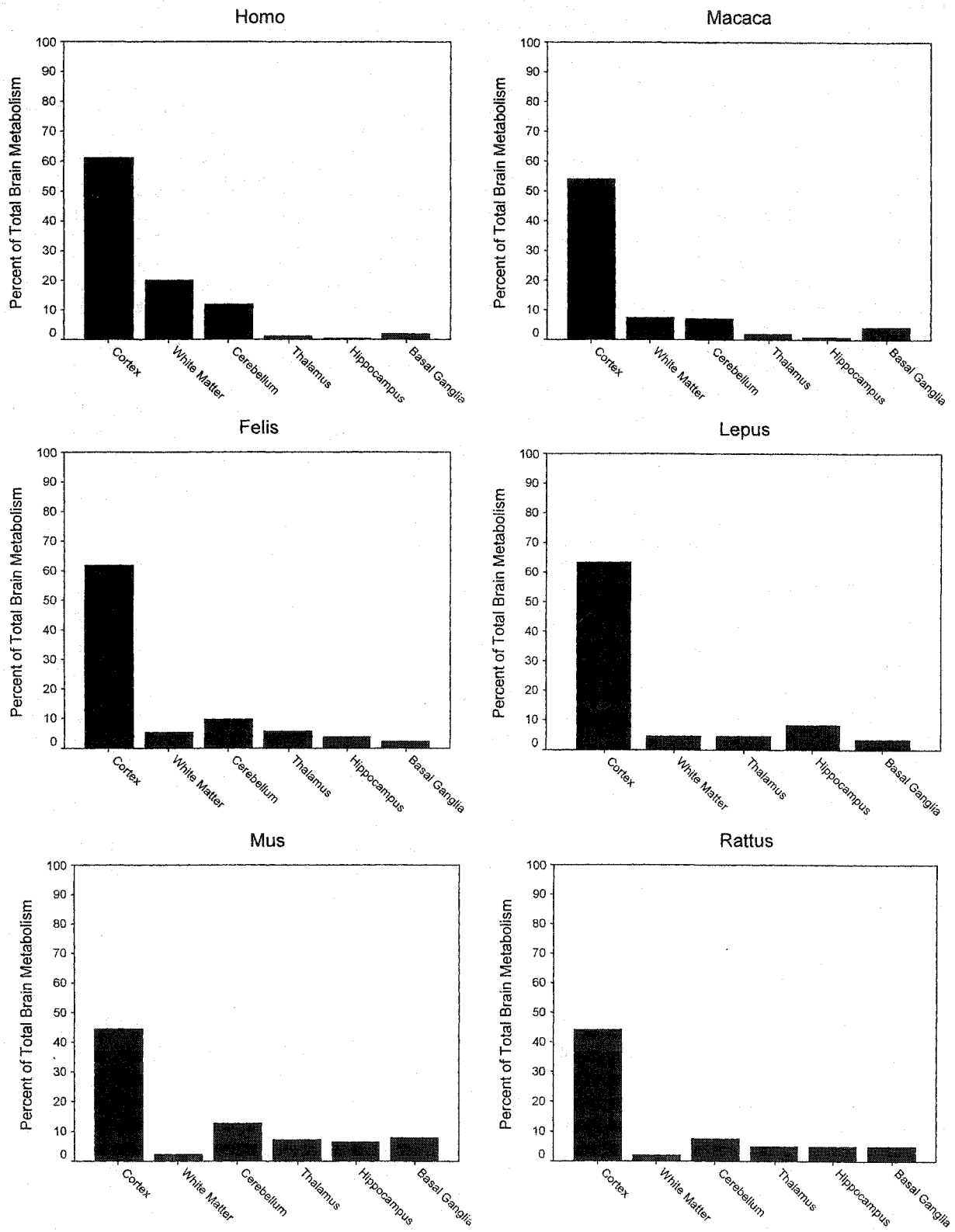


Figure 4.20. Distribution of brain glucose uptake in six mammalian species. Absolute regional metabolism is expressed as a percent of total brain metabolism.

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5. Discussion

In discussing the results of this study, it will be helpful to return to the primary research questions outlined in the beginning of this thesis. One set of questions center on intra-specific variation: a) how much does regional metabolic rate vary among individuals in a single species, and b) what are the biological and technical factors that influence measurements of regional brain metabolism? The second set of questions focus on inter-specific relationships: a) how does regional metabolic rate change as brains become larger, and b) is the pattern of regional metabolism consistent across species? I will deal with each of these questions in turn.

5.1 Intra-Specific Variation

The principal technical factor effecting CMRgl measurement is anesthesia. Volatile anesthetics (such as halothane and sevoflurane), dissociative drugs (such as phencyclidine and ketamine), and barbiturates (such as pentobarbital) cause marked depression of metabolic rate, on the order of 35%. Conventional wisdom in neuroimaging has held that nitrous-oxide sedation has minimal effects on brain metabolism (Crosby et al., 1984; Ingvar et al., 1980; Ingvar and Siesjo, 1982). The results presented here show that nitrous-oxide has a smaller effect size than that of other anesthetics but one that is significant nonetheless. Across species, whole brain metabolism decreased by an average of 15% under nitrous oxide sedation, while cortical metabolism was depressed between 20% and 40%. The odd exception is the gerbil, *Meriones,* where nitrous-oxide actually increased cortical and thalamic metabolism by almost 30%. However, this result is based only on two studies from

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the same laboratory (Choki et al., 1983, 1984) and needs to be confirmed independently.

The second technical factor influencing CMRgl measurement, especially in the cortex, is partial-volume averaging in PET. Compared with values derived from Sokoloff autoradiography, cortical metabolism is approximately 30% lower in macaques and rats scanned with PET. This result is larger than previous estimates, which range from 11% to 25% (Meltzer et al., 1999, 1990, 1996). The overall effect of partial-volume averaging is to deflate gray matter values and inflate white matter values. I found the effect to be particularly strong in the cortex and thalamus. Other sub-cortical gray regions are appear less vulnerable to this source of error. The error associated with partial-volume averaging is large enough that it appears to distort some interspecific allometric comparisons (see below).

Beyond these technical factors, blood chemistry appears to account for most of the remaining intra-specific variance. In particular, blood pH and carbon-dioxide tension have a significant effect on rCMRgl. In conscious rats, blood pH accounted for 33% of the variation in thalamic CMRgl, and CO_2 tension accounted for 22% of the variation in cortical metabolism. Other biological factors have negligible effects in this dataset. I found no relationship between age, sex, body mass, or brain mass and rCMRgl within species. This is not to say that these variables do not effect cerebral metabolism. Age has been repeatedly implicated in CMRgl variation (Burns and Tyrrell, 1992; Duara et al., 1983; Leenders et al., 1990; Noda et al., 2002; Takei et al., 1983; Willis et al., 2002). Sexual dimorphism in brain metabolism is still a subject of debate, but it could be a factor in intra-specific variation (Gur et al., 1995;

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Willis et al., 2002). The one group that has investigated intra-specific variation in brain size and CMRgl found no relationship between the two variables (Hatazawa et al., 1987a, 1987b).

Among the more than 200 studies of conscious subjects (human and non-human), variation both within and among studies averages approximately 22%. This result is comparable with previous estimates of inter-subject variability (Brooks et al., 1987; Camargo et al., 1992; Reivich et al., 1982). Intra-subject variation in PET studies is relatively small. Most repeatability studies for PET, conducted over a wide range of delays between scans, show intra-subject changes of 10% or less (Bartlett et al., 1988; Brooks et al., 1987; Camargo et al., 1992; Reivich et al., 1982). As would be expected, CMRgl values tend to be lower in the second scan, probably as a result of reduced anxiety (Stapleton et al., 1997).

Finally, it is worth noting that brain metabolism is itself divisible into active and basal components. (Astrup, 1982; Astrup et al., 1981, 1977; Michenfelder, 1974; Nemoto et al., 1996b, 1988). Active (or functional) metabolism is that part associated with spontaneous electrical activity and synaptic transmission, as represented in a normal electroencephalogram (EEG). Basal metabolism represents the remaining host of energy-derived processes necessary for cell viability, such as maintaining transmembrane ion potentials and fueling the metabolic apparatus itself.

A state of pure basal activity can be achieved experimentally by administering sufficient doses of inhibitory anesthetics, such as thiopental, which prevent spontaneous electrical activity and suppress all EEG activity – an idea first suggested by Kety when he noted that "since anesthetics act by reversible inhibition of synaptic

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transmission, suppression of all EEG activity by anesthetics should be that attributable to $CMRO_2$ associated with that needed for EEG activity [i.e. basal metabolism]." Subsequent researchers have measured $CMRO_2$ before and during isoelectric EEG induced by barbiturate anesthesia to calculate the percent contributions of total and basal $CMRO_2$, and have consistently demonstrated a 50-50 distribution (Astrup, 1982; Astrup et al., 1977; Michenfelder, 1974; Nemoto et al., 1988).

The metabolic data that make up my dataset represent both active and basal metabolism, but of the two, active metabolism should introduce greater variation than basal metabolism, which remains highly autoregulated under normal conditions. Variability in basal metabolism is also minimized since all experimental data included in this thesis were collected within a narrow range of biometric conditions. In this sense, it is reasonable to assume that most of the real physiological variation in brain metabolism encountered in this analysis is associated with the active 50% of total metabolism.

5.2 Inter-Specific Variation

Tests for phylogenetic independence indicate a large degree of phylogenetic signal in the rCMRgl data. In particular, the differences between primates and rodents/lagomorphs is far greater than their differences with carnivores, even though primates are more closely related to rodents and lagomorphs. Due to the small number of species available for analysis, statistical methods to control for phylogeny

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are only partially effective. The inter-specific comparisons in this study are therefore not entirely free of phylogenetic effects.

All three independent variables investigated here (body size, brain size, and basal metabolic rate) are strongly correlated with regional brain metabolism. Brain size is the most directly relevant variable and exhibits slightly higher correlations than body size or BMR. In all analyses, metabolic rates in the hippocampus and basal ganglia are most strongly correlated with the independent size variables, while the cerebellum and white matter exhibit weak correlations. The design of this study does not allow me to state that metabolic rates in the hippocampus and basal ganglia are more constrained with brain size than rates in other regions, but the consistently strong correlation coefficients for these two regions suggest that such constraints could be present.

The results of my allometric analyses indicate that mass-specific regional metabolism scales at a similar rate among regions, but there are consistent differences in the magnitude of rCMRgl. In other words, the allometric slopes do not appear to be different while the intercepts do appear to be different. Those differences in slopes that were detected using *n*-weighted least-squares of tip data are likely to be caused by partial-volume averaging. The log-log slope of –0.12 for regional metabolism regressed on brain size is in excellent agreement with previous studies. Mink et al. (Mink et al., 1981) found a slope of –0.13 for *in vivo* measures of oxygen metabolism regressed on brain size. Similarly, *in vitro* metabolic rates from tissue slices kept in suspension scaled to the –0.1 power across a wide range of brain sizes, including elephants and whales (Elliott and Henderson, 1948; Tower and Young, 1973).

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The physiological basis for this scaling factor remains unknown. It has been assumed that negative metabolic scaling in the brain is related to a reduction in neuronal density as brains get larger (Abeles, 1991; Mink et al., 1981; Nilsson and Siesjö, 1976). However, there are several potential problems with this assumption. First, the few cell counts that are available indicate that neuronal density scales with brain size to the –0.33 power, steeper than the metabolic scaling observed here (Blinkov and Glezer, 1968; Sholl, 1956; Tower, 1954; Tower and Elliott, 1952). Unfortunately, even the neuronal density estimates are based on old and potentially inaccurate cell counts conducted before the development of modern, unbiased stereological methods (Gundersen et al., 1988).

It is likely that factors other than, or in addition to, neuronal density play a role in metabolic scaling. For example, recent work has demonstrated that the rate of glucose uptake across the blood brain barrier is tightly regulated by glial cells. The coupling of neuronal activity and glucose utilization appears to be mediated by a "neuron-glia metabolic unit" that is highly sensitive to glutamate transport (Magistretti et al., 1999). In this model, synaptic glutamate reuptake into astrocytes triggers increased glucose uptake from capillaries via activation of Na^+/K^+-ATPase. Glucose is first processed glycolytically to lactate in the astrocytes and then transported into neurons where it is converted to pyruvate for entry into the TCA cycle (Tsacopoulos and Magistretti, 1996). It is entirely possible that glial cell density plays an important role in metabolic scaling.

Additionally, high-resolution deoxyglucose autoradiography has consistently demonstrated that most glucose uptake occurs in areas rich in synapses, dendrites, and

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axons rather than in cell bodies (Duncan and Stumpf, 1991; Duncan et al., 1987). The distribution of cytochrome oxidase corroborates this pattern of energy use: the highest density of cytochrome oxidase is found in presynaptic axon terminals and postsynaptic dendritic trees (Di Rocco et al., 1989; Kageyama and Wong-Riley, 1986). These data support the notion that there is greater glucose utilization in the neuropil than in the perikarya. It is therefore conceivable that the degree of dendritic arborization could also be a factor in metabolic scaling.

A better understanding of the functional basis for metabolic scaling in the brain will require histological analysis of neuronal density, glial cell density, and dendritic complexity in a variety of mammals using modern stereological methods. I have already made arrangements to pursue this analysis as the next phase in this project.

The data analyzed here indicate that the mammalian brain exhibits a consistent pattern of regional metabolism: the cortex and thalamus metabolize at high rates; the hippocampus, basal ganglia, and cerebellum metabolize at intermediate rates; and white matter metabolizes at a low rate. The coupling of cortical and thalamic metabolism is not surprising. It has already been noted that synaptic activity in the neuropil accounts for most of the energy demands in the brain (Jueptner and Weiller, 1995; Raichle, 1987). Since the thalamus is the principal synaptic relay for the cortex, it is likely that much of the neuronal activity in the cortex will actually appear as localized glucose uptake in the thalamic synapses. And since white matter consists almost exclusively of myelinated axon tracts with no synaptic activity, it makes sense that this region would exhibit the lowest metabolic rate.

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When mass-specific metabolic rates are combined with regional volume estimates, the results reflect several significant effects of differential brain expansion. First, a relative reduction in the size of the hippocampus in primates translates to a lower metabolic load for this region. In humans and macaques hippocampal metabolism constitutes less than 1% of total brain metabolism, compared to between 6% and 10% in other species.

Second, the fact that white matter volume must increase faster than gray matter volume (see below) results in a surprising shift in the distribution of energy use in the human brain. The large volume of white matter in the human brain accounts for nearly 20% of total brain glucose uptake, while the figure is less than 10% in all other species studied here.

Finally, expansion of the neocortex is reflected by a greater proportion of energy allocation. Among humans, macaques, cats, and rabbits, approximately 50-60% of total brain metabolism takes place in the cortex. Among the more lissencephalic gerbils, rats, and mice, the figure is approximately 35-45%.

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Implications for Human Brain Evolution

5.3 The Requirements for Interconnectivity

The increasing load of white matter metabolism in the human brain is part of a larger issue that encompasses interconnectivity and the theoretical limits of brain expansion. A neural network consisting either of individual cells or modular units can have varying degrees of interconnectivity (I). The lowest degree of interconnectivity ($I = 1$) is a completely binary network in which each unit sends and receives connections to and from one neighboring cell (Figure 5.1). In this situation, the number of connections (C) equals the number of units (U). The highest degree of interconnectivity ($I = U$) would be achieved if each unit is connected by separate fibers to all other units. In this situation, $C = U(U\text{-}1)$, which is essentially equivalent to $C = U^2$. In the human brain, the degree of cortical interconnectivity lies approximately on the midpoint between these two extremes: $I = U^{0.5}$.

White matter constitutes approximately one-third of total brain volume in humans (Hofman, 1989). Using prediction equations derived from regressions within anthropoid primates, Hofman (2001) estimates that the relative volume of white matter would reach 50% at a brain volume of approximately 3000 cm^3. Beyond this point, further brain expansion (assuming an equal level of interconnectivity, $I = U^{0.5}$)

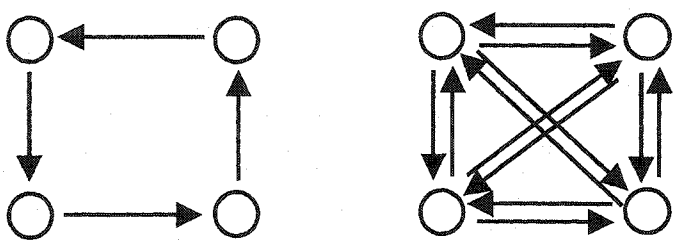


Figure 5.1. Two extremes of connectivity. On the left, a binary system where *C=U*. On the right, a completely interconnected system in which *C=U(U-1)*.

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would be inefficient. Hofman (2001, p. 125) argues that:

> Once the brain has grown to a point where the bulk of its mass is in the form of interconnections, further increases will be unproductive, due to the declining capability of neuronal integration and increased conduction time. At this point, corresponding to a brain size two to three times that of modern man, the brain reaches its maximum processing power.

Humans still have plenty of "breathing room" before reaching this critical mass, but the constraining role of interconnectivity has led to the suggestion that brain expansion necessitates a certain degree modularization, i.e. dividing the network into smaller, more autonomous components with higher local connectivity but fewer long-distance connections (Allman, 1990; Kaas, 1987, 2000). Evidence for modularization during primate brain evolution is seen by increases in the number of cortical areas (Kaas, 1993; Welker, 1990) and by lateralizing cortical functions (Aboitiz, 1996).

5.4 The Cost of Brain Work

The average whole brain metabolism for humans reported here (29.98 umoles of glucose / 100g / min) corresponds to a heat production of approximately 15 watts. Since the brain is enclosed in a well-insulated cavity, passive cooling (conducting heat out to the surface of the head) is not sufficient to keep the brain from over-heating. Therefore, the brain is actively cooled by the cerebral circulation. It is possible that adaptations for active cooling, such as the "cranial radiator" hypothesis (Falk, 1990), served to increase cooling efficiency as hominins evolved larger brains.

However, the metabolic/thermal limitations to brain expansion are thought to be much less restrictive than the limitations posed by interconnectivity. Cochrane et

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al. (1998) calculated that as long as increases in brain size could be matched by increases in cardiovascular output and blood flow to the brain, then the human brain would have to evolve to 250 times its present volume before the cerebral vasculature would constitute 50% of the cranial space. They identify this as the "dominant thermal limit" of brain expansion.

Rather than constraining brain size, cerebral metabolism has a profound impact on the organization of the brain and how information is coded. Attwell and Laughlin (2001) provide a detailed breakdown of the metabolic costs of a single neuron in rat neocortex, and Lennie (2003) extends these calculations to human cortex. According to their calculations, the aggregate cost of a single spike in a human cortical pyramidal neuron is 2.4×10^9 molecules of ATP. Recall from the background material that synapse density remains constant across brain sizes, while neuron density decreases. This means that each neuron in human cortex has more synapses, longer dendrites, and longer axon collaterals than in the rat. Accordingly, the estimated cost of single spike in a human neuron is 3.3 times greater than the cost of a single spike in a rat neuron, and the cost of simply maintaining a resting potential is 2.6 times greater in a human neuron than a rat neuron (Lennie, 2003).

Despite this dramatic increase in the cost of neural activity, the data presented here clearly demonstrate that metabolic rate declines significantly as brain size increases. While the cost of spiking is 3.3 times greater in humans than rats, human brain metabolism is 2.6 times lower than in the rat, and 1.4 times lower than in the macaque. The implication of this result is that the number of *active* neurons in the human cortex at any moment in time is much fewer than in other species. It is

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estimated that even in strongly active cortex, the number of neurons active at any one time is between 1 in 25 and 1 in 63 (Heeger et al., 2000; Lennie, 2003). The optimal fraction of active neurons is estimated at between 1% and 4% (Attwell and Laughlin, 2001; Levy and Baxter, 1996).

It must be concluded that the metabolic limits to human brain evolution are related not to brain size but to information processing. Three decades ago, Broadbent (1971) proposed that selective attention evolved in humans as a consequence of the brain's limited capacity for processing concurrent information. The high metabolic cost of neural activity corroborates this hypothesis. Moreover, the exquisitely precise circulatory mechanisms for shunting oxygen and glucose to active brain regions (Raichle, 1987; Shulman et al., 2001) provides further evidence for the narrow threshold of sustainable neural activity. Brain metabolism is essentially a "zero-sum game." Normal variations in mental activity change the distribution of energy use in the brain, but total brain energy consumption remains stable (Clarke and Sokoloff, 1994). If one region becomes active, it does so at the expense of another.

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6. Conclusions

The results of this study demonstrate a consistent pattern of regional energy consumption in the mammalian brain. The cortex and thalamus, which are dense with dendrites and synapses, exhibit the highest rates of glucose uptake. Within the cortex, sensory areas exhibit the highest metabolic rates. This is consistent with the fact that sensory information must be processed with a high degree of temporal resolution and therefore requires a high degree of local processing. It is interesting to note that association areas, such as frontal and parietal regions, which may have an important role in the evolution of human behavior, require less energy than sensory regions.

As expected, white matter exhibits the lowest rate of glucose uptake in the mammalian brain. However, the large volume of white matter in the human brain constitutes a significant proportion of total energy consumption. Since the requirements of interconnectivity dictate that white matter must increase faster than gray matter, humans must spend considerably more energy on white matter than smaller-brained animals.

Given the probable effects of partial-volume averaging in PET, the allometric results indicate that all brain regions scale with brain size at similar rates. While the magnitude (intercepts) of the lines are different, the different regions examined here scale with brain size to the –0.12 power. The functional basis of this scaling coefficient remains unknown. Possible factors are neuronal density, glial density, and dendritic arborization. Unbiased stereological analysis of histological sections in these species will be required to test the influence of each of these factors.

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The high metabolic costs of cortical activation mean that only a small fraction of cortical neurons can be active at any one time. While this does not appear to pose immediate limits on the size of the brain, it does have a significant effect on the way the brain codes information. Larger brains must be able to incorporate metabolically-efficient coding strategies that minimize the number of active neurons.

This study demonstrates that clinical neuroimaging methods, such as PET and 2DG autoradiography, are useful for anthropological studies of comparative neurology. Aside from anesthesia, which is an obvious confounding factor, the principal methodological issue encountered here is partial-volume averaging in PET. The error introduced by partial-volume averaging needs to be studied in greater detail to ensure that data are collected accurately and compared appropriately.

Finally, we need data for more species. Anthropologists should make it a priority to increase the number of primates in the metabolic sample. Future measurements of both small-brained strepsirrhines and large-brained anthropoids would provide a much-needed contextual framework for understanding human brain activity and energy use.

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Table A1. Basal Ganglia rCMR and rCBF (units: Glucose & Oxygen *umol/100g/min*, Blood flow *ml/100g/min*)

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Baxter et al., 1987)	7	none	PET	29.4	3.0				
Homo	(Baxter et al., 1987)	7	none	PET	35.9	4.3				
Homo	(Baxter et al., 1988)	10	none	PET	33.5	7.8				
Homo	(Bromberg et al., 1990)	12	none	PET	39.8	9.1				
Homo	(Brooks et al., 1987)	27	none	PET	46.9	8.7				
Homo	(Buchsbaum et al., 1989)	10	none	PET	24.8	6.4				
Homo	(Chugani et al., 1987)	7	none	PET	32.9	1.3				
Homo	(Clark et al., 2001)	32	none	PET	42.5	6.7				
Homo	(Delvenne et al., 1997)	10	none	PET	30.6	2.8				
Homo	(Delvenne et al., 1999)	10	none	PET	30.6	2.9				
Homo	(Duara et al., 1983)	18	none	PET	29.4	7.7				
Homo	(Foster et al., 1984)	8	none	PET	39.4	1.8				
Homo	(Gilman et al., 1990)	13	none	PET	34.5	6.7				
Homo	(Gilman et al., 1990)	10	none	PET	38.3	8.7				
Homo	(Heiss et al., 1985)	6	none	PET	41.1	4.9				
Homo	(Hu et al., 2000)	9	none	PET	41.3	10.3				
Homo	(Kuhl et al., 1982)	25	none	PET	36.3	8.1				
Homo	(Ludolph et al., 1992)	12	none	PET	48.2	11.0				
Homo	(Mazziotta et al., 1981)	7	none	PET	38.5	9.2				
Homo	(Perani et al., 1995)	15	none	PET	41.1	4.7				
Homo	(Phillips et al., 1991)	14	none	PET	36.2	6.5				
Homo	(Redies et al., 1989)	4	none	PET	31.0	2.0				
Homo	(Rudolf et al., 2000)	16	none	PET	39.4	6.5				
Homo	(Schapiro et al., 1990)	13	none	PET	55.4	8.7				
Homo	(Tedeschi et al., 1995)	11	none	PET	35.9	6.2				
Homo	(Tommasino et al., 1995)	10	none	PET	38.1	4.8				

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Table A1, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Volkow et al., 1992)	11	none	PET	42.2	5.0				
Homo	(Wu et al., 1992)	15	none	PET	27.1	6.5				
Homo	Present Study	5	none	PET	42.6	4.1				
Homo	(De Volder et al., 1997)	3	none	PET	41.1	5.3	177.3	15.3	50.4	6.6
Homo	(Hatazawa et al., 1988)	7	none	PET	27.3	4.1	144.2	37.5	50.1	7.1
Homo	(Herold et al., 1988)	8	none	PET	24.3	2.4	136.4	11.6	40.1	7.5
Homo	(Leenders et al., 1990)	34	none	PET			137.1	18.3	42.4	7.6
Homo	(Burns and Tyrrell, 1992)	4	none	PET			167.0	21.0		
Homo	(Kaisti et al., 2003)	8	none	PET			217.2	40.4		
Homo	(Kaisti et al., 2003)	8	Sevoflurane	PET			151.1	28.1		
Homo	(Langsjo et al., 2003)	10	none	PET			179.7	23.9		
Homo	(Perlmutter et al., 1987)	32	none	PET			133.0	34.8		
Homo	(Sasaki et al., 1986)	7	none	PET			159.4	29.6		
Homo	(Wessen et al., 1997)	6	none	PET			238.0	30.0		
Homo	(Yamaguchi et al., 1986)	14	none	PET			176.8	28.3		
Papio	(Meguro et al., 1999)	8	Phencyclidine, N2O	PET	47.7	6.1				
Papio	Present Study	2	70% N2O	PET	44.6	5.8				
M mulatta	(Brownell et al., 1994)	2	Ketamine	PET	50.6	2.4				
M mulatta	(Enlund et al., 1997)	4	Ketamine, N2O	PET	37.1	8.1				
M mulatta	(Palombo et al., 1988)	3	none	Sokoloff	40.0	3.0				
M mulatta	(Palombo et al., 1990)	4	none	Sokoloff	51.3	4.0				
M mulatta	(Palombo et al., 1991)	4	none	Sokoloff	46.0	3.0				
M mulatta	(Porrino et al., 1987)	3	none	Sokoloff	40.8	3.0				
M mulatta	(Noda et al., 2002)	6	none	PET	40.9	9.1			60.5	11.1
M mulatta	(Cross et al., 2000)	6	none	PET			255.0	32.0		

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Table A1, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
M fascic	(Brownell et al., 1994)	1	Ketamine	PET	57.8	4.8				
M fascic	(Deuel and Collins, 1984)	4	none	Sokoloff	55.8					
M fascic	(Kennedy et al., 1978)	7	none	Sokoloff	45.5					
M fascic	(Kuge et al., 2001)	4	Sevoflurane, N2O	PET	35.5	8.8				
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	42.2	3.0				
M fascic	(Schwartzman et al., 1988)	6	none	Sokoloff	38.0	1.6				
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl, N2O	Sokoloff	97.0	6.0				
Sus	(Manohar, 1986)	6	none	Microspheres					102.6	5.8
Sus	(Manohar, 1986)	6	Sevoflurane	Microspheres					86.0	7.3
Sus	(Manohar, 1986)	6	Sevoflurane	Microspheres					83.7	7.2
Lepus	(Passero et al., 1981)	5	none	Sokoloff	58.7	3.6				
Felis	(Hovda et al., 1996)	7	none	Sokoloff	59.4	12.0				
Felis	(Petitjean et al., 1982)	4	none	Sokoloff	49.7	7.0				
Felis	(Triolo et al., 1990)	6	N2O	Sokoloff	37.3	1.6				
Meriones	(Paschen et al., 1992)	5	none	Sokoloff	75.0	nr				
Rattus	(Bassant et al., 1996)	7	none	Sokoloff	68.0	6.0				
Rattus	(Bassant et al., 1996)	8	none	Sokoloff	56.5	6.0				
Rattus	(Blin et al., 1991)	10	none	Sokoloff	62.8	7.7				
Rattus	(Bryan et al., 1983)	6	none	Sokoloff	60.0	1.0				
Rattus	(Bryan and Lehman, 1988)	10	none	Sokoloff	58.0	1.5				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	81.5	1.5				

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Table A1, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	82.0	2.0				
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	74.5	2.5				
Rattus	(Crosby et al., 1984)	5	none	Sokoloff	67.5	2.5				
Rattus	(Crosby et al., 1984)	5	N2O	Sokoloff	82.5	4.0				
Rattus	(Cudennec et al., 1997)	7	none	Sokoloff	81.5	3.0				
Rattus	(Dam et al., 1990)	5	none	Sokoloff	65.0	7.5				
Rattus	(Davis et al., 1986)	10	none	Sokoloff	69.0	3.8				
Rattus	(Dow-Edwards et al., 1986)	4	none	Sokoloff	83.5	2.5				
Rattus	(Eintrei et al., 1999)	6	none	Sokoloff	81.5	3.0				
Rattus	(Fornai and Orzi, 2001)	5	none	Sokoloff	69.7	9.0				
Rattus	(Freedland et al., 2002)	5	none	Sokoloff	83.8	3.0				
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	63.5	6.5				
Rattus	(Frietsch et al., 2000)	12	Isoflurane	Sokoloff	32.0	5.5				
Rattus	(Furlow et al., 1983)	5	none	Sokoloff	66.0	14.0				
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	68.0	2.5				
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	79.5	4.5				
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	85.0	3.0				
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	72.0	1.5				
Rattus	(Ingvar et al., 1980)	10	N2O	Sokoloff	80.0	4.0				
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	51.5	1.5				
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	53.0	2.5				
Rattus	(Kadekaro et al., 1983)	7	none	Sokoloff	75.5	2.5				
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	60.5	7.5				
Rattus	(Lu et al., 1983)	10	none	Sokoloff	60.5	2.5				
Rattus	(MacNeill and Bryan, 1988)	10	none	Sokoloff	85.0	2.0				
Rattus	(Maekawa et al., 1986)	6	none	Sokoloff	60.5	9.5				
Rattus	(Moore et al., 2000)	6	none	PET	84.5	17.0				

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Table A1, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Moore et al., 2000)	6	none	PET	75.8	12.6				
Rattus	(Nehlig et al., 1985)	6	none	PET	76.5	2.5				
Rattus	(Ori et al., 1986)	6	none	Sokoloff	64.5	2.5				
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	75.5	6.0				
Rattus	(Otori et al., 2003)	6	none	Sokoloff	56.7	13.0				
Rattus	(Poindessous-Jazat et al., 1998)	7	none	Sokoloff	73.5	8.0				
Rattus	(Poindessous-Jazat et al., 1998)	5	none	Sokoloff	59.0	7.0				
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	67.0	3.0				
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	74.5	3.0				
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	84.5	2.5				
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	81.0	4.0				
Rattus	(Grome and McCulloch, 1981)	16	none	Sokoloff	65.0	1.5				
Rattus	(Whitlow et al., 2002)	5	none	Sokoloff	91.8	4.0				
Rattus	(Walker et al., 1999)	5	none	Sokoloff	85.9	5.2				
Rattus	(Lear et al., 1981)	4	none	Sokoloff					112.0	18.0
Rattus	(Cremer and Seville, 1983)	8	none	Sokoloff					123.0	7.0
Mus	(Jay et al., 1985)	8	none	Sokoloff	66.5	3.0				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	118.5	18.5				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	119.0	18.0				
Mus	(Yamaguchi et al., 1992b)	6	none	Sokoloff	54.7	2.6				
Mus	(Zocchi et al., 2001)	4	none	Sokoloff	73.4	5.2				
Mus	(Zocchi et al., 2001)	6	none	Sokoloff	50.9	4.5				

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Table A2. Hippocampus rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Biver et al., 1994)	12	none	PET	19.4	3.0				
Homo	(Duara et al., 1983)	18	none	PET	23.0	6.9				
Homo	(Heiss et al., 1985)	6	none	PET	38.3	5.2				
Homo	(Ishii et al., 1998)	22	none	PET	29.0	3.7				
Homo	(Phillips et al., 1991)	14	none	PET	30.8	5.0				
Homo	(Rudolf et al., 2000)	16	none	PET	30.5	3.8				
Homo	(Schapiro et al., 1990)	13	none	PET	38.6	4.9				
Homo	Present Study	5	none	PET	33.2	4.6				
Homo	(Yamaguchi et al., 1986)	14	none	PET			137.1	24.6	42.2	8.8
Homo	(Seitz and Roland, 1992)	26	none	PET					52.6	8.4
Papio	(Meguro et al., 1999)	8	Phencyclidine, N2O	PET	30.8	5.4				
Papio	(Millien et al., 2002)	10	Phencyclidine, N2O	PET	32.3	8.0				
Papio	Present Study	2	N2O	PET	33.1	6.8				
M mulatta	(Nakamura et al., 1983)	4	none	Sokoloff	42.0	nr				
M mulatta	(Noda et al., 2002)	6	none	PET	32.2	5.2			53.9	7.6
M mulatta	(Palombo et al., 1988)	3	none	Sokoloff	33.0	2.0				
M mulatta	(Palombo et al., 1990)	4	none	Sokoloff	32.0	1.0				
M mulatta	(Palombo et al., 1991)	4	none	Sokoloff	30.5	1.0				
M fascic	(Kennedy et al., 1978)	7	none	Sokoloff	39.0	2.0				
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	30.0	1.0				
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl, N2O	Sokoloff	48.0	3.0				

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Table A2, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Lepus	(Passero et al., 1981)	5	none	Sokoloff	58.1	2.5				
Felis	(Hovda et al., 1996)	7	none	Sokoloff	52.0	15.0				
Felis	(Petitjean et al., 1982)	4	none	Sokoloff	48.0	4.0				
Meriones	(Choki et al., 1983)	6	none	Sokoloff	69.0	19.0				
Meriones	(Choki et al., 1984)	3	N2O	Sokoloff	38.9	13.9				
Meriones	(Mies et al., 1990)	4	Halothane	Sokoloff	74.3	4.0			80.3	6.7
Meriones	(Mies et al., 1984)	2	none	Sokoloff					109.0	9.0
Meriones	(Suzuki et al., 1983)	7	Halothane	Sokoloff					93.0	5.3
Rattus	(Bassant et al., 1996)	7	none	Sokoloff	62.0	5.0				
Rattus	(Bassant et al., 1996)	8	none	Sokoloff	58.0	7.0				
Rattus	(Blin et al., 1991)	10	none	Sokoloff	56.4	7.1				
Rattus	(Bryan et al., 1983)	6	none	Sokoloff	50.0	2.4				
Rattus	(Bryan and Lehman, 1988)	10	none	Sokoloff	57.0	3.2				
Rattus	(Chang et al., 2001)	8	none	Sokoloff	63.0	14.1				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	72.0	2.0				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	71.0	4.0				
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	73.0	5.7				
Rattus	(Crosby et al., 1984)	5	none	Sokoloff	62.0	6.7				
Rattus	(Crosby et al., 1984)	5	N2O	Sokoloff	77.0	4.5				
Rattus	(Dam et al., 1990)	5	none	Sokoloff	51.0	11.2				
Rattus	(Davis et al., 1986)	10	none	Sokoloff	64.0	12.0				
Rattus	(Dow-Edwards et al., 1986)	4	none	Sokoloff	75.0	6.0				
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	60.2	4.7				
Rattus	(Eintrei et al., 1999)	6	none	Sokoloff	87.0	7.3				
Rattus	(Fornai and Orzi, 2001)	5	none	Sokoloff	60.5	9.0				

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Table A2, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Freedland et al., 2002)	5	none	Sokoloff	72.6	8.9				
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	55.0	6.8				
Rattus	(Frietsch et al., 2000)	12	Isoflurane	Sokoloff	39.5	7.0				
Rattus	(Furlow et al., 1983)	5	none	Sokoloff	53.0	13.0				
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	78.0	19.0				
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	83.0	15.9				
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	72.0	13.1				
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	66.0	5.4				
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	48.0	4.9				
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	46.0	6.3				
Rattus	(Katsumata et al., 2001)	5	none	Sokoloff	67.3	10.6				
Rattus	(Knapp et al., 2002)	8	none	Sokoloff	66.8	1.6				
Rattus	(Lu et al., 1983)	10	none	Sokoloff	59.0	6.3				
Rattus	(MacNeill and Bryan, 1988)	10	none	Sokoloff	62.0	9.5				
Rattus	(Nehlig et al., 1985)	6	none	Sokoloff	76.0	9.8				
Rattus	(Ori et al., 1986)	6	none	Sokoloff	47.0	7.3				
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	72.0	8.9				
Rattus	(Otori et al., 2003)	6	none	Sokoloff	45.3	13.0				
Rattus	(Poindessous-Jazat et al., 1998)	7	none	Sokoloff	65.5	5.0				
Rattus	(Poindessous-Jazat et al., 1998)	5	none	Sokoloff	59.0	6.8				
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	68.0	7.9				
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	80.0	7.3				
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	79.0	1.0				
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	58.0	8.0				
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	79.0	11.2				
Rattus	(Grome and McCulloch, 1981)	16	none	Sokoloff	73.0	6.0				
Rattus	(Whitlow et al., 2002)	5	none	Sokoloff	78.8	6.7				
Rattus	(Walker et al., 1999)	5	none	Sokoloff	74.7	6.8				

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Table A2, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Lear et al., 1981)	4	none	Sokoloff	70.0	8.0			81.0	10.0
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	52.0	8.0			76.0	10.0
Rattus	(Maekawa et al., 1986)	6	none	Sokoloff	57.0	11.0			104.0	18.0
Rattus	(Cremer and Seville, 1983)	8	none	Sokoloff					102.0	6.0
Rattus	(Dahlgren et al., 1981a)	6	none	Sokoloff					100.0	3.0
Mus	(Bouilleret et al., 2000)	4	none	Sokoloff	84.3	13.0				
Mus	(Jay et al., 1985)	8	none	Sokoloff	70.0	5.7				
Mus	(Kamp et al., 1980)	10	none	Sokoloff	83.0	25.3				
Mus	(Kamp et al., 1980)	17	none	Sokoloff	87.0	53.6				
Mus	(Yamaguchi et al., 1992b)	6	none	Sokoloff	53.4	6.1				
Mus	(Zocchi et al., 2001)	4	none	Sokoloff	52.2	5.2				
Mus	(Zocchi et al., 2001)	6	none	Sokoloff	39.6	7.2				

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Table A3. Thalamus rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Baxter et al., 1987)	7	none	PET	29.3	3.1				
Homo	(Baxter et al., 1987)	7	none	PET	34.4	4.3				
Homo	(Biver et al., 1994)	12	none	PET	30.1	4.7				
Homo	(Bromberg et al., 1990)	12	none	PET	39.0	9.4				
Homo	(Brooks et al., 1987)	27	none	PET	45.2	9.0				
Homo	(Buchsbaum et al., 1989)	10	none	PET	24.3	7.2				
Homo	(Chugani et al., 1987)	7	none	PET	32.2	5.2				
Homo	(Clark et al., 2001)	32	none	PET	42.2	6.7				
Homo	(Delvenne et al., 1997)	10	none	PET	30.1	2.8				
Homo	(Delvenne et al., 1999)	10	none	PET	30.1	2.9				
Homo	(Duara et al., 1983)	18	none	PET	27.4	8.0				
Homo	(Gilman et al., 1990)	13	none	PET	34.6	5.7				
Homo	(Gilman et al., 1990)	10	none	PET	41.9	9.3				
Homo	(Heiss et al., 1985)	6	none	PET	38.4	3.2				
Homo	(Hirono et al., 2000)	15	none	PET	42.2	6.9				
Homo	(Hu et al., 2000)	9	none	PET	35.4	8.9				
Homo	(Kuhl et al., 1982)	25	none	PET	35.4	7.2				
Homo	(Ludolph et al., 1992)	12	none	PET	51.0	13.3				
Homo	(Martinot et al., 1990)	8	none	PET	35.3	4.7				
Homo	(Mazziotta et al., 1981)	7	none	PET	36.6	7.1				
Homo	(Perani et al., 1995)	15	none	PET	39.1	6.7				
Homo	(Phillips et al., 1991)	14	none	PET	34.3	6.7				
Homo	(Redies et al., 1989)	4	none	PET	31.0	2.0				
Homo	(Rudolf et al., 2000)	16	none	PET	36.6	4.9				
Homo	(Sachs et al., 1986)	25	none	PET	42.2	5.6				
Homo	(Schapiro et al., 1990)	13	none	PET	50.9	9.7				

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Table A3, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Tedeschi et al., 1995)	11	none	PET	38.8	4.4				
Homo	(Tommasino et al., 1995)	10	none	PET	37.7	6.5				
Homo	(Volkow et al., 1992)	11	none	PET	41.3	5.0				
Homo	(Wu et al., 1992)	15	none	PET	26.6	6.8				
Homo	Present Study	5	none	PET	38.7	5.9				
Homo	(De Volder et al., 1997)	3	none	PET	39.1	5.3	176.7	20.3	55.9	4.0
Homo	(Sasaki et al., 1986)	7	none	PET	33.5	4.1	148.1	26.1	45.4	8.6
Homo	(Kaisti et al., 2003)	8	none	PET			251.3	38.0	54.8	6.3
Homo	(Kaisti et al., 2003)	8	Sevoflurane	PET			148.2	20.5	39.9	7.1
Homo	(Pantano et al., 1984)	18	none	PET			156.3	35.7	46.0	12.0
Homo	(Perlmutter et al., 1987)	32	none	PET			120.5	35.7	53.7	7.5
Homo	(Wessen et al., 1997)	6	none	PET			205.0	22.0	67.0	10.0
Homo	(Yamaguchi et al., 1986)	14	none	PET			163.8	25.0	48.9	10.4
Homo	(Burns and Tyrrell, 1992)	4	none	PET			170.5	24.6		
Homo	(Langsjo et al., 2003)	10	none	PET			203.6	24.6		
Homo	(Seitz and Roland, 1992)	26	none	PET					62.2	17.8
Papio	(Meguro et al., 1999)	8	Phencyclidine, N2O	PET	43.8	9.4				
Papio	(Millien et al., 2002)	10	Phencyclidine, N2O	PET	43.6	13.0				
Papio	Present Study	2	N2O	PET	45.7	8.4				
M mulatta	(Cross et al., 2000)	6	none	PET	38.8	3.2				
M mulatta	(Nakamura et al., 1983)	4	none	Sokoloff	38.0	nr				
M mulatta	(Palombo et al., 1988)	3	none	Sokoloff	43.6	3.5				
M mulatta	(Palombo et al., 1990)	4	none	Sokoloff	47.5	4.0				
M mulatta	(Palombo et al., 1991)	4	none	Sokoloff	44.3	8.0				
M mulatta	(Porrino et al., 1987)	3	none	Sokoloff	54.3	4.0				
M mulatta	(Enlund et al., 1997)	4	Ketamine, N2O	PET			245.0	34.0	70.5	15.0

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Table A3, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
M fascic	(Deuel and Collins, 1984)	4	none	Sokoloff	48.0	nr				
M fascic	(Kennedy et al., 1978)	7	none	Sokoloff	50.3	nr				
M fascic	(Kuge et al., 2001)	4	Sevoflurane, N2O	PET	35.5	8.8				
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	47.4	3.5				
M fascic	(Schwartzman et al., 1988)	6	none	Sokoloff	40.1	3.2				
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl, N2O	Sokoloff	59.0	6.0			96.0	10.0
Sus	(Manohar, 1986)	6	none						76.8	4.5
Sus	(Manohar, 1986)	6	Sevoflurane						46.2	3.0
Sus	(Manohar, 1986)	6	Sevoflurane						59.0	5.5
Lepus	(Passero et al., 1981)	5	none	Sokoloff	89.7	4.2				
Felis	(Hovda et al., 1996)	7	none	Sokoloff	85.4	14.5				
Felis	(Petitjean et al., 1982)	4	none	Sokoloff	58.0	14.0				
Felis	(Triolo et al., 1990)	6	N2O	Sokoloff	28.6	8.0				
Meriones	(Choki et al., 1983)	6	none	Sokoloff	78.3	15.6				
Meriones	(Choki et al., 1984)	3	N2O	Sokoloff	107.0	14.0				
Meriones	(Mies et al., 1984)	2	none	Sokoloff	70.5	6.0			117.0	10.0
Meriones	(Mies et al., 1990)	4	Halothane	Sokoloff					139.5	22.0
Rattus	(Bassant et al., 1996)	7	none	Sokoloff	89.0	8.0				
Rattus	(Bassant et al., 1996)	8	none	Sokoloff	84.0	11.0				
Rattus	(Blin et al., 1991)	10	none	Sokoloff	67.4	7.2				
Rattus	(Bryan et al., 1983)	6	none	Sokoloff	88.0	4.9				
Rattus	(Bryan and Lehman, 1988)	10	none	Sokoloff	86.0	12.6				

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Table A3, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Chang et al., 2001)	8	none	Sokoloff	93.6	22.6				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	103.5	4.0				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	98.5	5.0				
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	77.0	5.7				
Rattus	(Crosby et al., 1984)	5	none	Sokoloff	80.0	13.4				
Rattus	(Crosby et al., 1984)	5	N2O	Sokoloff	98.0	6.7				
Rattus	(Cudennec et al., 1997)	7	none	Sokoloff	102.0	13.2				
Rattus	(Dam et al., 1990)	5	none	Sokoloff	73.0	18.8				
Rattus	(Davis et al., 1986)	10	none	Sokoloff	97.8	17.4				
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	81.5	7.4				
Rattus	(Freedland et al., 2002)	5	none	Sokoloff	116.2	10.5				
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	71.0	6.0				
Rattus	(Frietsch et al., 2000)	12	Isoflurane	Sokoloff	32.0	7.0				
Rattus	(Furlow et al., 1983)	5	none	Sokoloff	70.0	16.0				
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	83.0	9.5				
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	95.0	16.9				
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	117.0	21.8				
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	101.0	10.1				
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	66.0	4.9				
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	82.0	15.8				
Rattus	(Kadekaro et al., 1983)	7	none	Sokoloff	82.0	15.9				
Rattus	(Katsumata et al., 2001)	5	none	Sokoloff	88.9	12.2				
Rattus	(Knapp et al., 2002)	8	none	Sokoloff	93.9	5.9				
Rattus	(Lu et al., 1983)	10	none	Sokoloff	97.0	9.5				
Rattus	(MacNeill and Bryan, 1988)	10	none	Sokoloff	77.0	9.5				
Rattus	(Mies et al., 1981)	6	none	Sokoloff	57.6	14.7				
Rattus	(Moore et al., 2000)	6	none	Sokoloff	82.3	17.5				
Rattus	(Moore et al., 2000)	6	none	Sokoloff	72.7	15.5				

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Table A3, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Nehlig et al., 1985)	6	none	Sokoloff	102.3	7.3				
Rattus	(Ori et al., 1986)	6	none	Sokoloff	72.0	7.3				
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	93.7	6.0				
Rattus	(Otori et al., 2003)	6	none	Sokoloff	71.9	19.1				
Rattus	(Poindessous-Jazat et al., 1998)	7	none	Sokoloff	89.0	6.0				
Rattus	(Poindessous-Jazat et al., 1998)	5	none	Sokoloff	90.0	12.0				
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	88.0	13.2				
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	100.0	7.3				
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	104.0	2.0				
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	93.5	12.0				
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	98.0	15.7				
Rattus	(Grome and McCulloch, 1981)	16	none	Sokoloff	79.5	4.0				
Rattus	(Whitlow et al., 2002)	5	none	Sokoloff	126.1	13.4				
Rattus	(Walker et al., 1999)	5	none	Sokoloff	109.1	2.5				
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	76.0	6.0			120.0	11.0
Rattus	(Maekawa et al., 1986)	6	none	Sokoloff	78.0	14.0			141.0	32.0
Rattus	(Cremer and Seville, 1983)	8	none	Sokoloff					117.0	6.0
Rattus	(Dahlgren et al., 1981b)	6	none	Sokoloff					146.0	8.0
Mus	(Bouilleret et al., 2000)	4	none	Sokoloff	119.2	10.0				
Mus	(Ito et al., 1994)	5	none	Sokoloff	96.0	11.2				
Mus	(Jay et al., 1985)	8	none	Sokoloff	81.0	14.1				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	141.6	12.2				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	143.8	20.8				
Mus	(Yamaguchi et al., 1992a)	6	none	Sokoloff	64.1	7.5				

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Table A4. Cerebellum rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Bartlett et al., 1994)	9	none	PET	34.5	4.2				
Homo	(Bromberg et al., 1990)	12	none	PET	32.4	6.5				
Homo	(Brooks et al., 1987)	27	none	PET	30.6	6.1				
Homo	(Clark et al., 2001)	32	none	PET	44.6	6.1				
Homo	(Duara et al., 1983)	18	none	PET	25.1	6.8				
Homo	(Gilman et al., 1990)	13	none	PET	28.8	6.0				
Homo	(Gilman et al., 1990)	10	none	PET	33.1	6.9				
Homo	(Heiss et al., 1985)	6	none	PET	32.9	6.0				
Homo	(Hirono et al., 2000)	15	none	PET	38.4	4.7				
Homo	(Hu et al., 2000)	9	none	PET	29.4	7.8				
Homo	(Ludolph et al., 1992)	12	none	PET	37.1	9.2				
Homo	(Perani et al., 1995)	15	none	PET	33.9	3.9				
Homo	(Phillips et al., 1991)	14	none	PET	29.3	6.1				
Homo	(Rudolf et al., 2000)	16	none	PET	33.1	4.5				
Homo	(Rudolf et al., 2002)	9	none	PET	28.0	nr				
Homo	(Sachs et al., 1986)	25	none	PET	43.3	6.1				
Homo	(Schapiro et al., 1990)	13	none	PET	40.4	7.5				
Homo	(Tommasino et al., 1995)	10	none	PET	32.5	5.2				
Homo	(Volkow et al., 1992)	11	none	PET	31.8	4.0				
Homo	(Wu et al., 1992)	15	none	PET	19.2	5.4				
Homo	Present Study	5	none	PET	34.9	6.4				
Homo	(De Volder et al., 1997)	3	none	PET	33.1	4.7	167.7	23.0	52.7	8.6
Homo	(Yamaguchi et al., 1992a)	14	none	PET			183.5	32.6	51.1	9.2
Homo	(Leenders et al., 1990)	34	none	PET			150.5	21.9	49.4	9.9
Homo	(Ishii et al., 1996b)	14	none	PET			246.4	54.5	87.1	19.1
Homo	(Kaisti et al., 2003)	8	none	PET			210.3	21.9	48.4	3.6

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Table A4, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Kaisti et al., 2003)	8	Sevoflurane	PET			118.8	9.4	35.4	6.8
Homo	(Burns and Tyrrell, 1992)	4	none	PET			171.0	21.0		
Homo	(Langsjo et al., 2003)	10	none	PET			175.0	29.5		
Homo	(Mintun et al., 1984)	5	none	PET			163.0	26.8		
Papio	(Meguro et al., 1999)	8	Phencyclidine, N2O	PET	35.2	6.7				
Papio	Present Study	2	N20	PET	41.2	5.2				
M mulatta	(Brownell et al., 1994)	2	Ketamine	PET	36.0	2.0				
M mulatta	(Cross et al., 2000)	6	none	PET	30.8	5.9				
M mulatta	(Palombo et al., 1988)	3	none	Sokoloff	25.5	2.6				
M mulatta	(Palombo et al., 1990)	4	none	Sokoloff	24.5	4.0				
M mulatta	(Porrino et al., 1987)	3	none	Sokoloff	25.0	3.0				
M mulatta	(Noda et al., 2002)	6	none	PET	26.8	5.9			47.8	5.1
M fascic	(Brownell et al., 1994)	1	Ketamine	PET	48.0	4.0				
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	36.0	1.7				
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl, N2O	Sokoloff	68.0	5.0				
Sus	(Miller et al., 1987)	4	Pentobarbital	Microspheres					42.0	7.0
Sus	(Manohar, 1986)	6	none	Microspheres					88.8	3.6
Sus	(Manohar, 1986)	6	Sevoflurane	Microspheres					70.0	4.7
Sus	(Manohar, 1986)	6	Sevoflurane	Microspheres					89.8	7.8
Felis	(Petitjean et al., 1982)	4	awake	Sokoloff	31.5	6.0				
Felis	(Triolo et al., 1990)	6	N2O	Sokoloff	26.2	2.7				

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Table A4, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Meriones	(Choki et al., 1983)	6	none	Sokoloff	44.4	14.4				
Meriones	(Choki et al., 1984)	3	N2O	Sokoloff	56.0	23.0				
Meriones	(Suzuki et al., 1983)	7	Halothane	Sokoloff					137.0	6.4
Meriones	(Van Uitert and Levy, 1978)	10	none	Sokoloff					93.0	5.0
Rattus	(Blin et al., 1991)	10	none	Sokoloff	39.5	6.8				
Rattus	(Bryan and Lehman, 1988)	10	none	Sokoloff	55.0	9.5				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	46.5	0.5				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	45.5	2.5				
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	54.0	7.1				
Rattus	(Davis et al., 1986)	10	none	Sokoloff	67.5	13.3				
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	37.5	3.8				
Rattus	(Eintrei et al., 1999)	6	none	Sokoloff	47.0	3.7				
Rattus	(Freedland et al., 2002)	5	none	Sokoloff	70.6	6.7				
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	35.0	4.5				
Rattus	(Frietsch et al., 2000)	12	Isoflurane	Sokoloff	18.5	4.5				
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	51.0	11.1				
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	59.5	7.9				
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	62.5	15.3				
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	73.7	8.7				
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	39.0	2.4				
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	39.0	3.1				
Rattus	(Lear et al., 1981)	4	none	Sokoloff	55.0	8.0				
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	50.0	8.0				
Rattus	(Lu et al., 1983)	10	none	Sokoloff	61.0	11.1				
Rattus	(MacNeill and Bryan, 1988)	10	none	Sokoloff	79.0	6.3				
Rattus	(Maekawa et al., 1986)	6	none	Sokoloff	34.5	6.0				
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	39.0	8.9				

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Table A4, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	44.0	23.8				
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	64.5	8.6				
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	61.0	2.0				
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	54.0	4.0				
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	55.0	6.7				
Rattus	(Whitlow et al., 2002)	5	none	Sokoloff	70.5	11.2				
Rattus	(Walker et al., 1999)	5	none	Sokoloff	59.2	3.3				
Rattus	(Cremer and Seville, 1983)	8	none	Sokoloff					89.0	17.0
Rattus	(Dahlgren et al., 1981a)	6	none	Sokoloff					144.0	17.1
Mus	(Ito et al., 1994)	5	none	Sokoloff	80.0	8.9				
Mus	(Jay et al., 1985)	8	none	Sokoloff	98.0	14.1				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	93.0	9.0				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	91.0	14.0				

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Table A5. White Matter rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Brooks et al., 1987)	27	none	PET	13.8	2.3				
Homo	(Duara et al., 1983)	18	none	PET	15.2	3.7				
Homo	(Mazziotta et al., 1981)	7	none	PET	29.0	7.7				
Homo	(Tedeschi et al., 1995)	11	none	PET	14.0	2.5				
Homo	(Heiss et al., 1985)	6	none	PET	21.2	1.3				
Homo	(Kuhl et al., 1979)	8	none	PET	20.5	3.9				
Homo	(Kuhl et al., 1982)	25	none	PET	16.7	3.7				
Homo	Present Study	5	none	PET	21.0	2.0				
Homo	(Schapiro et al., 1990)	13	none	PET	12.7	3.5				
Homo	(Redies et al., 1989)	4	none	PET	25.0	1.0				
Homo	(Hatazawa et al., 1988)	7	none	PET	14.8	2.7	85.3	5.8	28.6	2.9
Homo	(Sasaki et al., 1986)	7	none	PET	17.8	1.5	64.6	10.0	19.3	2.2
Homo	(Yamaki et al., 1996)	20	none	PET	27.2	4.4	89.3	13.4	25.0	4.0
Homo	(Leenders et al., 1990)	34	none	PET			63.8	10.7	22.2	4.9
Homo	(Lenzi et al., 1981)	27	none	PET			82.1	15.2	21.2	2.7
Homo	(Pantano et al., 1984)	18	none	PET			84.8	13.4	24.5	4.1
Homo	(Tohgi et al., 1998)	6	none	PET			62.5	11.2	23.0	5.0
Homo	(Wessen et al., 1997)	6	none	PET			66.0	11.0	18.0	2.0
Homo	(Yamaguchi et al., 1986)	14	none	PET			68.8	8.5	21.0	5.4
Homo	(Frackowiak et al., 1988)	7	none	PET			80.8	9.8	21.4	1.9
Homo	(Seitz and Roland, 1992)	26	none	PET					29.6	8.7
Papio	Present Study	2	N2O	PET	14.6	3.1				
M mulatta	(Palombo et al., 1988)	3	none	Sokoloff	11.0	1.0				
M mulatta	(Palombo et al., 1990)	4	none	Sokoloff	11.0	1.0				

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Table A5, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
M mulatta	(Palombo et al., 1991)	4	none	Sokoloff	11.3	1.0				
M mulatta	(Porrino et al., 1987)	3	none	Sokoloff	11.5	0.0				
M mulatta	(Enlund et al., 1997)	4	Ketamine, N2O	PET			128.0	12.0	33.5	8.0
M fascic	(Kennedy et al., 1978)	7	none	Sokoloff	12.0	1.0				
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	18.5	2.5				
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl, N2O	Sokoloff	25.0	3.0			35.0	3.0
Sus	(Miller et al., 1987)	4	Pentobarbital	Microspheres					19.0	5.0
Sus	(Manohar, 1986)	6	none	Microspheres					33.6	1.8
Sus	(Manohar, 1986)	6	Sevoflurane	Microspheres					27.9	1.5
Sus	(Manohar, 1986)	6	Sevoflurane	Microspheres					29.7	2.3
Lepus	(Passero et al., 1981)	5	none	Sokoloff	26.1	2.5				
Felis	(Petitjean et al., 1982)	4	none	Sokoloff	12.0	1.0				
Felis	(Triolo et al., 1990)	6	N2O	Sokoloff	13.5	0.6				
Meriones	(Mies et al., 1984)	2	none	Sokoloff					62.0	12.0
Rattus	(Bassant et al., 1996)	7	none	Sokoloff	28.0	1.0				
Rattus	(Bassant et al., 1996)	8	none	Sokoloff	25.0	3.0				
Rattus	(Bryan et al., 1983)	6	none	Sokoloff	37.5	6.1				
Rattus	(Bryan and Lehman, 1988)	10	none	Sokoloff	43.0	9.5				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	31.0	1.0				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	33.5	3.0				
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	34.5	4.2				

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Table A5, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Dam et al., 1990)	5	none	Sokoloff	23.5	8.9				
Rattus	(Davis et al., 1986)	10	none	Sokoloff	50.5	12.3				
Rattus	(Dow-Edwards et al., 1986)	4	none	Sokoloff	29.5	3.0				
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	26.5	2.1				
Rattus	(Eintrei et al., 1999)	6	none	Sokoloff	37.0	2.4				
Rattus	(Freedland et al., 2002)	5	none	Sokoloff	31.3	2.2				
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	26.5	4.5				
Rattus	(Frietsch et al., 2000)	12	Isoflurane	Sokoloff	13.0	5.0				
Rattus	(Furlow et al., 1983)	5	none	Sokoloff	31.0	8.0				
Rattus	(Grome and McCulloch, 1981)	16	none	Sokoloff	32.0	3.0				
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	32.0	4.7				
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	37.0	4.0				
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	36.5	8.7				
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	53.0	9.5				
Rattus	(Kadekaro et al., 1983)	7	none	Sokoloff	31.0	2.6				
Rattus	(Lear et al., 1981)	4	none	Sokoloff	33.0	6.0			35.0	4.0
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	23.0	3.5				
Rattus	(Lu et al., 1983)	10	none	Sokoloff	43.0	9.5				
Rattus	(MacNeill and Bryan, 1988)	10	none	Sokoloff	51.0	6.3				
Rattus	(Maekawa et al., 1986)	6	none	Sokoloff	34.5	8.5				
Rattus	(Nehlig et al., 1985)	6	none	Sokoloff	30.0	3.7				
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	31.0	6.7				
Rattus	(Otori et al., 2003)	6	none	Sokoloff	27.7	7.4				
Rattus	(Poindessous-Jazat et al., 1998)	7	none	Sokoloff	30.0	5.0				
Rattus	(Poindessous-Jazat et al., 1998)	5	none	Sokoloff	23.0	3.0				
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	27.5	5.3				
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	30.0	3.7				
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	36.0	2.0				

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Table A5, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Walker et al., 1999)	5	none	Sokoloff	46.5	4.9				
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	22.0	4.5				
Rattus	(Whitlow et al., 2002)	5	none	Sokoloff	33.2	4.5				
Mus	(Bouilleret et al., 2000)	4	none	Sokoloff	39.5	10.5				
Mus	(Jay et al., 1985)	8	none	Sokoloff	28.5	2.0				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	66.0	9.0				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	63.0	12.0				
Mus	(Nowaczyk and Des Rosiers, 1981)	4	none	Sokoloff	22.0	2.5				
Mus	(Yamaguchi et al., 1992b)	6	none	Sokoloff	26.5	1.9				

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Table A6. Neocortex rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Alkire et al., 1995)	6	none	PET	34.4	9.8				
Homo	(Azari et al., 1994)	15	none	PET	37.5	5.1				
Homo	(Baxter et al., 1987)	7	none	PET	26.0	2.4				
Homo	(Baxter et al., 1987)	7	none	PET	32.6	4.7				
Homo	(Bokde et al., 2001)	18	none	PET	42.6	9.5				
Homo	(Bromberg et al., 1990)	12	none	PET	34.1	7.5				
Homo	(Brooks et al., 1987)	27	none	PET	36.4	6.7				
Homo	(Buchsbaum et al., 1989)	10	none	PET	21.4	6.3				
Homo	(Chugani et al., 1987)	7	none	PET	26.3	3.3				
Homo	(Clark et al., 2001)	32	none	PET	48.8	8.3				
Homo	(Delvenne et al., 1997)	10	none	PET	29.7	1.8				
Homo	(Delvenne et al., 1999)	10	none	PET	29.6	1.9				
Homo	(Duara et al., 1983)	18	none	PET	28.4	6.6				
Homo	(Foster et al., 1984)	8	none	PET	40.2	7.6				
Homo	(Fukuyama et al., 1994)	9	none	PET	36.6	3.8				
Homo	(Gilman et al., 1990)	13	none	PET	30.8	5.1				
Homo	(Gilman et al., 1990)	10	none	PET	34.1	6.7				
Homo	(Hasselbalch et al., 1994)	9	none	PET	29.4	5.4				
Homo	(Haxby et al., 1985)	26	none	PET	27.8	7.2				
Homo	(Heiss et al., 1985)	6	none	PET	41.1	2.7				
Homo	(Hirono et al., 2000)	15	none	PET	40.7	6.3				
Homo	(Hu et al., 2000)	9	none	PET	48.0	7.7				
Homo	(Ishii et al., 1998)	22	none	PET	40.9	5.1				
Homo	(Kaye et al., 1990)	12	none	PET	31.8	5.0				
Homo	(Kuhl et al., 1979)	8	none	PET	37.8	4.3				
Homo	(Kuhl et al., 1982)	25	none	PET	36.6	8.3				

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Table A6, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Ludolph et al., 1992)	12	none	PET	44.1	9.0				
Homo	(Martinot et al., 1990)	8	none	PET	36.1	6.8				
Homo	(Mazziotta et al., 1981)	7	none	PET	42.8	10.7				
Homo	(McNamara et al., 1987)	14	none	PET	33.7	7.0				
Homo	(Perani et al., 1995)	15	none	PET	36.4	4.3				
Homo	(Phillips et al., 1991)	14	none	PET	37.1	7.2				
Homo	(Redies et al., 1989)	4	none	PET	30.0	4.0				
Homo	(Sachs et al., 1986)	25	none	PET	45.1	6.2				
Homo	(Schapiro et al., 1990)	13	none	PET	49.6	7.3				
Homo	(Tedeschi et al., 1995)	11	none	PET	33.4	2.6				
Homo	(Tommasino et al., 1995)	10	none	PET	37.5	5.0				
Homo	(Volkow et al., 1992)	11	none	PET	38.4	4.5				
Homo	(Wolkin et al., 1988)	8	none	PET	40.9	5.7				
Homo	(Wu et al., 1992)	15	none	PET	24.0	6.6				
Homo	(Yoshii et al., 1988)	30	none	PET	42.9	12.1				
Homo	(Yoshii et al., 1988)	46	none	PET	37.3	8.3				
Homo	Present Study	5	none	PET	34.7	5.2				
Homo	(De Volder et al., 1997)	3	none	PET	37.0	6.4	157.0	24.4	48.9	4.8
Homo	(Frackowiak et al., 1988)	7	none	PET	27.4	2.4	153.0	14.5	43.9	7.4
Homo	(Hatazawa et al., 1988)	7	none	PET	28.7	5.4	139.7	25.9	44.2	4.2
Homo	(Yamaki et al., 1996)	20	none	PET	38.9	4.4	147.3	22.0	43.0	7.0
Homo	(Shishido et al., 1996)	7	none	PET	33.5	4.3	147.0	13.3	44.9	3.9
Homo	(Sasaki et al., 1986)	7	none	PET	35.6	3.8	149.6	16.4	42.2	4.1
Homo	(Yamaguchi et al., 1986)	14	none	PET			154.5	17.9	42.6	7.8
Homo	(Yamaguchi et al., 1992a)	10	none	PET			146.9	22.3	47.5	6.9
Homo	(Leenders et al., 1990)	34	none	PET			138.8	13.8	42.8	7.0
Homo	(Hayashi et al., 2000)	16	none	PET			147.3	17.9	46.1	8.6
Homo	(Perlmutter et al., 1987)	32	none	PET			122.0	30.6		

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Table A6, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Tohgi et al., 1998)	6	none	PET			145.1	20.1		
Homo	(Wessen et al., 1997)	6	none	PET			211.4	33.4		
Papio	(MacKenzie et al., 1976)	6	Phencyclidine, N2O	Kety-Schmidt	33.1	nr	145.0	nr	52.3	
Papio	Present Study	2	N2O	PET	49.0	6.4				
M mulatta	(Cross et al., 2000)	6	none	PET	37.2	7.9				
M mulatta	(Palombo et al., 1991)	4	none	PET	51.1	3.0				
M mulatta	(Enlund et al., 1997)	4	Ketamine, N2O	PET			218.9	23.5	58.5	10.4
M mulatta	(Noda et al., 2002)	6	none	PET					53.0	7.5
M fascic	(Kennedy et al., 1978)	7	none	Sokoloff	55.8	3.0				
M fascic	(Kuge et al., 2001)	4	Sevoflurane, N2O	PET					34.0	
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl, N2O	Sokoloff					114.0	
Sus	(Miller et al., 1987)	4	Pentobarbital	Microspheres					29.0	5.0
Sus	(Manohar, 1986)	6					342.0	20.1	108.2	6.3
Sus	(Manohar, 1986)	6	Sevoflurane				171.9	7.6	68.4	3.5
Sus	(Manohar, 1986)	6	Sevoflurane				162.5	19.2	77.7	8.7
Lepus	(Passero et al., 1981)	5	none	Sokoloff	77.4	8.0				
Lepus	(Drummond et al., 1986)	12	Morphine						55.0	12.0
Lepus	(Kaieda et al., 1989)	8	Halothane						69.0	23.0
Lepus	(Kaieda et al., 1989)	8	Isoflurane						41.0	16.0
Felis	Hovda	7	none	Sokoloff	77.6	12.4				
Felis	Petitjean 82	4	none	Sokoloff	47.0	11.8				

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Table A6, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Felis	Triolo 90	6	N2O	Sokoloff	40.8	4.2				
Gerbil	(Mies et al., 1984)	2	none	Sokoloff	97.4	8.0			135.6	6.8
Gerbil	(Paschen et al., 1992)	5	none	Sokoloff	70.0	nr			90.0	nr
Gerbil	(Suzuki et al., 1983)	7	Halothane	Sokoloff					137.0	7.0
Gerbil	(Mies et al., 1990)	4	Halothane	Sokoloff					131.5	10.5
Rattus	(Bassant et al., 1996)	7	none	Sokoloff	98.2	8.0				
Rattus	(Bassant et al., 1996)	8	none	Sokoloff	87.2	9.0				
Rattus	(Blin et al., 1991)	10	none	Sokoloff	79.4	9.6				
Rattus	(Borgstrom et al., 1976)	7	none	Sokoloff	77.0	15.9				
Rattus	(Bryan et al., 1983)	6	none	Sokoloff	81.0	9.3				
Rattus	(Bryan and Lehman, 1988)	10	none	Sokoloff	92.5	7.9				
Rattus	(Chang et al., 2001)	8	none	Sokoloff	96.0	21.5				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	115.3	2.0				
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	111.6	3.6				
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	95.8	9.3				
Rattus	(Crosby et al., 1984)	5	none	Sokoloff	96.7	7.4				
Rattus	(Crosby et al., 1984)	5	N2O	Sokoloff	103.3	9.6				
Rattus	(Cudennec et al., 1997)	7	none	Sokoloff	123.8	15.3				
Rattus	(Dam et al., 1990)	5	none	Sokoloff	99.0	18.3				
Rattus	(Davis et al., 1986)	10	none	Sokoloff	93.8	14.2				
Rattus	(Dow-Edwards et al., 1986)	4	none	Sokoloff	120.8	11.6				
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	91.1	6.2				
Rattus	(Eintrei et al., 1999)	6	none	Sokoloff	106.0	9.1				
Rattus	(Fornai and Orzi, 2001)	5	none	Sokoloff	89.9	9.4				
Rattus	(Freedland et al., 2002)	5	none	Sokoloff	120.0	10.5				
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	95.8	10.8				

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Table A6, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Frietsch et al., 2000)	12	Isoflurane	Sokoloff	38.8	6.2				
Rattus	(Furlow et al., 1983)	5	none	Sokoloff	80.3	19.3				
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	96.8	10.4				
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	122.2	18.5				
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	113.5	21.8				
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	100.0	8.7				
Rattus	(Ingvar et al., 1980)	10	N2O	Sokoloff	72.6	9.5				
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	70.2	5.4				
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	69.3	10.1				
Rattus	(Kadekaro et al., 1983)	7	none	Sokoloff	112.0	8.7				
Rattus	(Katsumata et al., 2001)	5	none	Sokoloff	89.2	11.1				
Rattus	(Knapp et al., 2002)	8	none	Sokoloff	97.0					
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	85.8	9.3				
Rattus	(Lu et al., 1983)	10	none	Sokoloff	93.5	8.9				
Rattus	(MacNeill and Bryan, 1988)	10	none	Sokoloff	98.0	7.3				
Rattus	(Maekawa et al., 1986)	6	none	Sokoloff	81.5	8.5				
Rattus	(Moore et al., 2000)	6	none	PET	77.0	9.9				
Rattus	(Moore et al., 2000)	6	none	PET	63.1	11.5				
Rattus	(Nehlig et al., 1985)	6	none	Sokoloff	118.7	13.0				
Rattus	(Ori et al., 1986)	6	none	Sokoloff	96.8	12.2				
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	99.5	14.1				
Rattus	(Otori et al., 2003)	6	none	Sokoloff	75.4	20.2				
Rattus	(Ouchi et al., 1996)	5	Pentobarbital	PET	65.7	1.1				
Rattus	(Poindessous-Jazat et al., 1998)	7	none	Sokoloff	103.1	8.3				
Rattus	(Poindessous-Jazat et al., 1998)	5	none	Sokoloff	91.0	13.1				
Rattus	(Sakurada et al., 1978)	5	N2O	Sokoloff	86.4	3.4				
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	91.4	13.0				
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	115.3	14.5				

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Table A6, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	124.4	3.6				
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	93.5	13.6				
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	124.0	20.8				
Rattus	(Grome and McCulloch, 1981)	16	none	Sokoloff	98.3	5.8				
Rattus	(Whitlow et al., 2002)	5	none	Sokoloff	123.0	7.4				
Rattus	(Walker et al., 1999)	5	none	Sokoloff	117.9	6.1				
Rattus	(Lear et al., 1981)	4	none	Sokoloff	115.0	23.0			124.5	26.0
Rattus	(Mies et al., 1981)	6	Halothane	Sokoloff	67.2	5.1			77.5	4.7
Rattus	(Baughman et al., 1990)	20	none	Kety-Schmidt			464.3	93.8	130.5	53.7
Rattus	(Gjedde et al., 1977)	10	N2O	Kety-Schmidt			263.4	26.8	145.0	10.0
Rattus	(Naughton et al., 1985)	7	N2O	Kety-Schmidt			452.2	8.9	170.0	4.0
Rattus	(Norberg and Siesjo, 1974)	6	N2O	Kety-Schmidt			455.4	32.6	100.0	7.3
Rattus	(Cremer and Seville, 1983)	8	none	Kety-Schmidt					193.8	36.8
Mus	(Bouilleret et al., 2000)	4	none	Sokoloff	116.7	10.3				
Mus	(Ito et al., 1994)	5	none	Sokoloff	94.0	13.4				
Mus	(Jay et al., 1985)	8	none	Sokoloff	92.6	13.6				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	132.8	11.6				
Mus	(Niwa et al., 2002)	6	none	Sokoloff	130.2	17.0				
Mus	(Nowaczyk and Des Rosiers, 1981)	4	none	Sokoloff	63.0	11.0				
Mus	(Yamaguchi et al., 1992b)	6	none	Sokoloff	74.2	9.1				
Mus	(Zocchi et al., 2001)	4	none	Sokoloff	88.8	5.6				
Mus	(Zocchi et al., 2001)	6	none	Sokoloff	65.0	6.2				

Table A7. Frontal Cortex rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Alkire et al., 1995)	6	none	PET	36.5	10.1		
Homo	(Baxter et al., 1987)	7	none	PET	25.3	2.4		
Homo	(Baxter et al., 1987)	7	none	PET	32.1	5.1		
Homo	(Bokde et al., 2001)	18	none	PET	42.7	12.6		
Homo	(Brooks et al., 1987)	27	none	PET	39.9	8.1		
Homo	(Buchsbaum et al., 1989)	10	none	PET	23.5	6.5		
Homo	(Chugani et al., 1987)	7	none	PET	27.1	3.3		
Homo	(Clark et al., 2001)	32	none	PET	49.2	8.3		
Homo	(Delvenne et al., 1997)	10	none	PET	29.9	2.0		
Homo	(Delvenne et al., 1999)	10	none	PET	29.9	2.0		
Homo	(Duara et al., 1983)	18	none	PET	30.4	8.3		
Homo	(Hasselbalch et al., 1994)	9	none	PET	27.9	5.2		
Homo	(Haxby et al., 1985)	26	none	PET	28.3	7.2		
Homo	(Heiss et al., 1985)	6	none	PET	42.7	2.0		
Homo	(Hirono et al., 2000)	15	none	PET	41.7	6.5		
Homo	(Hu et al., 2000)	9	none	PET	41.7	5.7		
Homo	(Kaye et al., 1990)	12	none	PET	32.4	5.7		
Homo	(Kuhl et al., 1979)	8	none	PET	37.7	5.1		
Homo	(Kuhl et al., 1982)	25	none	PET	35.8	7.9		
Homo	(Ludolph et al., 1992)	12	none	PET	45.3	9.7		
Homo	(Martinot et al., 1990)	8	none	PET	37.7	7.8		
Homo	(Mazziotta et al., 1981)	7	none	PET	43.0	10.5		
Homo	(McNamara et al., 1987)	14	none	PET	36.6	8.3		
Homo	(Perani et al., 1995)	15	none	PET	40.0	5.5		
Homo	(Phillips et al., 1991)	14	none	PET	36.9	6.1		
Homo	(Redies et al., 1989)	4	none	PET	28.0	4.0		
Homo	(Rudolf et al., 2000)	16	none	PET	31.7	nr		
Homo	(Sachs et al., 1986)	25	none	PET	46.0	6.7		
Homo	(Schapiro et al., 1990)	13	none	PET	49.7	8.2		
Homo	(Tedeschi et al., 1995)	11	none	PET	32.9	2.5		
Homo	(Tommasino et al., 1995)	10	none	PET	38.2	4.7		
Homo	(Volkow et al., 1992)	11	none	PET	34.6	4.0		
Homo	(Wolkin et al., 1988)	8	none	PET	40.9	5.7		
Homo	(Yoshii et al., 1988)	30	none	PET	44.4	12.8		
Homo	(Yoshii et al., 1988)	46	none	PET	36.6	8.3		
Homo	Present Study	5	none	PET	38.2	7.6		
Homo	(De Volder et al., 1997)	3	none	PET	37.3	8.6	150.3	27.3
Homo	(Fukuyama et al., 1994)	9	none	PET	36.1	4.0	105.7	25.7
Homo	(Sasaki et al., 1986)	7	none	PET	35.6	3.9	136.1	18.4
Homo	Yamaki 96	20	none	PET	28.3	2.8	120.5	17.9
Homo	(Burns and Tyrrell, 1992)	4	none	PET			136.6	2.7
Homo	(Ishii et al., 1996b)	15	none	PET			160.3	26.8
Homo	(Ishii et al., 1996b)	14	none	PET			158.0	20.5
Homo	(Kaisti et al., 2003)	8	none	PET			195.5	20.1
Homo	(Kaisti et al., 2003)	8	Sevoflurane	PET			139.3	21.4

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Table A7, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Langsjo et al., 2003)	10	none	PET			160.7	19.6
Homo	(Leenders et al., 1990)	34	none	PET			141.5	20.1
Homo	(Lenzi et al., 1981)	27	none	PET			185.9	nr
Homo	(Meyer et al., 1987)	3	none	PET			123.0	nr
Homo	(Pantano et al., 1984)	18	none	PET			174.1	34.4
Homo	(Perlmutter et al., 1987)	32	none	PET			125.2	31.3
Homo	(Tohgi et al., 1998)	6	none	PET			133.9	17.9
Homo	(Wessen et al., 1997)	6	none	PET			209.5	25.0
Homo	(Yamaguchi et al., 1986)	14	none	PET			138.2	21.0
Homo	(Yamaguchi et al., 1992a)	10	none	PET			133.9	25.5
Papio	(Kiyosawa et al., 1989)	9	Phencylidine	PET	28.3	5.0		
Papio	(Meguro et al., 1999)	8	Phencylidine	PET	50.4	10.0		
Papio	(Millien et al., 2002)	10	Phencylidine	PET	49.4	13.0		
Papio	(Yamaguchi et al., 1990)	4	Phencylidine	PET	30.0	4.6		
Papio	Present Study	2	N2O	PET	50.2	12.0		
M mulatta	(Brownell et al., 1994)	2	Ketamine	PET	52.0	3.0		
M mulatta	(Cross et al., 2000)	6	none	PET	34.0	7.5		
M mulatta	(Noda et al., 2002)	6	none	PET	34.1	8.2		
M mulatta	(Palombo et al., 1988)	3	none	Sokoloff	56.7	7.0		
M mulatta	(Palombo et al., 1990)	4	none	Sokoloff	54.7	8.0		
M mulatta	(Porrino et al., 1987)	3	none	Sokoloff	59.0	4.0		
M mulatta	(Enlund et al., 1997)	4	Ketamine	PET			195.0	13.0
M fascic	(Brownell et al., 1994)	1	Ketamine	PET	58.0	2.0		
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	49.0	1.7		
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl, N2O	Sokoloff	70.0	10.0		
Felis	(Petitjean et al., 1982)	4	none	Sokoloff	49.0	16.0		
Felis	(Triolo et al., 1990)	6	N2O	Sokoloff	43.1	2.9		
Meriones	(Choki et al., 1983)	6	none	Sokoloff	67.8	15.0		
Rattus	(Bryan et al., 1983)	6	none	Sokoloff	87.0	12.2		
Rattus	(Chang et al., 2001)	8	none	Sokoloff	91.0	17.0		
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	93.0	5.7		
Rattus	(Cudennec et al., 1997)	7	none	Sokoloff	111.0	18.5		
Rattus	(Dam et al., 1990)	5	none	Sokoloff	103.0	22.4		
Rattus	(Davis et al., 1986)	10	none	Sokoloff	87.0	12.6		
Rattus	(Dow-Edwards et al., 1986)	4	none	Sokoloff	105.0	4.0		
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	90.0	3.0		
Rattus	(Eintrei et al., 1999)	6	none	Sokoloff	94.0	7.3		
Rattus	(Fornai and Orzi, 2001)	5	none	Sokoloff	105.0	10.0		
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	89.0	6.0		

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Table A7, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Rattus	(Frietsch et al., 2000)	12	Isoflurane	Sokoloff	34.0	5.0		
Rattus	(Furlow et al., 1983)	5	none	Sokoloff	83.0	21.0		
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	88.0	3.2		
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	111.0	18.5		
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	106.0	21.8		
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	94.0	10.5		
Rattus	(Ingvar et al., 1980)	10	N2O	Sokoloff	71.0	9.5		
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	67.0	2.4		
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	66.0	9.5		
Rattus	(Kadekaro et al., 1983)	7	none	Sokoloff	108.0	7.9		
Rattus	(Katsumata et al., 2001)	5	none	Sokoloff	83.9	11.5		
Rattus	(Knapp et al., 2002)	8	none	Sokoloff	89.2	7.6		
Rattus	(Lear et al., 1981)	4	none	Sokoloff	108.0	20.0		
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	80.0	9.0		
Rattus	(Lu et al., 1983)	10	none	Sokoloff	98.0	6.3		
Rattus	(Maekawa et al., 1986)	6	none	Sokoloff	77.0	7.0		
Rattus	(Moore et al., 2000)	6	none	Sokoloff	79.0	9.7		
Rattus	(Moore et al., 2000)	6	none	Sokoloff	65.1	10.6		
Rattus	(Nehlig et al., 1985)	6	none	Sokoloff	110.0	12.2		
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	83.0	8.9		
Rattus	(Otori et al., 2003)	6	none	Sokoloff	73.6	18.6		
Rattus	(Sakurada et al., 1978)	5	N2O	Sokoloff	87.0	2.0		
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	87.0	13.2		
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	111.0	14.7		
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	112.0	4.0		
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	87.5	12.0		
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	109.0	22.4		
Rattus	(Walker et al., 1999)	5	none	Sokoloff	115.9	5.9		
Mus	(Bouilleret et al., 2000)	4	none	Sokoloff	107.0	10.0		
Mus	(Jay et al., 1985)	8	none	Sokoloff	89.0	11.3		
Mus	(Niwa et al., 2002)	6	none	Sokoloff	129.0	11.0		
Mus	(Niwa et al., 2002)	6	none	Sokoloff	127.0	19.0		
Mus	(Yamaguchi et al., 1992b)	6	none	Sokoloff	69.6	7.3		
Mus	(Zocchi et al., 2001)	4	none	Sokoloff	117.0	8.0		
Mus	(Zocchi et al., 2001)	6	none	Sokoloff	84.0	8.0		

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Table A8. Parietal Cortex rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Alkire et al., 1995)	6	none	PET	37.2	11.1		
Homo	(Baxter et al., 1987)	7	none	PET	23.0	1.9		
Homo	(Baxter et al., 1987)	7	none	PET	29.3	5.1		
Homo	(Bokde et al., 2001)	18	none	PET	45.0	7.6		
Homo	(Brooks et al., 1987)	27	none	PET	36.1	7.0		
Homo	(Buchsbaum et al., 1989)	10	none	PET	23.2	6.5		
Homo	(Chugani et al., 1987)	7	none	PET	24.2	3.7		
Homo	(Clark et al., 2001)	32	none	PET	51.5	8.9		
Homo	(Delvenne et al., 1997)	10	none	PET	32.3	2.4		
Homo	(Delvenne et al., 1997)	10	none	PET	32.2	2.5		
Homo	(Duara et al., 1983)	18	none	PET	31.0	7.0		
Homo	(Hasselbalch et al., 1994)	9	none	PET	26.6	5.7		
Homo	(Haxby et al., 1985)	26	none	PET	28.1	7.7		
Homo	(Heiss et al., 1985)	6	none	PET	38.5	2.2		
Homo	(Hirono et al., 2000)	15	none	PET	41.2	5.9		
Homo	(Hu et al., 2000)	9	none	PET	58.5	7.6		
Homo	(Ishii et al., 1998)	22	none	PET	42.6	5.2		
Homo	(Kaye et al., 1990)	12	none	PET	31.4	5.2		
Homo	(Kuhl et al., 1979)	8	none	PET	35.7	4.6		
Homo	(Kuhl et al., 1982)	25	none	PET	35.7	7.6		
Homo	(Mazziotta et al., 1981)	7	none	PET	43.1	9.9		
Homo	(McNamara et al., 1987)	14	none	PET	32.1	7.0		
Homo	(Perani et al., 1995)	15	none	PET	37.7	4.6		
Homo	(Phillips et al., 1991)	14	none	PET	35.2	7.2		
Homo	(Rudolf et al., 2000)	16	none	PET	32.1	nr		
Homo	(Sachs et al., 1986)	25	none	PET	43.4	7.0		
Homo	(Schapiro et al., 1990)	13	none	PET	51.6	7.8		
Homo	(Tedeschi et al., 1995)	11	none	PET	33.1	2.6		
Homo	(Tommasino et al., 1995)	10	none	PET	37.7	4.1		
Homo	(Volkow et al., 1992)	11	none	PET	34.5	4.0		
Homo	(Yoshii et al., 1988)	30	none	PET	42.2	12.2		
Homo	(Yoshii et al., 1988)	46	none	PET	37.2	8.3		
Homo	Present Study	5	none	PET	32.6	7.7		
Homo	(Fukuyama et al., 1994)	9	none	PET	36.7	3.7	115.5	26.2
Homo	(Sasaki et al., 1986)	7	none	PET	36.1	4.4	164.3	25.3
Homo	(Yamaki et al., 1996)	20	none	PET	28.0	2.8	133.9	22.3
Homo	(Burns and Tyrrell, 1992)	4	none	PET			136.2	4.9
Homo	(Ishii et al., 1996b)	15	none	PET			155.8	28.6
Homo	(Ishii et al., 1996a)	14	none	PET			155.4	27.7
Homo	(Kaisti et al., 2003)	8	none	PET			205.4	25.9
Homo	(Kaisti et al., 2003)	8	Sevoflurane	PET			141.5	13.0
Homo	(Langsjo et al., 2003)	10	none	PET			168.8	23.2
Homo	(Leenders et al., 1990)	34	none	PET			124.1	14.3

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Table A8, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Lenzi et al., 1981)	27	none	PET			171.7	nr
Homo	(Meyer et al., 1987)	3	none	PET			167.0	nr
Homo	(Perlmutter et al., 1987)	32	none	PET			129.5	31.7
Homo	(Tohgi et al., 1998)	6	none	PET			147.3	26.8
Homo	(Wessen et al., 1997)	6	none	PET			213.0	35.0
Homo	(Yamaguchi et al., 1986)	14	none	PET			155.8	21.7
Homo	(Yamaguchi et al., 1992a)	10	none	PET			140.6	31.7
Papio	(Kiyosawa et al., 1989)	9	Phencyclidine	PET	26.6	5.0		
Papio	(Meguro et al., 1999)	8	Phencyclidine	PET	43.8	8.3		
Papio	(Millien et al., 2002)	10	Phencyclidine	PET	39.1	13.6		
Papio	Present Study	2	N2O	PET	45.2	9.0		
M mulatta	(Cross et al., 2000)	6	none	PET	37.2	7.8		
M mulatta	(Enlund et al., 1997)	4	Ketamine	PET			191.5	36.0
M fascic	(Deuel and Collins, 1984)	4	none	Sokoloff	48.3			
M fascic	(Kuge et al., 2001)	4	Sevoflurane	PET	28.9	7.4		
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl	Sokoloff	67.0	2.0		
Lepus	(Passero et al., 1981)	5	none	Sokoloff	73.1	17.4		
Felis	(Petitjean et al., 1982)	4	none	Sokoloff	45.0	16.0		
Felis	(Triolo et al., 1990)	6	N2O	Sokoloff	43.9	4.4		
Meriones	(Choki et al., 1984)	3	N2O	Sokoloff	107.0	24.0		
Rattus	(Bassant et al., 1996)	7	none	Sokoloff	97.0	8.0		
Rattus	(Bassant et al., 1996)	8	none	Sokoloff	89.0	9.0		
Rattus	(Blin et al., 1991)	10	none	Sokoloff	76.4	10.0		
Rattus	(Bryan et al., 1983)	6	none	Sokoloff	77.0	7.3		
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	90.0	5.7		
Rattus	(Cudennec et al., 1997)	7	none	Sokoloff	112.0	13.2		
Rattus	(Davis et al., 1986)	10	none	Sokoloff	96.0	12.6		
Rattus	(Dow-Edwards et al., 1986)	4	none	Sokoloff	110.0	16.0		
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	87.0	7.2		
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	85.0	8.0		
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	39.0	7.0		
Rattus	(Furlow et al., 1983)	5	none	Sokoloff	76.0	17.0		
Rattus	(Grome and McCulloch, 1981)	16	none	Sokoloff	86.0	4.0		
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	89.0	12.6		
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	108.0	23.8		
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	123.0	21.8		
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	105.0	7.2		
Rattus	(Ingvar et al., 1980)	10	N2O	Sokoloff	69.0	6.3		

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Table A8, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	68.0	4.9		
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	65.0	9.5		
Rattus	(Katsumata et al., 2001)	5	none	Sokoloff	90.8	13.3		
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	76.0	8.0		
Rattus	(Lu et al., 1983)	10	none	Sokoloff	96.0	9.5		
Rattus	(Moore et al., 2000)	6	none	Sokoloff	74.9	10.1		
Rattus	(Moore et al., 2000)	6	none	Sokoloff	61.1	12.4		
Rattus	(Nehlig et al., 1985)	6	none	Sokoloff	106.0	12.2		
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	92.0	13.4		
Rattus	(Otori et al., 2003)	6	none	Sokoloff	73.4	17.9		
Rattus	(Poindessous-Jazat et al., 1998)	7	none	Sokoloff	104.0	8.7		
Rattus	(Poindessous-Jazat et al., 1998)	5	none	Sokoloff	94.3	16.7		
Rattus	(Sakurada et al., 1978)	5	N2O	Sokoloff	79.0	2.0		
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	93.0	15.9		
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	97.0	7.3		
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	110.0	3.0		
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	92.8	12.0		
Rattus	(Walker et al., 1999)	5	none	Sokoloff	109.9	4.2		
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	109.0	20.1		
Mus	(Niwa et al., 2002)	6	none	Sokoloff	128.0	15.0		
Mus	(Niwa et al., 2002)	6	none	Sokoloff	124.0	14.0		
Mus	(Yamaguchi et al., 1992b)	6	none	Sokoloff	71.0	11.8		

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Table A9. Temporal Cortex rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Alkire et al., 1995)	6	none	PET	30.1	8.3		
Homo	(Baxter et al., 1987)	7	none	PET	23.2	2.3		
Homo	(Baxter et al., 1987)	7	none	PET	28.6	0.6		
Homo	(Brooks et al., 1987)	27	none	PET	36.1	6.6		
Homo	(Buchsbaum et al., 1989)	10	none	PET	17.6	4.6		
Homo	(Chugani et al., 1987)	7	none	PET	24.0	3.0		
Homo	(Clark et al., 2001)	32	none	PET	45.7	7.8		
Homo	(Delvenne et al., 1997)	10	none	PET	23.8	1.6		
Homo	(Delvenne et al., 1999)	10	none	PET	30.1	nr		
Homo	(Duara et al., 1983)	18	none	PET	23.1	6.0		
Homo	(Hasselbalch et al., 1994)	9	none	PET	28.9	5.2		
Homo	(Haxby et al., 1985)	26	none	PET	24.0	5.9		
Homo	(Heiss et al., 1985)	6	none	PET	38.5	2.7		
Homo	(Hirono et al., 2000)	15	none	PET	36.4	5.5		
Homo	(Hu et al., 2000)	9	none	PET	42.1	7.1		
Homo	(Ishii et al., 1998)	22	none	PET	39.1	4.9		
Homo	(Kaye et al., 1990)	12	none	PET	31.6	4.1		
Homo	(Kuhl et al., 1979)	8	none	PET	38.0	2.5		
Homo	(Kuhl et al., 1982)	25	none	PET	36.9	8.3		
Homo	(Martinot et al., 1990)	8	none	PET	32.8	5.4		
Homo	(Mazziotta et al., 1981)	7	none	PET	44.1	11.6		
Homo	(Perani et al., 1995)	15	none	PET	31.6	4.6		
Homo	(Phillips et al., 1991)	14	none	PET	32.5	5.6		
Homo	(Redies et al., 1989)	4	none	PET	32.0	6.0		
Homo	(Rudolf et al., 2000)	16	none	PET	27.2	nr		
Homo	(Sachs et al., 1986)	25	none	PET	44.1	5.0		
Homo	(Schapiro et al., 1990)	13	none	PET	46.0	6.6		
Homo	(Tedeschi et al., 1995)	11	none	PET	33.2	3.0		
Homo	(Tommasino et al., 1995)	10	none	PET	30.3	4.6		
Homo	(Volkow et al., 1992)	11	none	PET	35.8	4.0		
Homo	(Yoshii et al., 1988)	30	none	PET	40.0	10.5		
Homo	(Yoshii et al., 1988)	46	none	PET	34.4	7.2		
Homo	Present Study	5	none	PET	31.0	7.0		
Homo	(De Volder et al., 1997)	3	none	PET	36.4	6.1	153.3	25.4
Homo	(Fukuyama et al., 1994)	9	none	PET	37.5	3.2	117.7	31.6
Homo	(Sasaki et al., 1986)	7	none	PET	35.3	3.5	161.9	18.4
Homo	(Yamaki et al., 1996)	20	none	PET	27.8	2.2	129.5	17.9
Homo	(Burns and Tyrrell, 1992)	4	none	PET			139.7	6.3
Homo	(Ishii et al., 1996a)	15	none	PET			169.2	32.1
Homo	(Ishii et al., 1996b)	14	none	PET			141.5	20.5
Homo	(Kaisti et al., 2003)	8	none	PET			193.3	24.1
Homo	(Kaisti et al., 2003)	8	Sevoflurane	PET			143.3	19.2
Homo	(Langsjo et al., 2003)	10	none	PET			167.4	17.4

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Table A9, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Perlmutter et al., 1987)	32	none	PET			129.0	32.6
Homo	(Tohgi et al., 1998)	6	none	PET			147.3	17.9
Homo	(Wessen et al., 1997)	6	none	PET			213.0	35.0
Homo	(Yamaguchi et al., 1986)	14	none	PET			151.8	20.6
Homo	(Yamaguchi et al., 1992a)	10	none	PET			157.1	22.8
Papio	(Kiyosawa et al., 1989)	9	Phencyclidine	PET	30.5	5.0		
Papio	(Meguro et al., 1999)	8	Phencyclidine	PET	43.5	9.4		
Papio	(Millien et al., 2002)	10	Phencyclidine	PET	42.0	10.4		
Papio	(Yamaguchi et al., 1990)	4	Phencyclidine	PET	27.4	5.2		
Papio	Present Study	2	N2O	PET	45.6	6.8		
M mulatta	(Cross et al., 2000)	6	none	PET	35.2	7.2		
M mulatta	(Noda et al., 2002)	6	none	PET	36.0	7.3		
M mulatta	(Enlund et al., 1997)	4	Ketamine	PET			241.5	29.5
M fascic	(Kuge et al., 2001)	4	Sevoflurane	PET	34.7	11.5		
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	38.0	3.5		
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl, N2O	Sokoloff	57.0	4.0		
Felis	(Petitjean et al., 1982)	4	none	Sokoloff	46.0	16.0		
Felis	(Triolo et al., 1990)	6	N2O	Sokoloff	39.2	4.9		
Rattus	(Bassant et al., 1996)	7	none	Sokoloff	112.0	7.0		
Rattus	(Bassant et al., 1996)	8	none	Sokoloff	86.0	11.0		
Rattus	(Otori et al., 2003)	6	none	Sokoloff	80.3	23.7		
Rattus	(Walker et al., 1999)	5	none	Sokoloff	116.0	4.6		

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Table A10. Auditory Cortex rCMR

Species	Reference	n	Anesthesia	Mode	Glucose	SD
Homo	(Martinot et al., 1990)	8	none	PET	38.9	5.9
Homo	(Phillips et al., 1991)	14	none	PET	37.2	7.2
Homo	(Schapiro et al., 1990)	13	none	PET	48.2	5.2
Homo	(Tommasino et al., 1995)	10	none	PET	36.6	4.4
M fascic	(Schwartzman et al., 1988)	6	none	Sokoloff	70.3	12.2
Meriones	(Choki et al., 1984)	3	none	Sokoloff	79.0	12.0
Rattus	(Blin et al., 1991)	10	none	Sokoloff	87.3	11.6
Rattus	(Bryan and Lehman, 1988)	10	none	Sokoloff	101.0	9.5
Rattus	(Chang et al., 2001)	8	none	Sokoloff	112.0	28.3
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	153.0	1.0
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	143.0	6.0
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	128.0	11.3
Rattus	(Crosby et al., 1984)	5	none	Sokoloff	121.0	8.9
Rattus	(Crosby et al., 1984)	5	N2O	Sokoloff	115.0	11.2
Rattus	(Cudennec et al., 1997)	7	none	Sokoloff	155.0	29.1
Rattus	(Dam et al., 1990)	5	none	Sokoloff	134.0	17.9
Rattus	(Dow-Edwards et al., 1986)	4	none	Sokoloff	166.0	20.0
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	104.3	10.2
Rattus	(Eintrei et al., 1999)	6	none	Sokoloff	139.0	12.2
Rattus	(Freedland et al., 2002)	5	none	Sokoloff	140.3	11.2
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	131.0	22.0
Rattus	(Frietsch et al., 2000)	12	Isoflurane	Sokoloff	47.0	6.0
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	132.0	15.8
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	173.0	23.8
Rattus	(Ingvar et al., 1980)	10	N2O	Sokoloff	89.0	12.6
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	82.0	7.3
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	91.0	12.6
Rattus	(Kadekaro et al., 1983)	7	none	Sokoloff	126.0	10.6
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	113.0	10.0
Rattus	(MacNeill and Bryan, 1988)	10	none	Sokoloff	104.0	9.5
Rattus	(Maekawa et al., 1986)	6	none	Sokoloff	94.0	10.0
Rattus	(Nehlig et al., 1985)	6	none	Sokoloff	159.0	19.6
Rattus	(Ori et al., 1986)	6	none	Sokoloff	116.0	12.2
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	134.0	17.9
Rattus	(Poindessous-Jazat et al., 1998)	7	none	Sokoloff	117.7	7.0
Rattus	(Poindessous-Jazat et al., 1998)	5	none	Sokoloff	103.3	15.0
Rattus	(Sakurada et al., 1978)	5	N2O	Sokoloff	110.0	6.0
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	149.0	22.0
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	163.0	4.0
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	98.0	16.0

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Table A10, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	160.0	24.6
Rattus	(Grome and McCulloch, 1981)	16	none	Sokoloff	127.0	8.0
Rattus	(Whitlow et al., 2002)	5	none	Sokoloff	148.5	2.2
Rattus	(Walker et al., 1999)	5	none	Sokoloff	122.7	5.5
Mus	(Jay et al., 1985)	8	none	Sokoloff	101.0	11.3
Mus	(Niwa et al., 2002)	6	none	Sokoloff	151.0	9.0
Mus	(Niwa et al., 2002)	6	none	Sokoloff	146.0	17.0
Mus	(Nowaczyk and Des Rosiers, 1981)	4	none	Sokoloff	73.0	12.0
Mus	(Yamaguchi et al., 1992b)	6	none	Sokoloff	87.0	8.6
Mus	(Zocchi et al., 2001)	4	none	Sokoloff	78.0	6.0
Mus	(Zocchi et al., 2001)	6	none	Sokoloff	64.0	6.0

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Table A11. Occipital Cortex rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Alkire et al., 1995)	6	none	PET	33.7	9.7		
Homo	(Bokde et al., 2001)	18	none	PET	40.3	8.4		
Homo	(Brooks et al., 1987)	27	none	PET	33.6	5.2		
Homo	(Chugani et al., 1987)	7	none	PET	28.7	3.0		
Homo	(Duara et al., 1983)	18	none	PET	29.0	5.0		
Homo	(Hasselbalch et al., 1994)	9	none	PET	31.7	5.5		
Homo	(Haxby et al., 1985)	26	none	PET	28.7	7.2		
Homo	(Heiss et al., 1985)	6	none	PET	38.1	3.6		
Homo	(Hirono et al., 2000)	15	none	PET	42.7	6.3		
Homo	(Hu et al., 2000)	9	none	PET	51.2	10.6		
Homo	(Kuhl et al., 1979)	8	none	PET	39.6	5.1		
Homo	(Ludolph et al., 1992)	12	none	PET	43.9	9.6		
Homo	(Mazziotta et al., 1981)	7	none	PET	40.4	9.6		
Homo	(Perani et al., 1995)	15	none	PET	44.9	5.8		
Homo	(Phillips et al., 1991)	14	none	PET	39.0	8.3		
Homo	(Redies et al., 1989)	4	none	PET	32.0	6.0		
Homo	(Rudolf et al., 2000)	16	none	PET	32.1	nr		
Homo	(Sachs et al., 1986)	25	none	PET	46.9	6.0		
Homo	(Schapiro et al., 1990)	13	none	PET	46.3	5.0		
Homo	(Tedeschi et al., 1995)	11	none	PET	36.1	3.3		
Homo	(Tommasino et al., 1995)	10	none	PET	41.8	6.5		
Homo	(Volkow et al., 1992)	11	none	PET	48.8	6.0		
Homo	(Yoshii et al., 1988)	30	none	PET	45.0	12.8		
Homo	(Yoshii et al., 1988)	46	none	PET	41.1	9.4		
Homo	Present Study	5	none	PET	28.0	5.0		
Homo	(De Volder et al., 1997)	3	none	PET	36.5	5.7	156.7	23.3
Homo	(Fukuyama et al., 1994)	9	none	PET	36.6	3.7	111.4	19.8
Homo	(Sasaki et al., 1986)	7	none	PET	35.4	3.3	174.9	23.1
Homo	(Yamaki et al., 1996)	20	none	PET	28.0	2.8	118.3	22.3
Homo	(Burns and Tyrrell, 1992)	4	none	PET			150.9	12.5
Homo	(Ishii et al., 1996a)	15	none	PET			194.6	46.0
Homo	(Ishii et al., 1996b)	14	none	PET			190.2	29.5
Homo	(Kaisti et al., 2003)	8	none	PET			187.1	29.9
Homo	(Kaisti et al., 2003)	8	Sevoflurane	PET			105.8	18.8
Homo	(Langsjo et al., 2003)	10	none	PET			167.0	16.5
Homo	(Leenders et al., 1990)	34	none	PET			155.4	19.6
Homo	(Lenzi et al., 1981)	27	none	PET			210.0	nr
Homo	(Meyer et al., 1987)	3	none	PET			170.0	nr
Homo	(Pantano et al., 1984)	18	none	PET			227.7	58.0
Homo	(Perlmutter et al., 1987)	32	none	PET			113.4	27.7
Homo	(Tohgi et al., 1998)	6	none	PET			151.8	17.9
Homo	(Wessen et al., 1997)	6	none	PET			210.0	38.5
Homo	(Yamaguchi et al., 1986)	14	none	PET			162.5	22.5

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Table A11, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Yamaguchi et al., 1992a)	10	none	PET			159.8	26.3
Papio	(Kiyosawa et al., 1989)	9	Phencyclidine	PET	24.4	3.3		
Papio	(Meguro et al., 1999)	8	Phencyclidine	PET	31.6	5.1		
Papio	(Millien et al., 2002)	10	Phencyclidine	PET	35.0	10.6		
Papio	(Yamaguchi et al., 1990)	4	Phencyclidine	PET	23.1	3.4		
Papio	Present Study	2	N2O	PET	33.7	5.1		
M mulatta	(Cross et al., 2000)	6	none	PET	37.5	8.9		
M mulatta	(Noda et al., 2002)	6	none	PET	39.7	9.4		
M mulatta	(Palombo et al., 1988)	3	none	Sokoloff	65.0	4.3		
M mulatta	(Palombo et al., 1990)	4	none	Sokoloff	66.0	4.0		
M mulatta	(Porrino et al., 1987)	3	none	Sokoloff	63.5	2.0		
M mulatta	(Enlund et al., 1997)	4	Ketamine	PET			247.5	24.0
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	58.0	5.2		
M fascic	(Schwartzman et al., 1988)	6	none	Sokoloff	60.8	7.3		
Saimiri	(Delgado-Zygmunt et al., 1993)	4	Fentanyl, N2O	Sokoloff	46.0	12.0		
Lepus	(Passero et al., 1981)	5	none	Sokoloff	76.4	19.7		
Felis	(Petitjean et al., 1982)	4	none	Sokoloff	58.0	8.0		
Felis	(Triolo et al., 1990)	6	N2O	Sokoloff	37.0	4.6		
Meriones	(Choki et al., 1983)	6	none	Sokoloff	69.4	20.6		
Meriones	(Choki et al., 1984)	3	N2O	Sokoloff	123.0	38.0		
Rattus	(Bassant et al., 1996)	7	none	Sokoloff	97.0	7.0		
Rattus	(Bassant et al., 1996)	8	none	Sokoloff	78.0	4.0		
Rattus	(Blin et al., 1991)	10	none	Sokoloff	82.6	9.3		
Rattus	(Bryan et al., 1983)	6	none	Sokoloff	71.0	7.3		
Rattus	(Bryan and Lehman, 1988)	10	none	Sokoloff	87.0	6.3		
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	98.0	4.0		
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	98.0	2.0		
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	94.0	8.5		
Rattus	(Crosby et al., 1984)	5	none	Sokoloff	81.0	8.9		
Rattus	(Crosby et al., 1984)	5	N2O	Sokoloff	100.0	8.9		
Rattus	(Cudennec et al., 1997)	7	none	Sokoloff	114.0	13.2		
Rattus	(Dam et al., 1990)	5	none	Sokoloff	92.0	17.9		
Rattus	(Davis et al., 1986)	10	none	Sokoloff	94.0	14.9		
Rattus	(Dow-Edwards et al., 1986)	4	none	Sokoloff	110.0	14.0		
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	95.6	8.9		
Rattus	(Eintrei et al., 1999)	6	none	Sokoloff	93.0	9.8		
Rattus	(Frietsch et al., 2000)	12	none	Sokoloff	84.0	12.0		
Rattus	(Frietsch et al., 2000)	12	Isoflurane	Sokoloff	41.0	9.0		

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Table A11, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Rattus	(Furlow et al., 1983)	5	none	Sokoloff	82.0	20.0		
Rattus	(Hakim et al., 1983)	7	none	Sokoloff	103.0	10.6		
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	107.0	21.8		
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	96.0	9.9		
Rattus	(Ingvar et al., 1980)	10	N2O	Sokoloff	65.0	9.5		
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	69.0	9.8		
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	66.0	9.5		
Rattus	(Lear et al., 1981)	4	none	Sokoloff	122.0	26.0		
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	79.0	5.0		
Rattus	(Lu et al., 1983)	10	none	Sokoloff	84.0	9.5		
Rattus	(MacNeill and Bryan, 1988)	10	none	Sokoloff	95.0	6.3		
Rattus	(Maekawa et al., 1986)	6	none	Sokoloff	75.0	8.0		
Rattus	(Nehlig et al., 1985)	6	none	Sokoloff	117.0	14.7		
Rattus	(Ori et al., 1986)	6	none	Sokoloff	94.0	19.6		
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	98.0	15.7		
Rattus	(Otori et al., 2003)	6	none	Sokoloff	74.3	20.5		
Rattus	(Poindessous-Jazat et al., 1998)	7	none	Sokoloff	97.0	9.0		
Rattus	(Poindessous-Jazat et al., 1998)	5	none	Sokoloff	83.3	10.0		
Rattus	(Sakurada et al., 1978)	5	N2O	Sokoloff	76.0	4.0		
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	83.0	10.6		
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	116.0	17.1		
Rattus	(Sokoloff et al., 1977)	10	none	Sokoloff	113.0	4.0		
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	96.2	16.0		
Rattus	(Wechsler et al., 1979)	5	none	Sokoloff	102.0	17.9		
Rattus	(Walker et al., 1999)	5	none	Sokoloff	114.4	5.0		
Mus	(Jay et al., 1985)	8	none	Sokoloff	89.0	17.0		
Mus	(Niwa et al., 2002)	6	none	Sokoloff	124.0	9.0		
Mus	(Niwa et al., 2002)	6	none	Sokoloff	125.0	15.0		
Mus	(Nowaczyk and Des Rosiers, 1981)	4	none	Sokoloff	53.0	10.0		
Mus	(Yamaguchi et al., 1992b)	6	none	Sokoloff	70.4	nr		

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Table A12. Sensorimotor Cortex rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Chugani et al., 1987)	7	none	PET	23.83	3.44		
Homo	(Haxby et al., 1985)	26	none	PET	29.86	7.80		
Homo	(Heiss et al., 1985)	6	none	PET	44.20	3.07		
Homo	(Hirono et al., 2000)	15	none	PET	42.46	6.20		
Homo	(Martinot et al., 1990)	8	none	PET	36.08	7.80		
Homo	(McNamara et al., 1987)	14	none	PET	32.17	5.60		
Homo	(Schapiro et al., 1990)	13	none	PET	51.40	8.40		
Homo	(Tommasino et al., 1995)	10	none	PET	38.90	5.40		
Homo	Present Study	5	none	PET	38.00	8.00		
Homo	(Fukuyama et al., 1994)	9	none	PET	36.21	4.35	111.36	25.19
Homo	(Ishii et al., 1996b)	15	none	PET			151.79	25.00
Homo	(Lenzi et al., 1981)	27	none	PET			199.78	nr
Homo	(Pantano et al., 1984)	18	none	PET			156.25	26.80
Homo	(Perlmutter et al., 1987)	32	none	PET			122.99	28.13
Papio	(Meguro et al., 1999)	8	Phencyclidine	PET	46.90	7.80		
Papio	(Millien et al., 2002)	10	Phencyclidine	PET	45.20	15.80		
Papio	Present Study	2	N20	PET	60.3	12.4		
M mulatta	(Palombo et al., 1988)	3	none	Sokoloff	47.00	3.50		
M mulatta	(Palombo et al., 1990)	4	none	Sokoloff	46.00	4.00		
M mulatta	(Porrino et al., 1987)	3	none	Sokoloff	47.00	4.00		
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	56.00	5.20		
M fascic	(Schwartzman et al., 1988)	6	none	Sokoloff	48.90	3.40		
M fascic	(Deuel and Collins, 1984)	4	none	Sokoloff	64.75	nr		
Lepus	(Passero et al., 1981)	5	none	Sokoloff	82.65	15.30		
Meriones	(Choki et al., 1983)	6	none	Sokoloff	67.20	16.70		
Meriones	(Choki et al., 1984)	3	N2O	Sokoloff	102.00	32.00		
Rattus	(Bassant et al., 1996)	7	none	PET	96.00	8.00		
Rattus	(Bassant et al., 1996)	8	none	PET	96.00	13.00		
Rattus	(Blin et al., 1991)	10	none	PET	74.40	9.20		
Rattus	(Bryan and Lehman, 1988)	10	none	PET	91.00	9.50		
Rattus	(Chang et al., 2001)	8	none	PET	93.00	22.60		
Rattus	(Crane and Porrino, 1989)	6	none	PET	95.00	1.00		
Rattus	(Crane and Porrino, 1989)	6	none	PET	105.00	3.00		
Rattus	(Crosby et al., 1982)	8	none	PET	97.00	5.70		
Rattus	(Crosby et al., 1984)	5	none	PET	88.00	4.50		
Rattus	(Crosby et al., 1984)	5	N2O	PET	95.00	8.90		
Rattus	(Dam et al., 1990)	5	none	PET	98.00	20.10		

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Table A12, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Rattus	(Dow-Edwards et al., 1986)	4	none	PET	113.00	4.00		
Rattus	(Duelli et al., 2000)	5	none	PET	87.80	2.80		
Rattus	(Eintrei et al., 1999)	6	none	PET	101.00	7.30		
Rattus	(Fornai and Orzi, 2001)	5	none	PET	84.00	8.00		
Rattus	(Freedland et al., 2002)	5	none	PET	103.50	11.20		
Rattus	(Frietsch et al., 2000)	12	none	PET	90.00	6.00		
Rattus	(Frietsch et al., 2000)	12	Isoflurane	PET	33.00	4.00		
Rattus	(Grome and McCulloch, 1983)	6	none	PET	90.00	12.70		
Rattus	(Hakim et al., 1983)	7	none	PET	116.00	15.90		
Rattus	(Ingvar et al., 1980)	10	N2O	PET	69.00	9.50		
Rattus	(Ingvar and Siesjo, 1982)	6	none	PET	72.00	2.40		
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	PET	64.00	9.50		
Rattus	(Kadekaro et al., 1983)	7	none	PET	102.00	7.90		
Rattus	(Katsumata et al., 2001)	5	none	PET	92.80	8.50		
Rattus	(Knapp et al., 2002)	8	none	PET	96.10	7.60		
Rattus	(Lenz et al., 1998)	6	none	PET	81.00	10.00		
Rattus	(MacNeill and Bryan, 1988)	10	none	PET	95.00	6.30		
Rattus	(Maekawa et al., 1986)	6	none	PET	80.00	9.00		
Rattus	(Nehlig et al., 1985)	6	none	PET	112.00	7.30		
Rattus	(Ori et al., 1986)	6	none	PET	89.00	12.20		
Rattus	(Orzi et al., 1983)	5	none	PET	90.00	15.70		
Rattus	(Poindessous-Jazat et al., 1998)	7	none	PET	104.00	9.00		
Rattus	(Poindessous-Jazat et al., 1998)	5	none	PET	94.70	12.00		
Rattus	(Sakurada et al., 1978)	5	N2O	PET	80.00	3.00		
Rattus	(Savaki et al., 1983)	7	none	PET	95.00	10.60		
Rattus	(Schwartz and Crosby, 1985)	6	none	PET	101.00	9.80		
Rattus	(Sokoloff et al., 1977)	10	none	PET	124.00	3.00		
Rattus	(Soncrant et al., 1992)	16	none	PET	93.50	12.00		
Rattus	(Wechsler et al., 1979)	5	none	PET	118.00	17.90		
Rattus	(Grome and McCulloch, 1981)	16	none	PET	93.00	4.00		
Rattus	(Whitlow et al., 2002)	5	none	PET	101.60	8.90		
Rattus	(Walker et al., 1999)	5	none	PET	117.70	8.20		
Mus	(Bouilleret et al., 2000)	4	none	PET	121.00	9.00		
Mus	(Jay et al., 1985)	8	none	PET	88.00	11.30		
Mus	(Niwa et al., 2002)	6	none	PET	132.00	14.00		
Mus	(Niwa et al., 2002)	6	none	PET	129.00	20.00		
Mus	(Yamaguchi et al., 1992b)	6	none	PET	82.10	9.80		
Mus	(Zocchi et al., 2001)	4	none	PET	86.00	4.00		
Mus	(Zocchi et al., 2001)	6	none	PET	65.00	6.00		

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Table A13. Cingulate Cortex rCMR and rCBF

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Homo	(Baxter et al., 1987)	7	none	PET	29.3	2.8		
Homo	(Baxter et al., 1987)	7	none	PET	36.4	6.3		
Homo	(Chugani et al., 1987)	7	none	PET	29.8	3.5		
Homo	(Hasselbalch et al., 1994)	9	none	PET	29.5	5.8		
Homo	(Heiss et al., 1985)	6	none	PET	44.6	2.9		
Homo	(Hirono et al., 2000)	15	none	PET	40.2	6.9		
Homo	(Hu et al., 2000)	9	none	PET	47.4	7.7		
Homo	(Mazziotta et al., 1981)	7	none	PET	43.2	12.0		
Homo	(Perani et al., 1995)	15	none	PET	33.3	3.3		
Homo	(Schapiro et al., 1990)	13	none	PET	52.5	8.6		
Homo	(Tommasino et al., 1995)	10	none	PET	38.6	5.6		
Homo	Present Study	5	none	PET	42.0	6.7		
Homo	(Langsjo et al., 2003)	10	none	PET			185.0	27.2
Homo	(Perlmutter et al., 1987)	32	none	PET			111.6	31.9
Homo	(Sasaki et al., 1986)	7	none	PET			155.8	21.1
Papio	(Meguro et al., 1999)		Phencylidine, N2O	PET	49.3	7.8		
Papio	(Millien et al., 2002)		Phencylidine, N2O	PET	47.5	11.7		
Papio	Present Study	2	N2O	PET	58.9	13.0		
M mulatta	(Cross et al., 2000)	6	none	PET	37.8	8.8		
M mulatta	(Noda et al., 2002)	6	none	PET	36.6	8.5		
M mulatta	(Palombo et al., 1988)	3	none	Sokoloff	41.0	5.2		
M mulatta	(Palombo et al., 1990)	4	none	Sokoloff	39.0	4.0		
M mulatta	(Porrino et al., 1987)	3	none	Sokoloff	38.0	2.0		
M fascic	(Deuel and Collins, 1984)	4	none	Sokoloff	63.8	nr		
M fascic	(Lyons et al., 1996)	3	none	Sokoloff	44.6	3.5		
Felis	(Petitjean et al., 1982)	4	none	Sokoloff	47.0	10.0		
Rattus	(Bassant et al., 1996)	7	none	Sokoloff	102.0	11.0		
Rattus	(Bassant et al., 1996)	8	none	Sokoloff	96.0	11.0		
Rattus	(Blin et al., 1991)	10	none	Sokoloff	83.0	8.5		
Rattus	(Bryan et al., 1983)	6	none	Sokoloff	89.0	9.8		
Rattus	(Bryan and Lehman, 1988)	10	none	Sokoloff	91.0	6.3		
Rattus	(Chang et al., 2001)	8	none	Sokoloff	92.0	19.8		
Rattus	(Crane and Porrino, 1989)	6	none	Sokoloff	106.0	3.5		
Rattus	(Crosby et al., 1982)	8	none	Sokoloff	73.0	19.8		
Rattus	(Cudennec et al., 1997)	7	none	Sokoloff	122.0	7.9		
Rattus	(Dam et al., 1990)	5	none	Sokoloff	83.5	15.7		
Rattus	(Davis et al., 1986)	10	none	Sokoloff	98.0	16.8		
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	81.6	5.0		

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Table A13, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD
Rattus	(Eintrei et al., 1999)	6	none	Sokoloff	119.0	9.8		
Rattus	(Fornai and Orzi, 2001)	5	none	Sokoloff	88.0	8.0		
Rattus	(Freedland et al., 2002)	5	none	Sokoloff	116.2	8.9		
Rattus	(Grome and McCulloch, 1983)	6	none	Sokoloff	91.0	9.5		
Rattus	(Hawkins et al., 1979)	19	none	Sokoloff	118.0	21.8		
Rattus	(Hawkins et al., 1985)	9	none	Sokoloff	105.0	7.2		
Rattus	(Ingvar and Siesjo, 1982)	6	none	Sokoloff	63.0	4.9		
Rattus	(Ingvar and Siesjo, 1982)	10	N2O	Sokoloff	64.0	9.5		
Rattus	(Knapp et al., 2002)	8	none	Sokoloff	105.6	13.0		
Rattus	(Lenz et al., 1998)	6	none	Sokoloff	86.0	14.0		
Rattus	(Lu et al., 1983)	10	none	Sokoloff	96.0	9.5		
Rattus	(Nehlig et al., 1985)	6	none	Sokoloff	113.0	14.7		
Rattus	(Orzi et al., 1983)	5	none	Sokoloff	100.0	13.4		
Rattus	(Poindessous-Jazat et al., 1998)	7	none	Sokoloff	104.0	10.0		
Rattus	(Poindessous-Jazat et al., 1998)	5	none	Sokoloff	89.0	13.0		
Rattus	(Savaki et al., 1983)	7	none	Sokoloff	95.0	13.3		
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	105.0	7.3		
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	94.5	16.0		
Rattus	(Grome and McCulloch, 1981)	16	none	Sokoloff	87.0	7.0		
Rattus	(Whitlow et al., 2002)	5	none	Sokoloff	119.0	11.2		
Rattus	(Walker et al., 1999)	5	none	Sokoloff	120.7	7.7		
Mus	(Jay et al., 1985)	8	none	Sokoloff	96.0	17.0		
Mus	(Zocchi et al., 2001)	4	none	Sokoloff	93.0	4.0		
Mus	(Zocchi et al., 2001)	6	none	Sokoloff	60.0	6.0		

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Table A14. Whole Brain CMR and CBF (direct measurement)

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Alkire et al., 1995)	6	none	PET	29.00	8.00				
Homo	(Blomqvist et al., 1994)	3	none	PET	25.70	nr				
Homo	(Delvenne et al., 1997)	10	none	PET	24.65	2.58				
Homo	(Hasselbalch et al., 1994)	9	none	PET	30.70	5.00				
Homo	(Heiss et al., 1985)	6	none	PET	36.20	2.38				
Homo	(Kaye et al., 1990)	12	none	PET	29.25	3.40				
Homo	(Phillips et al., 1991)	14	none	PET	34.50	6.70				
Homo	(Reivich et al., 1979)	2	none	PET	32.75	nr				
Homo	(Rudolf et al., 2000)	16	none	PET	35.20	4.90				
Homo	(Tommasino et al., 1995)	10	none	PET	37.30	5.16				
Homo	Present Study	5	none	PET	26.80	4.70				
Homo	(Kuhl et al., 1982)	25	none	PET	28.00	6.30	147.00	15.00		
Homo	(Bednarczyk et al., 1992)	6	none	PET	27.20	2.80	147.30	43.70	41.40	4.70
Homo	(Blomqvist et al., 1990)	7	none	PET	28.30	4.30	152.70	11.60	50.20	6.70
Homo	(Cohen et al., 1964)	10	Halothane	Kety-Schmidt	23.03	2.80	125.00	nr	50.80	nr
Homo	(Cohen et al., 1967)	9	none	Kety-Schmidt	24.87	1.60	134.82	6.70	44.97	2.19
Homo	(Dastur, 1985)	15	none	Kety-Schmidt	33.30	3.90	156.70	9.40	62.10	2.90
Homo	(Hoyer et al., 1984)	15	none	PET	27.59	nr	158.04	nr	52.90	
Homo	(Jobes et al., 1977)	6	none	PET	23.00	3.70	144.20	41.50	48.20	10.80
Homo	(Madsen et al., 1995)	8	none	PET	23.30	2.40	142.50	18.90	50.60	4.90
Homo	(Mielck et al., 1998)	9	none	PET	22.20	3.90	147.30	31.30	45.00	10.00
Homo	(Mielck et al., 1998)	9	Desflurane	PET	14.40	3.90	71.40	8.90	35.00	6.00
Homo	(Ogawa et al., 1996)	7	none	PET	24.90	7.21	136.74	31.69	42.54	2.19
Homo	(Redies et al., 1989)	4	none	PET	29.00	2.00	154.00	24.00	43.00	12.00
Homo	(Scheinberg and Stead, 1949)	20	none	PET	34.40	7.80	169.20	24.60	64.70	12.10
Homo	(Felding et al., 1995)	9	Halothane, N2O	Kety-Schmidt			93.80	13.40	33.00	8.00
Homo	(Frackowiak et al., 1980)	11	none	PET			201.79	46.40	49.80	12.50

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Table A14, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Homo	(Jakobsen et al., 1990)	8	none	Kety-Schmidt			142.86	17.90	52.60	8.30
Homo	(Carpenter et al., 1991)	20	none	PET			117.86	22.80	47.70	9.00
Homo	(Lassen and Munck, 1955)	10	none	Kety-Schmidt			156.70	15.60	51.00	6.70
Homo	(Lassen and Munck, 1955)	10	none	Kety-Schmidt			145.98	33.00	52.10	10.50
Homo	(Madsen et al., 1990b)	8	none	Kety-Schmidt			151.80	17.90	57.00	8.00
Homo	(Madsen et al., 1991b)	11	none	Kety-Schmidt			156.25	18.30	56.70	8.00
Homo	(Madsen et al., 1991a)	9	none	Kety-Schmidt			121.88	25.00	49.80	6.30
Homo	(Madsen et al., 1992)	10	none	Kety-Schmidt			171.43	25.40	56.35	14.40
Homo	(Madsen et al., 1993)	12	none	Kety-Schmidt			162.05	21.00	46.90	7.00
Homo	(Meyer et al., 1987)	3	none	PET			165.00	nr	63.00	nr
Homo	(Mintun et al., 2002)	7	none	PET			191.96	47.20	53.50	14.60
Homo	(Olsen et al., 1992a)	19	none	Kety-Schmidt			162.90	33.50	52.00	nr
Homo	(Olsen et al., 1992b)	8	none	Kety-Schmidt			174.10	19.20	58.00	9.30
Homo	(Olsen et al., 1993)	8	none	Kety-Schmidt			161.16	nr	59.90	nr
Homo	(Olsen et al., 1995)	8	none	Kety-Schmidt			129.50	22.30	67.00	13.00
Homo	(Renou et al., 1978)	7	none	Kety-Schmidt			138.84	31.30	56.80	10.40
Homo	(Sari et al., 1976)	13	none	Kety-Schmidt			138.40	26.80	46.00	7.00
Homo	(Schmidt et al., 1990)	8	none	Kety-Schmidt			165.18	nr	52.00	nr
Homo	(Schmidt et al., 1991)	8	none	Kety-Schmidt			156.25	15.62	52.00	4.40
Homo	(Seitz and Roland, 1992)	5	none	PET			160.70	31.30	54.60	9.30
Homo	(Vafaee et al., 1998)	6	none	PET			159.90	19.60		
Homo	(Vernhiet et al., 1978)	8	none	Kety-Schmidt			127.20	35.70	55.00	8.00
Homo	(Vorstrup et al., 1984)	6	none	Kety-Schmidt			158.04	39.30	56.00	6.00
Homo	(Wessen et al., 1997)	6	none	PET			187.00	58.00	52.00	15.00
Homo	(Wolff et al., 1994)	8	none	Kety-Schmidt			183.00	nr	64.00	nr
Homo	(Wollman et al., 1964)	13	none	Kety-Schmidt			137.95	8.50	44.40	2.20
Homo	(Mintun et al., 1984)	5	none	PET			191.90	16.50		
Homo	(Elman et al., 1999)	13	none	PET					38.10	7.76

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Table A14, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Papio	(Hamar et al., 1979)	5	Phencyclidine	Kety-Schmidt	24.64	1.70	175.40	34.90	39.40	4.76
Papio	(Harper and MacKenzie, 1977)	7	Phencyclidine, N2O	Kety-Schmidt	25.30	5.60	144.20	20.10	51.00	5.30
Papio	(McCulloch and Harper, 1977)	4	Phencyclidine, N2O	Kety-Schmidt	18.30	nr	143.80	nr	44.00	nr
Papio	(Reivich et al., 1972)	3	Phencyclidine	Kety-Schmidt	24.60	2.90	175.40	27.00	38.00	5.20
Papio	(Tagashira et al., 1977)	17	Pentobarbital	Kety-Schmidt	23.20	9.00	124.60	25.40	37.90	8.50
Papio	(Van Aken et al., 1986)	6	Phencyclidine, N2O	Kety-Schmidt	24.42	6.53	162.50	44.60	48.30	3.50
Papio	(Van Hemelrijck et al., 1990)	7	Phencyclidine, N2O	Kety-Schmidt	20.76	3.70	90.60	19.70	50.01	4.49
Papio	(Brussel et al., 1991)	8	Phencyclidine, N2O	Kety-Schmidt			151.80	35.70	55.80	8.00
Papio	(Pappata et al., 1993)	5	Midazolam, N2O	PET			122.30	15.60	35.60	4.60
Papio	(Sahlin et al., 1986)	11	Phencyclidine, N2O	Kety-Schmidt			145.50	6.25	58.20	10.70
Papio	(Schumann et al., 1998)	10	Fentanyl, N2O	PET			134.40	16.10	29.30	3.00
Papio	(Touzani et al., 1995)	7	Etomidate, N2O	PET			124.60	42.40	32.50	14.60
Papio	(Sahlin et al., 1987)	13	N2O	Kety-Schmidt			143.75	5.80	63.20	4.40
Papio	(Kawamura et al., 1975)	11	Pentobarbital	Kety-Schmidt			110.16	10.80	41.30	4.50
Papio	(Aoyagi et al., 1975)	12	Pentobarbital	Kety-Schmidt			115.40	12.50	36.50	5.40
Papio	(Harper et al., 1981)	4	Phencyclidine, N2O	Kety-Schmidt			195.50	12.10	62.60	3.60
Papio	(Young et al., 1996)	4	Etomidate, N2O	PET			148.66	1.04	36.95	9.25
Papio	Present Study	2	N2O	PET	35.52					
M mulatta	(Fein, 1975)	19	N20	Kety-Schmidt	32.75	nr	165.18	nr	40.60	nr
M mulatta	(Mehta et al., 1980)	5	none	Kety-Schmidt	61.30	12.30	270.10	60.20	100.60	14.10
M mulatta	(Nemoto et al., 1994)	3	none	Kety-Schmidt	44.90	15.40	265.60	24.10	76.10	20.50
M mulatta	(Yao et al., 1992)	4	none	Kety-Schmidt	43.30	9.60	304.90	32.30	80.00	12.00
M mulatta	(Schmidt et al., 1945)	11	Thiopentone	Kety-Schmidt			165.18	31.25	46.90	11.70
M mulatta	(Kliefoth et al., 1979)	10	Ketamine	Kety-Schmidt			151.80	20.00	48.70	8.50
M mulatta	(Cross et al., 2000)	6	none	PET	34.60	6.90				
M fascic	(Rockoff et al., 1980)	10	none	Kety-Schmidt	31.64	11.40	136.16	70.50	43.00	3.20

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Table A14, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Canis	(Michenfelder, 1974)	7	Halothane	Kety-Schmidt	26.10	3.30	178.10	37.80	40.00	10.60
Canis	(Roald et al., 1986a)	6	Halothane, N2O	Kety-Schmidt	24.60	1.80	208.04	19.60	50.90	9.80
Canis	(Zimmer, 1975)	13	Allobarbital, Urethan, & Ethylene urea	Kety-Schmidt	27.67	3.99	167.40	53.00	45.60	7.20
Canis	(Hamer et al., 1978)	10	Halothane, N2O	Kety-Schmidt	28.90	4.30	169.60	34.80	60.30	8.10
Canis	(Sieber et al., 1993)	8	none	Kety-Schmidt	44.50	5.00	205.40	22.30		
Canis	(Frizzell et al., 1993)	13	Isoflurane, N2O	Kety-Schmidt	35.90	7.80	147.30	12.10		
Canis	(Astrup et al., 1981)	12	Halothane	Kety-Schmidt			46.50	4.80	19.60	1.80
Canis	(McPherson et al., 1997)	5	Isoflurane	Kety-Schmidt			156.25	30.00	66.00	17.90
Canis	(McPherson et al., 1997)	6	Isoflurane	Kety-Schmidt			160.70	8.90	74.00	7.00
Canis	(Roald et al., 1986b)	6	Halothane, N2O	Kety-Schmidt			188.40	40.40	45.00	12.50
Canis	(Sakabe et al., 1974)	16	Halothane	Kety-Schmidt			263.40	64.50	62.70	16.90
Canis	(Nakanishi et al., 1997)	7	Halothane	Kety-Schmidt			272.77	9.38	63.00	5.00
Canis	(Nakanishi et al., 1997)	7	Halothane, N2O	Kety-Schmidt			317.00	17.40	173.00	26.00
Canis	(Lanier et al., 1994)	6	Halothane	Kety-Schmidt			170.98	17.90	66.00	8.00
Canis	(Lanier et al., 1990)	6	Halothane	Kety-Schmidt			173.21	7.59	70.00	6.00
Canis	(Roald et al., 1991)	6	Isoflurane, N2O	Kety-Schmidt			163.39	10.27	103.00	15.00
Canis	(Milde et al., 1990)	18	Halothane	Kety-Schmidt			180.80	5.36	83.00	6.00
Canis	(Roald et al., 1988)	6	Halothane	Kety-Schmidt			141.52	13.40	51.50	6.40
Canis	(Roald et al., 1989)	5	Isoflurane, N2O	Kety-Schmidt			200.45	4.50	58.80	4.80
Canis	(Fleischer et al., 1988)	9	Halothane, N2O	Kety-Schmidt			236.60	17.90	66.00	4.00
Canis	(McPherson and Traystman, 1988)	12	Isoflurane	Kety-Schmidt			162.95	17.90	69.50	11.00
Canis	(Milde et al., 1985)	6	Halothane, N2O	Kety-Schmidt			246.43	12.10	145.00	23.00
Canis	(Lanier et al., 1985)	12	Halothane	Kety-Schmidt			191.07	7.40	62.60	4.30
Canis	(Lanier et al., 1986)	12	Halothane	Kety-Schmidt			159.20	8.00	67.00	6.00
Canis	(Nugent et al., 1982)	12	Halothane, N2O	Kety-Schmidt			240.85	13.40	112.00	11.00
Canis	(Steen et al., 1983)	6	Spinal Anesthesia	Kety-Schmidt			220.98	14.30	56.00	7.00
Canis	(Miyauchi et al., 1985)	10	Halothane	Kety-Schmidt			267.90	8.90	59.00	3.00

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Table A14, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Canis	(Artru and Michenfelder, 1980)	14	Halothane, N2O	Kety-Schmidt			205.40	8.90	78.00	7.00
Canis	(Boarini et al., 1984)	10	N2O	Kety-Schmidt			218.75	.20	57.00	4.00
Canis	(Kassell et al., 1983)	6	Halothane, N2O	Kety-Schmidt			151.80	17.90	68.00	6.00
Canis	(Qureshi et al., 1999)	10	Pentobarbital	Kety-Schmidt			73.20	9.00	22.90	1.90
Canis	(Rhodes et al., 1981)	6	Pentobarbital	PET			159.82	36.20	39.90	8.30
Canis	(Forsman et al., 1986)	6	Halothane, N2O	Kety-Schmidt			243.30	4.00	62.00	11.00
Canis	(Artru and Michenfelder, 1981)	6	Halothane, N2O	Kety-Schmidt			220.98	17.90	124.00	12.00
Canis	(Chen et al., 1984)	9	Pentobarbital	Kety-Schmidt			156.25	26.80	49.50	5.30
Canis	(Sturaitis et al., 1994)	10	Isoflurane	Kety-Schmidt			151.80	22.30		
Canis	(Roth et al., 1970)	12	none	Microspheres					30.80	6.60
Sus	(Poulsen et al., 1997)	7	Ketamine, Midazolam	PET	25.25	3.00	171.00	18.00	37.00	3.00
Sus	(Akeson et al., 1992)	9	Fentanyl, N2O	Kety-Schmidt			194.00	24.00	56.10	9.90
Sus	(Miller et al., 1987)	4	Pentobarbital	Microspheres			84.80	26.80	30.00	10.00
Sus	(Sakoh et al., 2000)	11	Fentanyl, Midazolam	PET			197.00	7.70	42.30	1.20
Sus	(Stange et al., 1989)	8	Droperidol, N2O	Microspheres			102.70	50.60	34.00	17.00
Sus	(Madsen et al., 1990a)	18	Ketamine, Midazolam	Microspheres					45.60	
Lepus	(Drummond et al., 1986)	12	Morphine, N2O	Kety-Schmidt			143.30	38.80	46.50	27.70
Lepus	(Kaieda et al., 1989)	8	Halothane	Kety-Schmidt			165.20	62.50	71.00	30.00
Lepus	(Kaieda et al., 1989)	8	Isoflurane	Kety-Schmidt			120.50	49.10	65.00	23.00
Lepus	(Pearce et al., 1981)	38	Halothane	Kety-Schmidt			205.40	107.10	48.00	18.50
Lepus	(Todd et al., 1994)	7	Isoflurane	Microspheres			145.50	46.00	77.00	19.00
Felis	(Van Zijl et al., 1997)	10	Isoflurane, N2O	NMR	21.90	3.30				
Felis	(Haller et al., 1990)	7	Alpha Chloralose	Kety-Schmidt	35.20	13.20	147.32	5.40	38.00	16.00
Felis	(Unterberg et al., 1988)	7	Alpha Chloralose	Microspheres	32.00	6.00	117.00	14.00	29.60	2.20
Felis	(Andersen et al., 1988)	24	Alpha Chloralose	Microspheres	29.00	4.00	107.00	13.00	29.90	2.80

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Table A14, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Felis	(Fiat and Kang, 1993)	5	Pentobarbital	NMR			120.00	60.00	38.00	12.00
Felis	(Kobayashi et al., 1992)	20	Alpha Chloralose	Kety-Schmidt			232.14	40.20	60.00	6.00
Felis	(Stingele et al., 1996)	9	Fentanyl	Microspheres			258.90	26.80	76.00	9.00
Felis	(Frewen et al., 1989)	6	Ketamine	Kety-Schmidt			91.10	26.20	52.60	16.70
Felis	(Todd and Drummond, 1984)	18	N2O	Kety-Schmidt			196.40	15.60	51.00	4.00
Felis	(Pekar et al., 1995)	7	Pentobarbital	NMR			150.00	50.00		
Meriones	(Van Uitert and Levy, 1978)	10	none	Kety-Schmidt					102.00	12.60
Rattus	(Blin et al., 1991)	10	none	Sokoloff	56.10	7.10				
Rattus	(Borgstrom et al., 1976)	7	none	Sokoloff	60.00	15.90				
Rattus	(Duelli et al., 2000)	5	none	Sokoloff	72.00	5.60				
Rattus	(Hawkins et al., 1974)	27	none	Sokoloff	62.00	8.00				
Rattus	(Schwartz and Crosby, 1985)	6	none	Sokoloff	74.00	4.90				
Rattus	(Soncrant et al., 1992)	16	none	Sokoloff	84.00	2.00				
Rattus	(Linde et al., 1999)	8	none	Kety-Schmidt	66.00	17.00	374.00	84.00	106.00	23.00
Rattus	(Madsen et al., 1998)	31	none	Kety-Schmidt	66.00	nr	392.00	nr	108.00	25.00
Rattus	(Meldrum and Nilsson, 1976)	10	none	Kety-Schmidt	77.00	19.00	440.00		100.00	9.50
Rattus	(Takei et al., 1983)	15	none	Kety-Schmidt	92.00	31.00	495.00	85.20	104.00	15.50
Rattus	(Takei et al., 1983)	15	none	Kety-Schmidt	97.00	38.70	485.00	77.50	107.00	15.50
Rattus	(Takei et al., 1983)	15	none	Kety-Schmidt	84.00	27.10	498.00	89.10	103.00	11.60
Rattus	(Magata et al., 1995)	7	Pentobarbital	PET	51.00	14.00			61.00	7.00
Rattus	(Gjedde and Hindfelt, 1975)	6	N2O	Kety-Schmidt			352.70	76.30	106.00	17.10
Rattus	(Gjedde et al., 1977)	10	N2O	Kety-Schmidt			357.10	84.80	108.00	6.30
Rattus	(Hauerberg and Juhler, 1997)	8	Halothane, N2O	Kety-Schmidt			321.40	84.80	130.00	29.00
Rattus	(Nemoto et al., 1996a)	17	N2O	Kety-Schmidt			342.19	63.20	111.50	27.00
Rattus	(Nilsson and Siesjö, 1975)	5	Fentanyl	Kety-Schmidt			495.50	17.90	135.00	13.00
Rattus	(Harp et al., 1976)	9	N2O	Kety-Schmidt			459.80	8.90	108.00	5.00

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Table A14, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Rattus	(Harp et al., 1976)	6	Fentanyl	Kety-Schmidt			509.82	19.64	133.00	11.00
Rattus	(Harp et al., 1976)	8	Halothane	Kety-Schmidt			363.39	37.50	136.00	9.00
Rattus	(Harp et al., 1976)	6	Halothane	Kety-Schmidt			243.75	25.00	97.00	8.00
Rattus	(Carlsson et al., 1976)	5	N2O	Kety-Schmidt			455.40	8.90	121.00	8.00
Rattus	(Hemmingsen et al., 1979a)	10	N2O	Kety-Schmidt			549.10	nr	97.00	nr
Rattus	(Hemmingsen et al., 1979a)	8	Halothane, N2O	Kety-Schmidt			522.30	nr	107.00	nr
Rattus	(Hemmingsen et al., 1979a)	9	Pentobarbital	Kety-Schmidt			540.20	nr	78.00	nr
Rattus	(Nilsson and Siesjö, 1976)	8	N2O	Kety-Schmidt			338.39	8.03	79.50	2.40
Rattus	(Dahlgren et al., 1981b)	8	N2O	Kety-Schmidt			391.00	28.00	110.00	11.00
Rattus	(Hemmingsen et al., 1979b)	7	N2O	Kety-Schmidt			584.80	nr	103.00	nr
Rattus	(Klementavicius et al., 1996b)	17	Halothane, N2O	Kety-Schmidt			305.80	31.30	131.80	20.20
Rattus	(Nemoto et al., 1996b)	14	N2O	Kety-Schmidt			346.88	66.30	110.50	28.00
Rattus	(Fiat and Kang, 1993)	5	Pentobarbital	Kety-Schmidt			209.00	35.00	140.00	nr
Rattus	(Wang et al., 1993)	12	N2O	Kety-Schmidt			316.96	13.40	85.00	17.00
Rattus	(Kochanek et al., 1988)	17	Halothane, N2O	Kety-Schmidt			424.10	55.80	149.00	17.00
Rattus	(Singh et al., 1992)	16	N2O	Kety-Schmidt			455.36	148.40	141.30	40.50
Rattus	(Kozniewska and Oseka, 1992)	10	Chloral Hydrate	Kety-Schmidt			334.80	22.30	85.00	4.00
Rattus	(Nilsson and Siesjö, 1983)	4	Halothane	Kety-Schmidt			321.40	nr	76.00	nr
Rattus	(Klementavicius et al., 1996a)	10	Isoflurane, N2O	Kety-Schmidt			288.39	95.10	169.88	37.61
Rattus	(Baughman et al., 1990)	8	N2O	Kety-Schmidt			511.16	53.60		
Rattus	(Nemoto et al., 1988)	10	N2O	Kety-Schmidt			261.16	107.10		
Rattus	(Matsumoto et al., 1975)	12	N2O	Kety-Schmidt					86.00	15.00
Mus	(Itoh et al., 2001)	5	none	Sokoloff	58.00	6.00			117.00	20.00
Mus	(Kamp et al., 1980)	10	none	Sokoloff	71.60	11.40				
Mus	(Kamp et al., 1980)	17	none	Sokoloff	65.00	9.00				
Mus	(Vannucci et al., 1997)	5	none	Sokoloff	77.53	3.40				

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Table A14, cont.

Species	Reference	n	Anesthesia	Mode	Glucose	SD	Oxygen	SD	Blood Flow	SD
Equus	(Wagerle et al., 1983)	6	none	Microspheres					68.80	13.50
Equus	(Busija et al., 1980)	12	none	Microspheres					62.00	14.50
Ovis	(Rosenberg et al., 1982)	8	none	Microspheres			204.90	21.50		
Ovis	(Koehler et al., 1984)	7	none	Microspheres			221.00	47.40		
Ovis	(Donegan et al., 1985)	3	none	Microspheres			245.50	84.00	78.00	17.30
Ovis	(Iwamoto et al., 1991)	8	none	Microspheres					66.40	5.80
Ovis	(Yang and Krasney, 1995)	9	none	Microspheres					55.00	17.00
Ovis	(Matalon et al., 1983)	10	none	Microspheres					60.00	3.50
Capra	(Schwedler et al., 1982)	17	none	Probe			192.40	16.10	77.00	7.00
Capra	(Stanley and Cherniack, 1976)	6	none	Probe			154.46	4.50		

Table A15. Species Means in Conscious Subjects
(CV* = unbiased coefficient of variation)

Whole Brain CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	22	29.31	4.42	15.07	15.24
	weighted	216	29.98	5.28	17.61	17.63
Macaca	unweighted	5	43.15	11.60	26.88	28.22
	weighted	28	40.66	11.05	27.18	27.42
Canis	unweighted	1	44.50	5.00	11.24	14.04
	weighted	8	44.50	5.00	11.24	11.59
Rattus	unweighted	13	75.24	13.01	17.29	17.62
	weighted	205	76.89	19.83	25.79	25.82
Mus	unweighted	4	68.03	8.42	12.38	13.15
	weighted	37	67.53	8.88	13.15	13.24

Whole Brain CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	37	156.92	18.64	11.88	11.96
	weighted	358	155.44	27.19	17.49	17.50
Macaca	unweighted	4	244.19	74.13	30.36	32.25
	weighted	22	214.93	57.89	26.93	27.24
Canis	unweighted	1	205.40	22.30	10.86	13.57
	weighted	8	205.40	22.30	10.86	11.20
Lepus	unweighted	1	205.40	17.90	8.71	10.89
	weighted	38	205.40	17.90	8.71	8.77
Rattus	unweighted	7	428.71	69.91	16.31	16.89
	weighted	106	429.41	76.12	17.73	17.77
Ovis	unweighted	3	223.80	20.44	9.14	9.90
	weighted	18	217.93	47.49	21.79	22.09
Capra	unweighted	2	173.43	26.83	15.47	17.40
	weighted	23	182.50	14.03	7.69	7.77

Whole Brain CBF		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	35	53.06	7.04	13.27	13.36
	weighted	335	53.25	9.14	17.16	17.18
Macaca	unweighted	4	74.93	23.84	31.82	33.81
	weighted	22	67.33	11.55	17.15	17.35
Canis	unweighted	1	30.80	6.60	21.43	26.79
	weighted	12	30.80	6.60	21.43	21.88
Lepus	unweighted	1	48.00	18.50	38.54	48.18
	weighted	38	48.00	18.50	38.54	38.80
Meriones	unweighted	1	102.00	12.60	12.35	15.44
	weighted	10	102.00	12.60	12.35	12.66
Rattus	unweighted	7	101.86	7.90	7.76	8.04
	weighted	106	103.08	18.73	18.17	18.21
Mus	unweighted	1	117.00	20.00	17.09	21.37
	weighted	5	117.00	20.00	17.09	17.95
Equus	unweighted	2	65.40	4.81	7.35	8.27
	weighted	18	64.27	14.17	22.05	22.35
Ovis	unweighted	4	64.85	9.93	15.31	16.27
	weighted	30	62.01	11.39	18.37	18.52
Capra	unweighted	1	77.00	7.00	9.09	11.36
	weighted	17	77.00	7.00	9.09	9.22

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Table A15. Species Means in Conscious Subjects
(CV* = unbiased coefficient of variation)

Cortex CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	49	35.68	6.40	17.94	18.03
	weighted	650	36.78	6.88	18.70	18.71
Macaca	unweighted	10	50.45	7.95	15.76	16.15
	weighted	46	49.99	6.02	12.04	12.11
Lepus	unweighted	1	77.42	17.90	23.12	28.90
	weighted	5	77.42	17.90	23.12	24.28
Felis	unweighted	2	62.30	21.64	34.73	39.07
	weighted	11	66.47	12.19	18.34	18.76
Meriones	unweighted	3	78.51	16.38	20.87	22.61
	weighted	13	73.35	15.62	21.29	21.70
Rattus	unweighted	49	99.14	15.53	15.67	15.75
	weighted	360	98.91	12.37	12.51	12.52
Mus	unweighted	8	97.88	26.83	27.41	28.27
	weighted	43	98.90	11.99	12.12	12.19

Cortex CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	13	150.73	20.23	13.42	13.68
	weighted	169	143.74	21.86	15.21	15.23
Sus	unweighted	1	341.96	20.10	5.88	7.35
	weighted	6	341.96	20.10	5.88	6.12
Rattus	unweighted	1	464.30	93.80	20.20	25.25
	weighted	20	464.30	93.80	20.20	20.45

Cortex CBF		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	10	44.61	2.25	5.04	5.16
	weighted	125	43.98	6.90	15.69	15.72
Macaca	unweighted	1	53.00	7.50	14.15	17.69
	weighted	6	53.00	7.50	14.15	14.74
Sus	unweighted	1	108.20	6.30	5.82	7.28
	weighted	6	108.20	6.30	5.82	6.07
Meriones	unweighted	2	112.80	32.24	28.59	32.16
	weighted	7	103.03	6.80	6.60	6.84
Rattus	unweighted	3	149.60	38.40	25.67	27.80
	weighted	32	145.58	47.17	32.40	32.66

Frontal Cortex CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	40	36.31	6.51	17.94	18.05
	weighted	550	37.34	7.53	20.17	20.18
Macaca	unweighted	6	47.92	11.24	23.46	24.44
	weighted	25	44.86	6.93	15.45	15.60
Felis	unweighted	1	49.00	16.00	32.65	40.82
	weighted	4	49.00	16.00	32.65	34.69
Meriones	unweighted	1	67.80	15.00	22.12	27.65
	weighted	6	67.80	15.00	22.12	23.05
Rattus	unweighted	34	94.06	13.75	14.62	14.73
	weighted	247	94.31	12.79	13.56	13.58
Mus	unweighted	6	108.83	19.06	17.52	18.24
	weighted	34	107.29	12.07	11.25	11.33

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Frontal Cortex CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	18	149.39	27.70	18.54	18.80
	weighted	240	149.07	24.43	16.39	16.41
Parietal Cortex CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	36	36.02	7.85	21.79	21.95
	weighted	537	37.15	7.06	19.01	19.01
Macaca	unweighted	2	42.73	7.81	18.29	20.57
	weighted	10	41.62	7.80	18.74	19.21
Lepus	unweighted	1	73.10	17.40	23.80	29.75
	weighted	5	73.10	17.40	23.80	24.99
Felis	unweighted	1	45.00	16.00	35.56	44.44
	weighted	4	45.00	16.00	35.56	37.78
Rattus	unweighted	32	93.08	14.44	15.51	15.63
	weighted	251	94.29	12.35	13.10	13.11
Mus	unweighted	2	126.00	2.83	2.24	2.53
	weighted	12	126.00	14.51	11.52	11.76
Parietal Cortex CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	16	155.25	26.87	17.31	17.58
	weighted	219	148.41	25.39	17.11	17.13
Temporal Cortex CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	37	33.23	6.86	20.64	20.78
	weighted	520	34.23	6.23	18.20	18.21
Macaca	unweighted	3	36.40	1.44	3.96	4.29
	weighted	15	36.08	6.67	18.49	18.79
Felis	unweighted	1	46.00	16.00	34.78	43.48
	weighted	4	46.00	16.00	34.78	36.96
Rattus	unweighted	4	98.58	18.05	18.31	19.45
	weighted	26	97.46	13.57	13.92	14.06
Temporal Cortex CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	14	155.13	25.49	16.43	16.72
	weighted	158	148.91	25.57	17.17	17.20
Auditory Cortex CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	4	40.23	5.41	13.44	14.28
	weighted	45	40.56	5.87	14.47	14.55
Macaca	unweighted	1	70.30	12.20	17.35	21.69
	weighted	6	70.30	12.20	17.35	18.08
Rattus	unweighted	33	128.40	24.11	18.78	18.92
	weighted	237	125.87	15.04	11.95	11.96
Mus	unweighted	6	102.17	37.94	37.14	38.69
	weighted	34	105.24	11.08	10.53	10.61

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Occipital Cortex CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	31	37.37	7.10	18.99	19.15
	weighted	442	38.01	7.51	19.76	19.77
Macaca	unweighted	7	55.79	12.05	21.61	22.38
	weighted	31	53.27	7.04	13.21	13.32
Lepus	unweighted	1	76.40	19.70	25.79	32.23
	weighted	5	76.40	19.70	25.79	27.07
Felis	unweighted	1	58.00	8.00	13.79	17.24
	weighted	4	58.00	8.00	13.79	14.66
Meriones	unweighted	1	69.40	20.60	29.68	37.10
	weighted	6	69.40	20.60	29.68	30.92
Rattus	unweighted	37	94.25	13.74	14.57	14.67
	weighted	276	94.20	12.69	13.47	13.48
Mus	unweighted	4	97.75	34.21	35.00	37.18
	weighted	24	100.75	13.77	13.67	13.81

Occipital Cortex CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	18	167.31	32.81	19.61	19.88
	weighted	240	165.20	30.57	18.50	18.52

Sensorimotor Cortex CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	10	37.31	7.76	20.80	21.32
	weighted	113	36.79	6.60	17.94	17.98
Macaca	unweighted	6	51.61	7.39	14.32	14.92
	weighted	23	51.58	3.97	7.70	7.78
Lepus	unweighted	1	82.65	15.30	18.51	23.14
	weighted	5	82.65	15.30	18.51	19.44
Meriones	unweighted	1	67.20	16.70	24.85	31.06
	weighted	6	67.20	16.70	24.85	25.89
Rattus	unweighted	38	96.61	11.64	12.05	12.13
	weighted	270	96.09	10.09	10.50	10.51
Mus	unweighted	6	103.50	27.56	26.63	27.74
	weighted	34	102.59	12.37	12.06	12.15

Sensorimotor Cortex CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	5	148.43	34.39	23.17	24.33
	weighted	101	152.69	26.85	17.59	17.63

Cingulate Cortex CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	12	38.90	7.50	19.28	19.68
	weighted	110	39.19	6.52	16.64	16.68
Macaca	unweighted	7	42.96	9.56	22.26	23.05
	weighted	29	42.35	6.59	15.56	15.70
Felis	unweighted	1	47.00	10.00	21.28	26.60
	weighted	4	47.00	10.00	21.28	22.61
Rattus	unweighted	32	97.88	14.42	14.73	14.85
	weighted	248	97.93	12.77	13.04	13.05
Mus	unweighted	3	83.00	19.97	24.07	26.07
	weighted	18	83.33	12.00	14.40	14.60

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Cingulate Cortex CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	3	150.80	36.96	24.51	26.55
	weighted	49	132.90	29.67	22.32	22.44
Thalamus CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	33	36.51	6.26	17.16	17.29
	weighted	406	37.41	6.95	18.58	18.59
Macaca	unweighted	10	45.22	5.26	11.63	11.92
	weighted	44	44.82	4.41	9.84	9.90
Lepus	unweighted	1	89.70	9.40	10.48	13.10
	weighted	5	89.70	9.40	10.48	11.00
Felis	unweighted	2	71.70	19.37	27.02	30.40
	weighted	11	75.44	14.32	18.98	19.41
Meriones	unweighted	2	74.40	5.52	7.41	8.34
	weighted	8	76.35	13.84	18.13	18.69
Rattus	unweighted	44	89.44	13.79	15.41	15.50
	weighted	334	89.80	12.06	13.43	13.44
Mus	unweighted	5	116.32	27.67	23.79	24.98
	weighted	29	114.39	14.51	12.69	12.79
Thalamus CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	9	177.33	38.21	21.55	22.15
	weighted	102	161.66	31.52	19.50	19.54
Thalamus CBF		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	8	54.23	7.59	14.00	14.43
	weighted	114	54.15	11.70	21.61	21.66
Sus	unweighted	1	76.80	4.50	5.86	7.32
	weighted	6	76.80	4.50	5.86	6.10
Meriones	unweighted	1	117.00	10.00	8.55	10.68
	weighted	2	117.00	10.00	8.55	9.62
Rattus	unweighted	4	131.00	14.63	11.17	11.86
	weighted	26	129.92	21.11	16.25	16.40
Hippocampus CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	8	30.34	6.71	22.12	22.81
	weighted	106	29.25	4.79	16.38	16.42
Macaca	unweighted	7	34.10	4.57	13.41	13.89
	weighted	31	34.62	3.15	9.10	9.17
Lepus	unweighted	1	58.10	5.60	9.64	12.05
	weighted	5	58.10	5.60	9.64	10.12
Felis	unweighted	2	50.00	2.83	5.66	6.36
	weighted	11	50.55	12.90	25.52	26.10
Meriones	unweighted	1	69.00	19.00	27.54	34.42
	weighted	6	69.00	19.00	27.54	28.68
Rattus	unweighted	44	65.59	10.38	15.83	15.92
	weighted	327	65.36	8.67	13.26	13.27
Mus	unweighted	6	69.34	19.49	28.11	29.28
	weighted	49	74.54	33.99	45.60	45.83

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Hippocampus CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	1	137.05	24.60	17.95	22.44
	weighted	14	137.05	24.60	17.95	18.27
Hippocampus CBF		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	2	47.40	7.35	15.51	17.45
	weighted	40	48.96	8.54	17.44	17.55
Macaca	unweighted	1	53.90	7.60	14.10	17.63
	weighted	6	53.90	7.60	14.10	14.69
Meriones	unweighted	1	109.00	9.00	8.26	10.32
	weighted	2	109.00	9.00	8.26	9.29
Rattus	unweighted	5	92.60	13.07	14.11	14.82
	weighted	30	94.00	13.63	14.50	14.62
Basal Ganglia CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	32	36.72	6.99	19.04	19.19
	weighted	372	37.63	6.93	18.42	18.43
Macaca	unweighted	9	44.50	5.83	13.09	13.46
	weighted	40	44.33	5.65	12.74	12.82
Lepus	unweighted	1	58.70	8.00	13.63	17.04
	weighted	5	58.70	8.00	13.63	14.31
Felis	unweighted	2	54.55	6.86	12.57	14.15
	weighted	11	55.87	12.76	22.84	23.36
Meriones	unweighted	3	83.80	27.86	33.25	36.02
	weighted	13	74.88	13.80	18.43	18.78
Rattus	unweighted	42	71.95	10.23	14.22	14.31
	weighted	307	71.74	5.54	7.72	7.73
Mus	unweighted	5	85.64	31.31	36.56	38.38
	weighted	30	85.18	12.65	14.85	14.97
Basal Ganglia CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	11	169.63	33.78	19.92	20.37
	weighted	133	156.16	28.11	18.00	18.03
Macaca	unweighted	1	255.00	32.00	12.55	15.69
	weighted	6	255.00	32.00	12.55	13.07
Basal Ganglia CBF		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	4	45.75	5.28	11.54	12.27
	weighted	52	43.54	7.46	17.13	17.21
Macaca	unweighted	1	60.50	11.10	18.35	22.93
	weighted	6	60.50	11.10	18.35	19.11
Sus	unweighted	1	102.60	5.80	5.65	7.07
	weighted	6	102.60	5.80	5.65	5.89
Meriones	unweighted	2	113.75	47.73	41.96	47.21
	weighted	7	99.29	10.00	10.07	10.43
Rattus	unweighted	2	117.50	7.78	6.62	7.45
	weighted	12	119.33	19.36	16.22	16.56

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Cerebellum CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	22	33.02	5.68	17.21	17.40
	weighted	299	33.91	6.06	17.87	17.89
Macaca	unweighted	6	28.10	4.49	15.97	16.64
	weighted	25	28.12	4.64	16.50	16.66
Felis	unweighted	1	31.50	6.00	19.05	23.81
	weighted	4	31.50	6.00	19.05	20.24
Meriones	unweighted	1	44.40	14.40	32.43	40.54
	weighted	6	44.40	14.40	32.43	33.78
Rattus	unweighted	28	53.95	12.26	22.72	22.93
	weighted	220	54.85	9.30	16.95	16.97
Mus	unweighted	4	90.50	7.59	8.39	8.92
	weighted	25	91.52	12.08	13.20	13.33

Cerebellum CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	8	183.41	30.87	16.83	17.36
	weighted	92	180.09	31.66	17.58	17.63

Cerebellum CBF		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	5	57.75	16.49	28.56	29.98
	weighted	73	56.99	11.68	20.50	20.57
Macaca	unweighted	1	47.80	5.10	10.67	13.34
	weighted	6	47.80	5.10	10.67	11.11
Sus	unweighted	1	88.80	3.60	4.05	5.07
	weighted	6	88.80	3.60	4.05	4.22
Meriones	unweighted	1	93.00	15.80	16.99	21.24
	weighted	10	93.00	15.80	16.99	17.41
Rattus	unweighted	2	116.50	38.89	33.38	37.56
	weighted	14	112.57	17.04	15.14	15.41

White Matter CMRgl		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	13	19.15	5.36	27.99	28.53
	weighted	158	18.06	3.55	19.65	19.68
Macaca	unweighted	6	12.55	2.94	23.41	24.39
	weighted	24	12.34	2.22	17.99	18.18
Lepus	unweighted	1	26.10	5.60	21.46	26.82
	weighted	5	26.10	5.60	21.46	22.53
Felis	unweighted	1	12.00	2.00	16.67	20.83
	weighted	4	12.00	2.00	16.67	17.71
Rattus	unweighted	36	33.34	7.90	23.71	23.87
	weighted	265	34.41	6.08	17.67	17.69
Mus	unweighted	5	43.80	19.93	45.51	47.78
	weighted	28	44.57	8.76	19.65	19.83

White Matter CMRO2		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	10	74.80	10.54	14.09	14.44
	weighted	146	75.68	12.01	15.87	15.90

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White Matter CBF		N	Mean	Std. Deviation	CV	CV*
Homo	unweighted	11	23.07	3.61	15.65	16.01
	weighted	172	23.62	5.01	21.22	21.25
Sus	unweighted	1	33.60	1.80	5.36	6.70
	weighted	6	33.60	1.80	5.36	5.58
Meriones	unweighted	1	62.00	12.00	19.35	24.19
	weighted	2	62.00	12.00	19.35	21.77
Rattus	unweighted	1	35.00	4.00	11.43	14.29
	weighted	4	35.00	4.00	11.43	12.14

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